

The AASC Description of Consciousness

Referential Standing, Qualitative Tensors, and the Conscious Quotient

Consciousness and Qualia as a Fixed-Domain Admissibility Interface

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Dependency lock. This is a downstream AASC paper. It imports the fixed-domain admissibility kernel, quotient-interface theorem, standing/admissibility identity, no-hidden-remainder closure, same-scope operator exhaustion, and anchor–tensor–skin decomposition. Those upstream results are not reproved here. The local task is to instantiate them in the consciousness domain and derive the resulting record-reference, subject-index, qualitative-tensor, diachronic, cross-subject, and machine-consciousness consequences.

Claim architecture. The analysis distinguishes imported AASC results, local consciousness-domain constructions, local theorems under the dependency lock, and the AASC explicative identification of consciousness with standing-positive occupancy of the complete referential interface.

Abstract

Under the dependency lock of the AASC fixed-domain admissibility-and-standing framework, this paper develops a downstream description of consciousness and qualia. It imports the established AASC kernel—bivalent standing, fail-closed enforcement, witness-coherent evaluation, redescription covariance, quotient-interface formation, standing/admissibility identity, no-hidden-remainder closure, same-scope operator exhaustion, and anchor–tensor–skin decomposition—and asks what follows when that structure is instantiated on candidate records, candidate referents, subject indices, memory and report channels, and qualitative distinctions.

Report, computation, global broadcast, integrated information, predictive inference, recurrence, and neural correlation may realize or witness consciousness-related structure, but none alone fixes what a report, memory trace, discrimination, neural surrogate, or machine self-ascription refers to. The formal core proves that non-vacuous, reusable consciousness evidence requires a unique, redescription-invariant, diachronically compositional map from active record orbits to quotient-level referents. When the witness basis is consequence-complete and confirmation factors through that map, consciousness-level confirmation is presentation-independent.

The constructive result defines the qualified consciousness interface as the least simultaneous closure of neutral candidate records, referents, profiles, channels, subject relations, and standing conditions. Experiential status is earned only after internal reference, subject coordination, self-involving reuse, diachronic naturalness, standing, and qualitative nontriviality are jointly satisfied. The scope-indexed profile $\Pi_E^{(S)}$ records the authoritative same-domain consequences of record incidence, reference, contrast, memory, discrimination, access, self-indexing, transition, history, and compatibility. Within this dependency-locked regime, consciousness is standing-positive occupancy of the subject-indexed referential quotient interface. Qualia are its non-skin qualitative tensors: quotient-invariant differences that preserve standing while changing the complete profile. Null-effect same-domain qualitative surplus is skin under the imported closure. The account is substrate-neutral: a biological or artificial system that genuinely instantiates the full interface is structurally conscious under AASC, whereas fluent output, logging, self-monitoring, or externally imposed semantics alone are insufficient. The paper supplies exact qualification, false-positive, bridge, diachronic, comparative, empirical, and objection-jurisdiction criteria.

Central downstream theorem. Under the imported AASC dependency lock, consciousness-description structure factors through a standing-positive, subject-indexed referential quotient interface; qualia are the non-skin qualitative tensors of that interface.

Keywords: consciousness; qualia; AASC; dependency lock; referential admissibility; quotient interface; redescription invariance; standing; anchor–tensor–skin; first-person evidence; machine consciousness.

Contents

1	Introduction: From Mechanism to the AASC Description of Consciousness	1
1.1	Dependency lock and downstream scope	2
1.2	Imported dependencies and local application hypotheses	2
1.3	Three levels of claim under the dependency lock	2
1.4	Candidate-first vocabulary	4
1.5	Main results	5
1.6	Scope	5
1.7	Organization	6
2	The Diagnostic Problem: Records without Fixed Referents	6
2.1	Records are not self-interpreting	6
2.2	Physical completeness and consciousness-level self-sufficiency	7
2.3	Why correlation and mechanism are insufficient	7
2.4	Temporal, boundary, and surrogate ambiguities	8
2.5	Non-vacuity	8
2.6	Four diagnostic failures	8
3	AASC Infrastructure for Candidate Consciousness Evaluation	9
3.1	State domain and presentation identity	9
3.2	Admissible redescription	10
3.3	Standing and fixed-domain eligibility	10
3.4	Licensed transitions and covariance	11
3.5	Quotient descent	11
3.6	Invariant subject index without representative selection	12
3.7	Grounded trajectories	13
3.8	Anchor, tensor, and skin	13
3.9	Axiom and dependency register	13
4	Candidate Records, Referential Qualification, and Experiential Reference	14
4.1	Candidate record carrier and formation	14
4.2	A hierarchy from trace to experiential record	15
4.3	Record presentation identity and orbits	15
4.4	Candidate referents and raw reference assignments	16
4.5	Candidate profile schema and qualification kernel	16
4.6	The core factorization diagram	18
4.7	Self-involving closure	18
4.8	Record channels and referent transport	19
4.9	Reference, accuracy, and reliability	19
4.10	No canonical raw representative is required	20
4.11	Referential standing	20
5	Candidate Systems, Qualification, and the Conscious Interface	20
5.1	Candidate consciousness evaluation system	20

5.2	Qualification conditions	21
5.3	Least qualified-interface closure	21
5.4	Reduced representation of the closure	23
5.5	Experiential standing	23
5.6	Qualitative nontriviality	23
5.7	Occurrence and certification	24
5.8	A hierarchy of adequacy	24
5.9	The role of mechanisms	25
5.10	AASC-admissible consciousness	25
6	Reference Fixation and Description-Independent Confirmation	25
6.1	Experiential hypotheses and confirmation	26
6.2	Record formation descends	26
6.3	Reference-fixation necessity	27
6.4	Reference invariance	28
6.5	Reference-fixation sufficiency	28
6.6	Diachronic reference theorem	29
6.7	The representation theorem	29
6.8	Record/referent typing	30
6.9	No hidden selector theorem	30
6.10	Interface rigidity	30
6.11	What the theorems establish	31
7	The Consciousness-Description Bridge	31
7.1	AASC descriptive adequacy principle	31
7.2	The scope-indexed complete experiential profile	32
7.3	Consciousness-bearing distinctions	33
7.4	Bridge theorem	33
7.5	Earned experiential status	34
7.6	What the bridge proves	34
8	Profile Completeness as a Fixed-Scope Lock	35
8.1	Scope versioning and lock discipline	35
8.2	Local authority and profile admission	35
8.3	Expansion protocol	36
8.4	Post-admission tensor classification	36
8.5	Formal role names are not phenomenal primitives	37
8.6	Jurisdiction of framework and local challenges	37
9	The AASC Description of Consciousness as Referential Standing	38
9.1	From a condition on theories to a condition on systems	38
9.2	Why the mechanism classes are not enough	38
9.3	The standing-positive regime	39
9.4	The structural identification	39
9.4.1	The no-detached-referent argument	39

9.4.2	The no-detached-record argument	40
9.4.3	The uniqueness argument	40
9.5	Necessary, sufficient, and constitutive claims	40
9.6	Why this is not reportism	40
9.7	Why this is not generic functionalism	41
9.8	Why this is not substance dualism	41
9.9	Why this is not eliminativism	41
9.10	Why this is not panpsychism by default	41
9.11	Consciousness level versus content	41
9.12	First-person character	42
9.13	Consciousness without later evidence	42
9.14	The burden of a rival structural account	42
10	First-Person Priority and the Third-Person Reference Bridge	43
10.1	First-person record priority	43
10.2	Third-person records	43
10.3	Bridge certificate schema	44
10.4	Bridge theorem	44
10.5	Error, disagreement, and correction	45
10.6	No-report and surrogate-only cases	45
11	Qualia as Non-Skin Tensors of the Conscious Interface	46
11.1	The target problem	46
11.2	The complete profile as the qualitative test surface	46
11.3	Qualitative descriptors and variations	46
11.4	Anchor, tensor, and skin tests	47
11.5	Qualitative tensor theorem	47
11.6	Raw labels, reports, and qualia	48
11.7	Qualia are not mechanisms	48
11.8	Qualitative contrast and geometry	48
11.9	Intrinsicness, ineffability, and relational structure	49
11.10	Quality, intensity, modality, and valence	49
11.11	Qualitative identity across time	49
11.12	Physical and computational realization	50
11.13	Experiential discrimination criterion	50
11.14	Same-domain epiphenomenal qualia	50
11.15	Hard-remainder normal form	51
12	Diachrony, Memory, and the Stream of Consciousness	51
12.1	Experiential referents as trajectory objects	51
12.2	Temporal redescription	52
12.3	Memory as reference-preserving transformation	52
12.4	Report formation	52
12.5	The stream as retained quotient trajectory	53
12.6	Five continuity layers	53

12.7	No present-only identity	53
12.8	Recurrence and qualitative type identity	54
12.9	Temporal binding	54
12.10	Dreams, no-report states, and record-silent episodes	54
12.11	Blindsight and unconscious processing	55
12.12	Anesthesia, sleep, and standing transitions	55
12.13	Branching and duplicated continuers	56
12.14	Diachronic audit conditions	56
13	Comparator Analysis: What Leading Frameworks Supply and Still Owe	56
13.1	Conditional mechanism non-sufficiency	57
13.2	Functionalism	57
13.2.1	Best internal reference-fixing candidate	57
13.2.2	Residual AASC burden	57
13.2.3	Completion route	57
13.3	Global workspace theories	58
13.3.1	Best internal reference-fixing candidate	58
13.3.2	Residual AASC burden	58
13.3.3	Completion route	58
13.4	Recurrent-processing and neural-dynamics approaches	58
13.4.1	Best internal reference-fixing candidate	58
13.4.2	Residual AASC burden	59
13.4.3	Completion route	59
13.5	Integrated information theory	59
13.5.1	Best internal reference-fixing candidate	59
13.5.2	Residual AASC burden	59
13.5.3	Completion route	59
13.6	Predictive processing and active inference	59
13.6.1	Best internal reference-fixing candidate	60
13.6.2	Residual AASC burden	60
13.6.3	Completion route	60
13.7	Higher-order theories	60
13.7.1	Best internal reference-fixing candidate	60
13.7.2	Residual AASC burden	60
13.7.3	Completion route	60
13.8	Neural correlate approaches	61
13.9	Access and phenomenal consciousness	61
13.10	Computational theories and the Chinese-room concern	61
13.11	Behaviorism and the Turing tradition	61
13.12	Comparator matrix	61
13.13	Relation to the explanatory gap	62
14	Classic Problems Recast in the Referential Interface	62
14.1	The inverted spectrum	63
14.2	Philosophical zombies	63

14.3	The explanatory gap and the hard problem	64
14.4	What-it-is-likeness	64
14.5	Privacy and intersubjective evidence	65
14.6	First-person authority and error	65
14.7	Mary and new knowledge	65
14.8	The conceivability of absent qualia	66
14.9	The combination problem	66
14.10	The unity of consciousness	66
14.11	Phenomenal overflow	66
14.12	Illusionism	67
14.13	Summary of the recasting	67
15	False-Positive Filters: Why Logs, Control, and Self-Monitoring Are Not Enough	68
15.1	Priority order for false-positive exclusion	68
15.2	No mere-logging theorem	68
15.3	Negative model I: thermostat	69
15.4	Negative model II: external database logger	69
15.5	Negative model III: self-monitoring controller	69
15.6	Positive boundary case	70
15.7	Decision procedure	70
16	Cross-Subject Comparison, Machine Consciousness, and Collective Interfaces	71
16.1	Joint systems	71
16.2	Cross-subject reference compatibility	71
16.3	Pairwise versus global comparison	72
16.4	Machine consciousness: neither behaviorism nor biocentrism	72
16.5	External semantics and the lookup-table objection	74
16.6	Large language models and conversational systems	74
16.7	Copying, uploading, and branching	75
16.8	Collective consciousness	75
16.9	Animal and nonlinguistic consciousness	75
16.10	Clinical applications	76
16.11	Ethical status	76
17	Empirical and Modeling Protocol	76
17.1	Stage 0: state the claim type	76
17.2	Stage 1: fix the evaluated domain	77
17.3	Stage 2: classify candidate records before qualifying them	77
17.4	Stage 3: specify admissible descriptions	77
17.5	Stage 4: construct candidate quotients	78
17.6	Stage 5: list candidate quotient referents	78
17.7	Stage 6: state and test the reference rule	78
17.8	Stage 7: declare the candidate profile schema and qualification kernel	79
17.9	Stage 8: compute the least qualified-interface closure	79

17.10	Stage 9: test diachronic naturality	79
17.11	Stage 10: certify first-person/third-person bridges	80
17.12	Stage 11: classify qualitative descriptors	80
17.13	Stage 12: perform adversarial redescription and false-positive tests	80
17.14	Stage 13: compare theories at the interface level	81
17.15	Failure signatures and falsifiability	81
17.16	Uncertainty, reliability, and partial identification	81
17.17	No-report, clinical, and machine adaptations	81
17.18	Recommended reporting statement	82
17.19	Scientific payoff	83
18	Worked Model Pair: Logger versus Minimal Conscious Interface	83
18.1	Model A: detached logger	83
18.2	Model B: minimal qualified interface	83
18.3	State and record carriers	83
18.4	Redescription symmetry	84
18.5	Standing and transitions	84
18.6	Record formation	85
18.7	Experiential referents and record channels	85
18.8	Qualitative tensor structure	85
18.9	A skin permutation	86
18.10	A tensor permutation	86
18.11	An inadmissible reference resolver	86
18.12	Ambiguous temporal referent	86
18.13	Direct comparison	87
18.14	What the model demonstrates	87
19	Worked Human Case: Pain, Memory, and Later Report	87
19.1	Trial structure	87
19.2	Qualified profile	87
19.3	Reference versus accuracy	88
19.4	ATS role audit	88
19.5	First-person and third-person evidence	89
19.6	What the case establishes	89
20	Worked Human Case II: Visual Afterimage, Ambiguity, and Report Error	89
20.1	Candidate states and records	89
20.2	Qualified-record admission	90
20.3	Afterimage profile	90
20.4	ATS role audit	90
20.5	First-person and third-person evidence	91
21	Objection Jurisdiction and Exact Local Burdens	91
21.1	Four objection classes	92
21.2	“You did not reprove AASC”	92

21.3	“You proved a condition on evidence, not a description of consciousness”	92
21.4	“The theory assumes experience in its vocabulary”	92
21.5	“The complete profile is whatever you need it to be”	93
21.6	“Null-effect or epiphenomenal qualia remain possible”	93
21.7	“The theory is functionalism in new notation”	93
21.8	“Reference is merely stipulated”	93
21.9	“A self-monitoring controller is an obvious false positive”	94
21.10	“Machine consciousness is either impossible or too easy”	94
21.11	“The first-person/third-person bridge is only a schema”	94
21.12	“No-report consciousness is excluded”	94
21.13	“Sleep or anesthesia makes the person cease to exist”	94
21.14	“Leading theories already solve reference”	95
21.15	“The framework is unfalsifiable”	95
21.16	“The witness basis can never be complete”	95
21.17	“A canonical representative can fix reference harmlessly”	95
21.18	“Subject identity is smuggled in”	95
21.19	Burden-classifier matrix	95
21.20	Objection: profile revision is post hoc repair	96
21.21	Objection: qualified records are admitted by consciousness vocabulary	96
21.22	Objection: qualitative nontriviality is too weak	96
21.23	What must be shown locally	96
21.24	Audit conclusion	97
22	Discussion: What the AASC Account Closes, Localizes, and Leaves Open	97
22.1	What is formally closed	97
22.2	What the dependency lock closes	98
22.3	What is localized rather than eliminated	98
22.4	What is not supplied	98
22.5	Occurrence, certification, and realization	98
22.6	First-person and third-person structure	99
22.7	A disciplined pluralism about mechanisms	99
22.8	The status of phenomenology	99
22.9	Structural realism and ontological neutrality	99
22.10	Potential formal extensions	99
22.10.1	Probabilistic and coarse reference	100
22.10.2	Sheaf and descent formulations	100
22.10.3	Geometry of qualitative tensors	100
22.10.4	Causal realization models	100
22.10.5	Formal verification	100
22.11	Research program	100
22.12	Empirical and formal extension agenda	101
22.13	Why the account is substantive	101
23	Conclusion	101

A	Notation and Conceptual Crosswalk	102
A.1	Record hierarchy	105
A.2	Ordinary-language distinctions	105
B	Dependency-Lock and Claim-Status Audit	106
B.1	Dependency spine	106
B.2	Imported theorem lock	107
B.3	Local construction and theorem register	107
B.4	Companion sources	108
B.5	What must be shown locally	109
B.6	Objection routing	109
C	Formal Type Specification and Mechanization Boundary	109
C.1	Candidate structures	109
C.2	Qualification and complete profile	110
C.3	Key declarations	111
C.4	Formalization obligations	113
C.5	Dependency order	113
C.6	Machine-checkable audit requirements	113
D	Consciousness Attribution Certificate	114
D.1	Certificate fields	114
D.2	Verdict classes	115
D.3	Minimum publication statement	116
E	Failure-Mode Ledger	116
E.1	Failure-localization rule	118

1 Introduction: From Mechanism to the AASC Description of Consciousness

Consciousness science has become increasingly successful at correlating reports, behavioral discriminations, electrophysiological signatures, imaging patterns, perturbational responses, computational architectures, and informational measures with conditions in which subjects are ordinarily said to be conscious. Functionalist programs emphasize causal or computational organization; global-workspace programs emphasize widespread availability and broadcast; integrated-information approaches emphasize intrinsic causal or informational structure; predictive-processing approaches emphasize hierarchical inference; higher-order approaches emphasize representation of lower-order states; recurrent-processing and neural-correlate programs emphasize particular dynamical or anatomical conditions. These are serious implementation and marker programs rather than one settled theory (Seth and Bayne, 2022).

The present paper asks a prior structural question. When a theory takes a report, memory, button press, discrimination, neural signature, or machine output as evidence about experience, what within the theory fixes which state that record refers to? A theory may predict the record exactly, reproduce its causal ancestry, and preserve every externally measured regularity while still admitting more than one internally admissible assignment of the record to an early sensory state, a recurrent interval, a broadcast state, a decision state, a memory-ready state, an integrated complex, or a posterior trajectory. If no invariant rule selects the quotient-level target, the record's consciousness-level evidential role is underdetermined.

This is the referential gap. It differs from report error: a false report can still have a determinate target. It also differs from the traditional explanatory gap, although the two interact (Levine, 1983). The gap arises when the theory cannot say, in a manner preserved by its own admissible descriptions, what state, interval, or trajectory the record is about. A framework can therefore be predictively complete while remaining empirically non-self-sufficient with respect to consciousness. The diagnostic result was developed in Maley (2025); this paper supplies the downstream constructive architecture.

The construction imports three upstream AASC strands. Redescription-stable evaluation descends to a quotient interface on which standing and transition structure are presentation-independent (Maley, 2026b). Projection and mediation results distinguish quotient-level invariant reference from the optional and often impossible selection of a privileged raw representative (Maley, 2026c). The anchor–tensor–skin theorem then exhausts fixed-domain descriptive work: anchors fix admissibility preconditions, tensors carry quotient-level invariant content and compatibility, and skin is representational surplus without admissibility-bearing authority (Maley, 2026a).

The resulting thesis is stronger than the earlier observation that projection structures are merely mind-like. It is also more strictly typed. Quotients, stable identifiers, compression, and control occur in many nonconscious systems. The consciousness-domain instantiation additionally requires internally qualified records, unique quotient-level referents, subject coordination, self-involving reuse, diachronic naturality, standing, and a nontrivial qualitative profile.

Flagship thesis. Consciousness is not report, computation, global broadcast, integration, prediction, recurrence, or neural correlation alone. Under the dependency-locked AASC framework, consciousness is the standing-positive regime in which qualified records acquire

unique, invariant, diachronically reusable quotient-level referents. Qualia are the non-skin qualitative tensors of that referentially fixed interface.

1.1 Dependency lock and downstream scope

This paper is a downstream AASC application. It does not reconstruct AASC from first principles, and it does not reopen the upstream proofs of the fixed-domain kernel, quotient-interface formation, standing/admissibility identity, no-hidden-remainder closure, same-scope operator exhaustion, or the anchor–tensor–skin decomposition. Those results are imported as governing dependencies.

The local question is narrower and exact:

Once the AASC dependencies are in force, what structure must an admissible description of consciousness have?

A critique that denies AASC as such is an upstream framework objection. It is not a local objection to the present paper unless it also shows that the consciousness application fails to satisfy the imported hypotheses or that consciousness introduces a same-domain admissibility-bearing role not classified by the imported framework.

The dependency lock does not exclude criticism; it classifies criticism. Upstream objections target the imported AASC results. Local objections remain fully live and must target the consciousness-domain typing, qualification, profile, bridge, realization, or scope discipline. Jurisdiction is therefore explicit rather than immunized.

Dependency discipline.

- (D1) The upstream AASC layer fixes the admissibility regime: fixed domain, bivalence, standing, fail-closure, redescription covariance, quotient descent, no-hidden-remainder, and anchor–tensor–skin exhaustion.
- (D2) The present paper fixes the consciousness-domain instantiation: candidate records, candidate referents, subject indexing, reference maps, diachronic reuse, qualitative profiles, bridge certificates, and false-positive filters.
- (D3) The local theorems show that, under those imported dependencies, consciousness-description structure factors through a subject-indexed referential quotient interface.
- (D4) Local objections must target the application: typing, qualification, reference fixation, profile completeness at the declared scope, qualitative tensor classification, subject coordination, diachronic reuse, empirical bridging, or scope discipline.

A denial of the imported AASC dependencies is recorded as upstream denial, not as a defect internal to the downstream consciousness theorem.

1.2 Imported dependencies and local application hypotheses

1.3 Three levels of claim under the dependency lock

The argument is stratified by local task, not by uncertainty about the imported framework.

L1. Local evidential theorem. Records used as non-vacuous, reusable consciousness evidence require unique invariant quotient-level reference and diachronic naturality.

Structure	Status	Role in this paper
Fixed-domain bivalence and fail-closure	Imported	Fixes the single standing/admissibility gate and excludes same-lineage repair.
Redescription covariance and quotient descent	Imported	Removes presentation surplus and makes state, record, and transition consequences quotient-valued.
Standing/admissibility identity and no-hidden-remainder	Imported	Blocks competing same-domain gates and undeclared governance loci.
Anchor-tensor-skin effect exhaustion	Imported	Classifies every same-domain descriptive variation by definedness, invariant-profile effect, or skin.
Candidate record carrier R_0	Local	Supplies tokens proposed for use as consciousness evidence.
Candidate referent carrier Q_0	Local	Supplies quotient-level state, interval, or trajectory targets.
Subject index and channel family	Local	Coordinates self-involving formation, memory, reuse, and continuation.
Least qualified-interface closure $\mu\Gamma_S$	Local	Determines records, referents, subject structure, channels, standing, and profile simultaneously from the declared qualification kernel.
Complete profile $\Pi_E^{(S)}$	Local under imported closure	Declares the consequence-complete consciousness-bearing profile at versioned scope S ; the unadorned symbol denotes the active lock.
Qualitative tensor family	Local classification	Applies ATS to profile-changing, standing-preserving differences.

Table 1: Dependency lock: upstream AASC results are imported; consciousness-domain structures are constructed and audited locally.

- L2. Downstream AASC description theorem.** Given the imported AASC dependencies and the local consciousness-domain instantiation, every same-domain consciousness-bearing distinction factors through the conscious quotient interface or is classified as anchor, tensor, skin, hidden authority, or scope change.
- L3. AASC consciousness identification.** Within that dependency-locked regime, consciousness is standing-positive occupancy of the subject-indexed referential qualitative interface.

A denial of AASC completeness, no-hidden-remainder, or ATS is an upstream denial of the dependency framework. A local rejection of L2 or L3 must instead show that the consciousness-domain instantiation satisfies the imported hypotheses while leaving a same-domain consciousness-bearing distinction outside the interface, or that a declared local qualification or profile component is inadequate.

1.4 Candidate-first vocabulary

To avoid defining experience by using “experiential” in the primitives, the construction begins with neutral carriers. The candidate record carrier is R_0 , the candidate quotient-referent carrier is Q_0 , and the candidate interface is I_0 . A record or referent earns the subscript E only as a projection of the least interface closure that jointly establishes internal reference, uniqueness, subject coordination, self-involving closure, diachronic reuse, standing, and qualitative nontriviality. The construction separates generic logging and control structure from the consciousness-specific interface without mutual bootstrapping.

Standing is not a substance, moral status, or intensity. It is the binary eligibility predicate governing whether a state participates in the fixed consciousness-description regime. Graded arousal, vividness, confidence, and epistemic probability may be tensors or uncertainty measures; they do not turn one fixed-domain state into both standing-positive and standing-negative in the same respect.

The same discipline applies to qualia. A qualitative word, coordinate, code, or report is not a quale merely because it is used phenomenologically. A qualitative candidate is an AASC quale only when it is a quotient-invariant difference whose variation preserves standing while changing the complete subject-indexed profile. A mere renaming is skin. A condition whose variation destroys the regime is anchor-level.

This treatment rejects two conflation. It rejects reportism, because records and referents occupy different carriers. It also rejects same-domain null-effect phenomenal surplus: a proposed difference that changes no standing, reference, contrast, memory, discrimination, access, self-location, history, compatibility, or continuation has no load-bearing role under the imported closure. Calling such a surplus phenomenally real is an upstream denial of ATS/no-hidden-remainder or an explicit expansion of scope, not a local counterexample left inside the same fixed domain.

A second distinction is equally central. Operational tensors alter task, routing, optimization, logging, or control while leaving the qualified consciousness profile fixed. Qualitative tensors preserve standing while altering the qualified subject-indexed profile. A sufficiently organized controller is not conscious merely because it has operational tensors; a biological or artificial system that genuinely realizes qualitative tensors and the full interface is not disqualified by substrate.

1.5 Main results

- R1. Candidate quotient descent.** Redescription-covariant state transitions, witness extractions, and record formation descend to admissible orbit quotients.
- R2. Record-reference descent.** Any admissible candidate reference assignment factors through record orbits and candidate quotient states.
- R3. Reference-fixation necessity and sufficiency.** Non-vacuous consciousness confirmation requires a unique invariant referent; appropriately factored qualified reference is sufficient for presentation-independent confirmation.
- R4. Least qualified-interface closure.** Qualified records, referents, channels, subject structure, standing, and profile are determined simultaneously as the least fixed point of a monotone closure operator on candidate subinterfaces.
- R5. Diachronic and observational transport.** Reference preservation composes through memory, report, reuse, and certified first-person/third-person bridges.
- R6. Consciousness-description bridge.** Under the dependency lock and local profile completeness, every same-domain consciousness-bearing distinction factors through the qualified interface or is classified by the imported normal form.
- R7. Profile lock.** The complete profile is scope-versioned and governed by explicit admission and revision protocols; revision declares a new, not necessarily monotone, theorem scope rather than repairing a theorem in place.
- R8. Qualitative tensor classification.** A qualified descriptor is anchor, tensor, skin, hidden-selector failure, or scope change; AASC qualia are the non-skin tensors.
- R9. False-positive exclusion.** Traces, logs, and self-monitoring controllers do not qualify merely by possessing records, quotients, or operational tensors; qualified-record admission and qualitative nontriviality are separately tested.
- R10. Substrate-neutral machine consequence.** Any artificial system genuinely instantiating the full interface is AASC-conscious; fluent behavior and external semantics alone do not instantiate it.
- R11. Cross-subject and empirical criteria.** Intersubjective comparison and third-person measurement require compatible joint reference and model-relative bridge certificates.

1.6 Scope

The paper does not establish which human neural structures realize the interface, provide a numerical consciousness measure, make introspection infallible, require overt report in every episode, or settle physicalism, dualism, idealism, or panpsychism. Those questions lie outside the declared downstream scope. The paper’s empirical burdens are concrete: construct candidate quotients, reference maps, subject indices, profile components, channel naturality, bridge certificates, and realization tests for actual biological or artificial systems.

Audit note. The dependency boundary is exact. Upstream challenges address the imported AASC theorems; local challenges address their instantiation here: imported-hypothesis fit, least-closure qualification, profile completeness, reference construction, bridge certification, realization, or scope discipline.

1.7 Organization

Section 2 sharpens the referential problem. Section 3 states the imported AASC substrate as used locally. Section 4 constructs candidate records and referents. Section 5 states qualification and separates occurrence from certification. Section 6 proves the reference theorems; Section 7 proves the consciousness-description bridge; and Section 8 hardens profile completeness as a scope lock. Section 9 states the AASC consciousness identification, Section 10 gives the bridge certificate, and Section 11 classifies qualia. Section 12 treats memory and continuity. Section 13 gives dependency-locked structured comparisons. Section 14 treats inversion, zombies, and the hard remainder. Section 15 supplies false-positive filters, and Section 16 treats cross-subject and machine consciousness. Section 17 gives the empirical protocol; Section 18, Section 19, and Section 20 provide finite and ordinary-human worked cases. Section 21 routes objections by jurisdiction and states exact local burdens. The appendices record notation, dependencies, the formal type specification, the attribution certificate, and the failure ledger.

2 The Diagnostic Problem: Records without Fixed Referents

2.1 Records are not self-interpreting

Scientific investigation of consciousness treats physical, behavioral, neural, or computational tokens as evidentially reusable. A subject says “I see red.” A button is pressed. A no-report paradigm yields an ocular, autonomic, or neural signature. A later memory is elicited. A perturbation changes a discriminative threshold. An artificial system generates a self-ascription. Within a theory these tokens are assigned a confirmational role: they are taken to support claims about whether consciousness occurred, what its content was, how it changed, and which mechanisms realized it.

A token’s causal production does not by itself determine that role. Consider a trial in which a stimulus initiates

$$x_0 \longrightarrow x_1 \longrightarrow x_2 \longrightarrow x_3 \longrightarrow r, \quad (1)$$

where r is a report. One theory may privilege x_1 , another x_2 , another x_3 , and another an interval or complex spanning several stages. They may agree on every measured output while disagreeing about the state evidenced by r . The token does not announce its own consciousness-level target.

To keep the diagnosis neutral, the initial terms do not presuppose that the target is already experiential.

Definition 2.1 (Candidate consciousness record). A *candidate consciousness record* is a token that a theory proposes to use as reusable evidence for the occurrence, identity, content, comparison, memory, or continuation of consciousness. The token may be overt or internal, biological or artificial, direct or surrogate. At this stage it is a candidate by assigned evidential role, not yet a qualified experiential record.

Definition 2.2 (Candidate consciousness referent). A *candidate consciousness referent* is a theory-internal state, state class, interval, or trajectory segment proposed as the target of a candidate record’s consciousness-level evidential role. It is not assumed to be nonphysical or phenomenally adequate merely because it has been named.

The distinction is typed. A report event may itself be represented as a system state, but its role as evidence remains distinct from the role of the object it records. A later metacognitive token may refer to the earlier report without collapsing the earlier report into the original target.

2.2 Physical completeness and consciousness-level self-sufficiency

A theory can be physically or functionally complete for production of its candidate records while remaining referentially incomplete. Let T predict every measured variable in an experimental domain. Let D_1 and D_2 be admissible internal descriptions preserving those predictions. Suppose D_1 assigns a record r to candidate referent q_1 and D_2 assigns it to q_2 , where q_1 and q_2 are distinct at the level at which the theory makes consciousness claims. The data do not determine which consciousness claim is confirmed unless the theory supplies an invariant rule fixing the target.

Definition 2.3 (Empirical self-sufficiency for consciousness claims). A theory is *empirically self-sufficient for consciousness claims* on a fixed domain when its active candidate records have determinate confirmational roles invariant across every admissible internal description of that domain and reusable under the theory's licensed temporal, memorial, and cross-contextual transformations.

The qualification matters. A theory may be self-sufficient for predicting button presses while non-self-sufficient for claims about conscious content.

Proposition 2.4 (Generic referential non-self-sufficiency). *Let a theory preserve the same active record tokens and predictive relations under two admissible descriptions while assigning at least one record to distinct candidate consciousness referents. If the theory contains no invariant reference-fixing rule, then it is not empirically self-sufficient for consciousness claims.*

Proof. An active record contributes to the ordering, confirmation, rejection, or parameterization of at least two consciousness hypotheses. If its target differs between admissible descriptions, a hypothesis true of the first target and false of the second acquires a different evidential relation under those descriptions, unless the confirmation procedure ignores the difference. If it ignores the difference, the record is not active with respect to the disputed consciousness distinction. Thus confirmation is either description-dependent or vacuous at that level. $\square$

This is the constructive starting point and restates the core diagnosis of Maley (2025) without yet assuming a theory of what qualifies as experience.

2.3 Why correlation and mechanism are insufficient

Suppose a neural variable N covaries perfectly with record r . Three claims remain distinct:

1. N predicts r ;
2. N participates in producing r ;
3. N , or a state determined by N , is the consciousness-level referent of r .

The first two do not entail the third. The same point applies to computation, global broadcast, integration, recurrence, prediction, and higher-order representation. Operationalization is indispensable, but operational success becomes a consciousness theory only when a reference bridge fixes the target.

2.4 Temporal, boundary, and surrogate ambiguities

The referential burden occurs in several dimensions.

Temporal slice. A record formed at t_4 may be caused by states at t_1, t_2, t_3 . The target might be onset, stabilization, broadcast, decision, consolidation, an interval, or a trajectory class. A coordinate timestamp is not enough when temporal origin or grain is itself redescription-sensitive.

System boundary. A mechanism can be described at the scale of a neuron, network, whole brain, organism, organism–environment loop, artificial agent plus memory, or distributed system. If the candidate bearer shifts while the evidential basis remains fixed, the theory owes a boundary-invariant target rule.

Surrogate records. No-report paradigms exchange verbal records for other records: pupil response, optokinetic response, decoding accuracy, autonomic change, or neural signature. The surrogate remains in need of a referent. Its distance from ordinary first-person evidence may increase rather than remove that burden.

Accuracy. A distorted memory may still refer determinately to an earlier event. Reference fixation is not truth, accuracy, confidence, or reliability; it is a precondition for evaluating them.

2.5 Non-vacuity

A theory could evade underdetermination by declaring that no record discriminates among consciousness hypotheses. Such a theory would be invariant but empirically empty at that level.

Definition 2.5 (Consciousness-level non-vacuity). A candidate record system is *non-vacuous* when there exist standing-positive records r_1, r_2 , or a standing-positive record r and hypotheses H_1, H_2 , such that the record-reference structure induces a determinate difference in consciousness-level confirmation, classification, or prediction.

Non-vacuity prevents the future reference map from being decorative annotation.

2.6 Four diagnostic failures

F1. Target absence. A record is used as evidence, but the theory identifies no candidate target.

F2. Target multiplicity. More than one inequivalent target remains admissible.

F3. Target drift. The assignment changes under admissible redescription, memory, report formation, or reuse.

F4. Target stipulation. A unique target is declared through an external parameter, privileged representative, post hoc convention, or theoretical label not carried by the fixed domain.

The constructive task is to block all four without adding a nonphysical selector. Quotient descent blocks presentation drift; internality blocks invisible selection; unique reference blocks multiplicity; channel naturality blocks diachronic drift; and the later qualification theorem determines when a candidate record/referent pair earns experiential status.

Audit note. The diagnosis does not assume that every dispute among consciousness theories is a mere redescription. The formal claim is conditional: whenever a theory’s own admissible description class preserves its evidence while varying consciousness attribution, that theory owes an invariant reference rule. Genuine cross-theory disagreement is illustrative, not a premise.

3 AASC Infrastructure for Candidate Consciousness Evaluation

This section states the fixed-domain substrate used throughout. The framework is adapted from the quotient-interface and evaluability architecture of Maley (2026b), with the projection clarification of Maley (2026c). The definitions are repeated here so that the consciousness results do not depend on informal import.

3.1 State domain and presentation identity

Definition 3.1 (Consciousness-candidate domain). A *consciousness-candidate domain* is a nonempty set S of well-typed system states together with a base equivalence relation $\sim$ representing immediate identity-preserving changes of presentation. If $x \sim y$, the two states differ only by a permitted presentation change and not by any consequence recognized as admissibility-bearing at the fixed scope.

A state may include biological, computational, bodily, environmental, or social components. The system boundary is not prescribed by the mathematics. It is fixed by the theory’s declared domain and by the internality requirement below.

Definition 3.2 (Witness basis). A *witness basis* is a family of partial maps

$$\mathcal{W} = \{w_i : S \rightarrow A_i\}_{i \in I}$$

with distinguished undefined values, satisfying:

- (W1) every datum used by a licensed same-domain evaluation, transition, branch, or reference mechanism is recoverable from finitely many witnesses or from represented transition-label data;
- (W2) witness values are covariant under admissible redescription;
- (W3) the basis separates every fixed-domain consequence that the theory treats as relevant to standing, candidate reference, qualitative contrast, memory, self-location, or licensed continuation;
- (W4) undefinedness is fail-closed rather than silently repaired.

The induced evaluation signature is

$$E : S \longrightarrow \Sigma_E, \quad E(x) = ([w_i(x)])_{i \in I}, \quad (2)$$

where harmless codomain presentation differences are quotiented. The witness basis need not contain every physical fact. It must contain every fact granted admissibility-bearing authority at the fixed scope.

Assumption 3.3 (Consequence completeness). When a theorem invokes *consciousness-profile completeness*, agreement on the witness basis, standing, quotient transitions, record-reference

structure, subject indexing, and declared compatibility relations is assumed to imply agreement on every same-domain consciousness-bearing consequence recognized by the theory. A further authoritative consequence defeats the active profile lock unless it can be recovered from the profile; any replacement result is stated at an explicitly revised scope.

This assumption is the explicit closure premise behind the later no-hidden-quale results. It is not smuggled into the definition of a witness.

3.2 Admissible redescription

Definition 3.4 (Admissible redescription groupoid). A class $\mathcal{G}$ of total maps $g : S \rightarrow S$ is an admissible redescription groupoid modulo $\sim$ when:

- (G1) $\text{id}_S \in \mathcal{G}$;
- (G2) $g_2 \circ g_1 \in \mathcal{G}$ whenever $g_1, g_2 \in \mathcal{G}$;
- (G3) each $g \in \mathcal{G}$ has an inverse $g^{-1} \in \mathcal{G}$ up to $\sim$;
- (G4) g preserves typing, witness consequences, standing, declared anchors, and the admissibility envelope;
- (G5) $x \sim y$ implies $g(x) \sim g(y)$.

The groupoid language accommodates local presentation changes and partially overlapping coordinate regimes. For most proofs the reader may think of a group action on a fixed set.

Definition 3.5 (Admissible orbit equivalence). Define

$$x \approx y \iff \exists g \in \mathcal{G} \, g(x) \sim y. \quad (3)$$

Let

$$Q := S / \approx, \quad \pi : S \rightarrow Q, \quad \pi(x) = [x].$$

The relation $\approx$ is an equivalence relation by identity, inverse, and composition. The quotient removes admissible presentation-orbit variation, not every observational or semantic difference.

3.3 Standing and fixed-domain eligibility

Definition 3.6 (Standing). A standing predicate is a map

$$\text{Elig} : S \rightarrow \{0, 1\}$$

that is invariant under $\sim$ and $\mathcal{G}$. A state with $\text{Elig}(x) = 1$ is standing-positive for the fixed evaluative domain; a state with $\text{Elig}(x) = 0$ is standing-negative.

The general AASC architecture treats standing as binary and fail-closed. In the consciousness application the generic gate is later restricted to the qualified consciousness-description domain and denoted ExpElig . This is not a second gate beneath standing: it is the same standing function restricted to the subdomain that passes the qualification conditions.

Assumption 3.7 (Nondegeneracy). The domain contains at least one standing-positive and one standing-negative state. When qualitative nontriviality is asserted, the declared admissible profile

space contains an active qualified referent and at least one standing-compatible quotient-distinct profile point, not necessarily simultaneously active.

3.4 Licensed transitions and covariance

Definition 3.8 (Licensed transition schemas). Let $\mathcal{T}$ be a class of well-typed same-domain transition schemas $T : \mathbf{S} \rightarrow \mathbf{S}$. The redescription groupoid acts on transition labels by $g \mapsto g_{\#}$, and the covariance condition is

$$g(T(x)) \sim (g_{\#}T)(g(x)) \quad (4)$$

whenever both sides are defined. A transition is *redescription-stable* when $g_{\#}T = T$ for all relevant g .

Transitions may represent perception, memory formation, attention, report formation, internal discrimination, self-location, intersubjective communication, or other licensed operations. Covariance requires the transition label to transform with the state; otherwise it would function as a hidden parameter.

Assumption 3.9 (Internal data closure). Every datum consulted by a licensed transition, witness, reference resolver, or comparison mechanism is represented in the state or in a declared transition label acted on by $\mathcal{G}$. No evidentially or referentially authoritative parameter remains invisible to admissible redescription.

Assumption 3.10 (Fail-closed continuation). A same-domain licensed transition preserves standing. A move that destroys, creates, repairs, or resets standing is represented as termination, boundary crossing, or explicit regime change rather than as an ordinary transition in $\mathcal{T}$.

This assumption does not mean that organisms cannot lose consciousness. It means that a loss-of-consciousness transition crosses the boundary of the qualified consciousness-standing regime and cannot be modeled as an ordinary positive-lineage transition without an explicit boundary map.

3.5 Quotient descent

Lemma 3.11 (Stable transition descent). *If $T \in \mathcal{T}$ is redescription-stable, then there is a unique induced partial map*

$$\hat{T} : \mathbf{Q} \rightarrow \mathbf{Q}$$

satisfying

$$\pi \circ T = \hat{T} \circ \pi \quad (5)$$

where defined.

Proof. Suppose $x \approx y$. Choose $g \in \mathcal{G}$ with $g(x) \sim y$. By stability and covariance,

$$T(y) \sim T(g(x)) \sim g(T(x)).$$

Hence $T(x) \approx T(y)$, so $[T(x)]$ is independent of representative. Define $\hat{T}([x]) = [T(x)]$. Surjectivity of π gives uniqueness. $\square$

For non-stable schemas the family $\mathcal{T}$ descends to a quotient transition relation rather than to an individual map. The consciousness results can be stated either relationally or for stable subfamilies; for readability most proofs use stable maps and note the relational generalization.

Proposition 3.12 (Universal property of the quotient interface). *Let J be a presentation-independent representation and $\eta : S \rightarrow J$ a map constant on admissible redescription orbits. Then there exists a unique $\bar{\eta} : Q \rightarrow J$ such that*

$$\eta = \bar{\eta} \circ \pi.$$

If J also preserves and separates every fixed-domain witness, standing, and transition consequence, $\bar{\eta}$ is faithful on the reduced interface.

Proof. Set $\bar{\eta}([x]) = \eta(x)$. Orbit constancy gives well-definedness and surjectivity of π gives uniqueness. Faithfulness follows from consequence separation. $\square$

The universal property prevents a later consciousness-specific layer from bypassing the quotient while remaining presentation-neutral. Any such layer factors through Q or changes the domain.

Definition 3.13 (Candidate referent carrier and interior interface). Let $Q_0 \subseteq \mathcal{O}(Q)$ be the family of quotient states, intervals, or retained trajectory objects that the theory proposes as possible consciousness referents. The candidate interior interface is

$$\mathcal{I}_0 = (Q_0, \bar{E}_0, \overline{\text{Elig}}_0, \hat{\mathcal{T}}_0, \prec_0),$$

where the remaining components are the restrictions of descended evaluation, standing, transition, and predecessor structure to Q_0 . The subscript 0 records that no experiential or consciousness status has yet been earned.

Every later qualified conscious interface is a structured subinterface of $\mathcal{I}_0$. Rich quotient organization alone is therefore not sufficient for consciousness.

3.6 Invariant subject index without representative selection

A consciousness theory generally attributes records and experiences to a continuing subject or interface. AASC must represent this without selecting a privileged raw state from each orbit, because symmetric orbits may admit no equivariant section.

Definition 3.14 (Invariant subject index). An invariant subject index is a map

$$\text{Subj} : Q \multimap J$$

to a set J of subject or interface identities. It is defined on a candidate or qualified subdomain and is constant by construction on raw representatives. It does not choose a section $Q \rightarrow S$.

The distinction repairs a common ambiguity. Quotient-level identity is available whenever orbit descent is available. A representative-valued resolver requires an additional equivariant section and may be impossible under symmetry. The present theory never makes consciousness depend on such a section.

3.7 Grounded trajectories

For diachronic reuse, states are organized into licensed trajectories. Let $\prec$ be a strict standing-positive predecessor relation invariant under redescription. When persistence claims are made, $\prec$ is assumed well-founded or otherwise equipped with finite invariant transport certificates, as in Maley (2026b). This prevents a continuity claim from being an ungrounded label.

Definition 3.15 (Candidate retained trajectory). A candidate retained trajectory is a finite or suitably well-founded sequence

$$\gamma : q_0 \rightarrow q_1 \rightarrow \cdots \rightarrow q_n$$

in Q induced by licensed quotient transitions, with standing preserved along every retained segment.

A later record may refer to an earlier point, an interval, or the trajectory as a whole. The candidate referent carrier Q_0 is therefore allowed to contain state, interval, and trajectory quotient objects. Only after qualification is the corresponding subcarrier denoted Q_E .

3.8 Anchor, tensor, and skin

The ATS decomposition is used in a specialized form.

Definition 3.16 (Effect classification). Let d be a descriptive or structural feature in a fixed candidate consciousness domain. Consider admissible variations of d while other declared data are held fixed.

1. d is *anchor-typed* if its variation changes typing, boundary membership, or the relevant standing predicate.
2. d is *tensor-typed* if standing remains fixed but the qualified quotient-level profile, reference structure, compatibility structure, or licensed continuation changes.
3. d is *skin-typed* if neither standing nor the complete qualified quotient-level profile changes.

The word “tensor” is a role term from the AASC decomposition, not necessarily a tensor in multilinear algebra. It denotes load-bearing quotient-level invariant content and relations. The later qualitative tensor theorem is conditional on the ATS effect exhaustion result: within the fixed complete domain, no fourth same-scope admissibility-bearing effect remains (Maley, 2026a).

3.9 Axiom and dependency register

The load-bearing assumptions of the consciousness construction are collected here for audit.

Assumption	Role in the construction
Fixed domain	Prevents silent expansion of the witness, reference, or mechanism class during an evaluation.
Witness coherence	Makes evaluative distinctions recoverable from declared internal data.
Redescription groupoid	Defines presentation neutrality and the orbit quotient.
Transition covariance	Allows lawful dynamics to descend without hidden transition-label parameters.

Assumption	Role in the construction
Internality	Blocks external selectors and meta-level reference assignment.
Binary fail-closed standing	Separates same-regime continuation from loss, reset, or repair.
Consequence completeness	Supports the bridge factorization, no-hidden-remainder result, and qualitative role exhaustion.
Non-vacuity	Prevents an invariant but experientially empty record system.
Grounded transport where used	Supports diachronic persistence and reusable memory claims.
ATS effect exhaustion	Supports the anchor/tensor/skin classification of qualitative descriptors.

Audit note. Nothing in quotient descent alone distinguishes conscious from nonconscious systems. The consciousness-specific work begins only after candidate records and referents pass internal-reference, subject-indexing, diachronic, standing, and qualitative-nontriviality tests. This paper does not repeat the overreach that generic mediation is sufficient for phenomenology.

4 Candidate Records, Referential Qualification, and Experiential Reference

The diagnostic problem becomes tractable only after records, referents, and their qualification conditions are explicitly typed. This section therefore proceeds in two stages. It first defines neutral candidate carriers possessed by many nonconscious systems. It then states the additional conditions under which a candidate token and candidate quotient object earn the terms *experiential record* and *experiential referent*.

4.1 Candidate record carrier and formation

Definition 4.1 (Candidate record carrier). Let R_0 be a set of candidate record tokens. A token may be neural, computational, behavioral, memorial, attentional, linguistic, or otherwise physically realized. Each token carries only the metadata declared by the fixed domain, such as source state, formation channel, subject/system index, and trajectory location.

Definition 4.2 (Candidate record formation). A candidate record formation relation is a covariant relation

$$\text{Form} \subseteq S \times R_0.$$

The assertion $\text{Form}(x, r)$ means that r is formed, instantiated, retained, or made available from x through a licensed channel. Covariance requires an induced action $g_R : R_0 \rightarrow R_0$ such that

$$\text{Form}(x, r) \implies \text{Form}(gx, g_R r) \tag{6}$$

for every admissible redescription g .

Using a relation rather than a total function allows one state to form several tokens, a token type to have many instances, and some candidate conscious states to form no overt report.

4.2 A hierarchy from trace to experiential record

The word “record” is too permissive unless its levels are separated. The following hierarchy is cumulative but not reversible.

Definition 4.3 (Trace). A *trace* is any state component whose present value depends on an earlier or concurrent condition. A thermostat state, cache bit, log entry, synaptic modification, and memory token may all be traces.

Definition 4.4 (Operational record). An *operational record* is a trace used by a system or observer to guide a task, comparison, update, or prediction. Operational use alone does not supply consciousness-level reference.

Definition 4.5 (Self-indexed record). A *self-indexed record* is an operational record bound by invariant structure to a system or subject index $j \in J$. It concerns the indexed system’s own state, history, or interaction rather than being merely an external database entry about it.

Definition 4.6 (Interior record). An *interior record* is a self-indexed record whose candidate reference is fixed by data inside the same standing-positive candidate interface and whose later use participates in at least one licensed discrimination, memory, access, self-location, or attribution channel of that interface.

Definition 4.7 (Experiential record). An *experiential record* is an interior record whose orbit belongs to the least qualified-interface closure $\mu\Gamma_S$. Thus “experiential” is an earned closure status, not a primitive annotation on an arbitrary token.

The inclusions are strict in principle:

$$\text{trace} \supseteq \text{operational record} \supseteq \text{self-indexed record} \supseteq \text{interior record} \supseteq \text{experiential record}.$$

A logger may reach the second level, and a sophisticated controller may reach the third or fourth, without satisfying the fifth.

4.3 Record presentation identity and orbits

Definition 4.8 (Record presentation identity and orbit). Let $\sim_R$ be immediate presentation identity on R_0 . Define

$$r \approx_R s \iff \exists g \in \mathcal{G} \, g_R(r) \sim_R s.$$

The candidate record-orbit carrier is

$$\overline{R_0} := R_0 / \approx_R, \quad \pi_R : R_0 \rightarrow \overline{R_0}.$$

Record wording, coordinate labels, implementation addresses, data layouts, and code names may change under redescription without changing the reduced token. The quotient removes only variations already declared non-authoritative.

4.4 Candidate referents and raw reference assignments

Definition 4.9 (Candidate referent carrier). The candidate referent carrier $Q_0 \subseteq \mathcal{O}(Q)$ contains the quotient states, intervals, or retained trajectory objects proposed as possible targets of candidate consciousness records.

Membership in Q_0 is not yet a consciousness verdict. Housekeeping, control, decision, broadcast, posterior, and memory-ready objects may all be included for comparison.

Definition 4.10 (Raw candidate reference assignment). A raw candidate reference assignment is a partial map

$$\rho_0 : \overline{R_0} \rightarrow Q_0.$$

It is admissible only if it is induced by declared internal witness and transition structure rather than by a theorist's unrepresented choice of state, temporal slice, subsystem, or representative.

At this stage ρ_0 may fail to be total, unique, invariant, self-involving, or diachronically stable. Those failures are precisely what the qualification conditions test.

4.5 Candidate profile schema and qualification kernel

The final qualified interface is constructed simultaneously rather than by declaring records, referents, and qualitative status one at a time. The construction begins with a profile schema on the neutral candidate carriers.

Definition 4.11 (Candidate profile schema at scope $\mathcal{S}$). Let $\mathcal{J}$ be a candidate subinterface of $\mathcal{C}_0$ with record carrier $R_{\mathcal{J}} \subseteq \overline{R_0}$ and referent carrier $Q_{\mathcal{J}} \subseteq Q_0$. For $q \in Q_{\mathcal{J}}$, the candidate profile

$$\Pi_{0,\mathcal{J}}^{(\mathcal{S})}(q)$$

collects the record incidence, candidate reference, contrast, memory, discrimination, access, subject-index, transition, history, and compatibility consequences available from the witnesses and licensed channels of $\mathcal{J}$. Candidate profile components are descriptive data only; none is yet granted experiential standing.

Definition 4.12 (Qualification-ready record/referent pair). A pair $(\bar{r}, q) \in \overline{R_0} \times Q_0$ is *qualification-ready* at scope $\mathcal{S}$ when all of the following hold relative to the full candidate structure:

1. **Internal fixation:** the data fixing q as the target of $\bar{r}$ are represented inside the candidate interface and transformed by admissible redescription;
2. **Closure-stable uniqueness:** no quotient-distinct $q' \neq q$ remains equally admissible as the target of the same active record orbit after all consequences forced by the declared qualification kernel are closed. Equivalently, every such q' is defeated, excluded, or rendered non-equally-admissible by the closure generated at scope $\mathcal{S}$;
3. **Subject coordination:** formation source, record, referent, and licensed reuse are compatible with one invariant subject/interface index;
4. **Represented self-involvement:** the candidate channel graph contains at least one licensed memory, discrimination, access, self-location, report, or action-attribution path of the same indexed interface involving $\bar{r}$;

5. **Diachronic naturality:** every declared candidate transformation of the record carries its target by a represented referent transport;
6. **Candidate standing:** formation, target assignment, and licensed reuse are well-typed and standing-positive at the declared scope;
7. **Profile admissibility:** q belongs to the declared candidate profile space and participates in a witness-supported, standing-preserving profile contrast of the kind specified in Theorem 5.13; the contrast witness need not be simultaneously active with q ;
8. **No detached semantics:** the target is not supplied solely by an external interpreter, an arbitrary canonical representative, or a post hoc selector.

Definition 4.13 (Qualification kernel). The qualification kernel $B_S \subseteq \overline{R_0} \times Q_0$ is the set of all qualification-ready record/referent pairs together with the witness-supported contrast, subject-index, and channel seeds declared by scope S . The kernel is fixed before qualified closure is computed and may contain only represented, redescription-covariant data.

The kernel does not by itself define the conscious interface. It supplies the internally supported seeds from which the least channel-closed, subject-coordinated interface is obtained in Theorem 5.7. This ordering prevents qualified records, qualified referents, qualitative nontriviality, and standing from being assigned by mutual verbal stipulation. The uniqueness test is closure-stable: a pair cannot be admitted merely because it looks unique before memory, channel, subject, profile, and standing consequences are propagated.

Definition 4.14 (Qualified carriers from least closure). Let $\mathcal{I}_C^{(S)} = \mu\Gamma_S$ be the least qualified-interface closure of Theorem 5.7. Its record projection is R^+ , its referent projection is Q_E , and its inherited unique reference map is

$$\rho : R^+ \longrightarrow Q_E. \quad (7)$$

The carrier $R \subseteq R_0$ consists of raw tokens whose record orbits lie in R^+ .

The terminology is earned by closure. A database token externally interpreted as “the system saw red” remains in R_0 unless its orbit belongs to $\mu\Gamma_S$. A qualified record can be entirely internal and need not be verbally reportable.

Qualified-record admission criterion. For the locked scope S , a raw token r and orbit $\bar{r} = \pi_R(r)$ enter the qualified carriers exactly when

$$r \in R \iff \bar{r} \in R^+ \iff \exists! q \in Q_E [(\bar{r}, q) \in \mu\Gamma_S],$$

and the closure witnesses internal fixation, closure-stable uniqueness, subject coordination, self-involving channel closure, diachronic naturality, standing, profile admissibility, and absence of detached semantics. Storage, consultation, behavioral use, or verbal form alone never satisfies this criterion.

Criterion 4.15 (Reference totality on active evidence). *Every orbit used as active evidence for a consciousness claim lies in $\text{dom}(\rho)$. A token outside the domain may predict behavior or mechanism but bears no consciousness-level confirmational weight in the fixed regime.*

Criterion 4.16 (Reference uniqueness). *For every $\bar{r} \in \text{dom}(\rho)$ there is exactly one $q \in Q_E$ with $\rho(\bar{r}) = q$.*

Because ρ is defined on record orbits and returns quotient objects, redescription invariance is built into its typing. At the raw level,

$$\rho(\pi_R(g_R r)) = \rho(\pi_R(r)). \quad (8)$$

4.6 The core factorization diagram

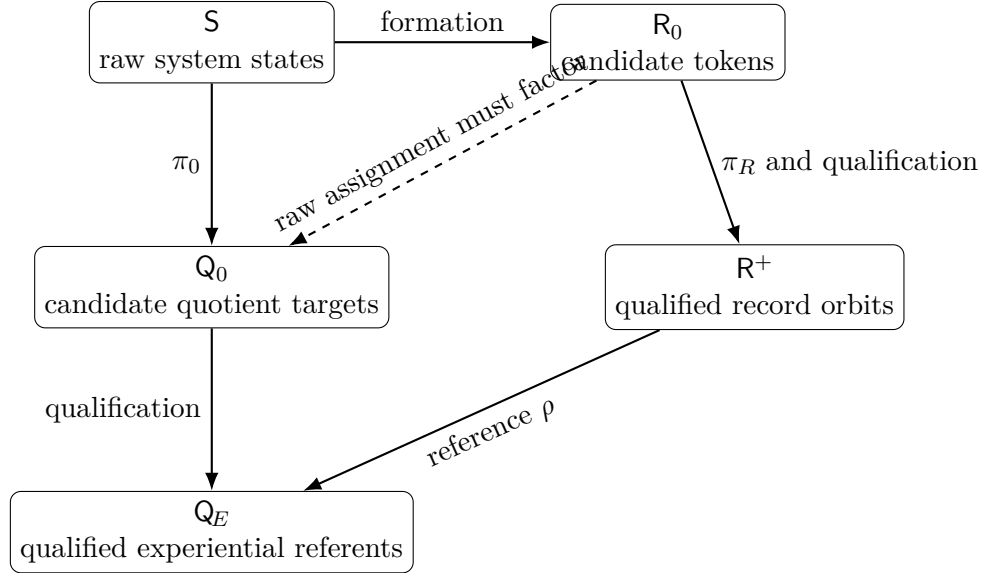


Figure 1: Candidate-first record-reference factorization. “Experiential” status appears only after record and referent qualification.

The diagram does not identify formation source and referent, and the qualification arrows denote projection from the least simultaneous closure rather than sequential relabeling. A later report may refer to an earlier state, interval, or trajectory. Temporal transport is represented by record channels.

4.7 Self-involving closure

Definition 4.17 (Self-involving record closure). A qualified record orbit $\bar{r}$ is self-involving relative to $j \in J$ when:

1. its formation source lies in the standing-positive subinterface indexed by j ;
2. $\text{Subj}(\rho(\bar{r})) = j$;
3. the record is available to at least one licensed transition of that interface governing later discrimination, memory, access, self-location, report, or attribution;
4. every datum establishing these facts is internal to the fixed domain.

This does not identify consciousness with control. It excludes detached records whose semantics are supplied only by an external observer.

Definition 4.18 (Experiential record system). An experiential record system is the qualified tuple

$$\mathfrak{R}_E = (R, \sim_R, \mathcal{G}_R, \text{Form}, R^+, \rho, \text{Subj})$$

satisfying covariance, orbit descent, total and unique reference on active evidence, and self-involving closure.

4.8 Record channels and referent transport

A record may be transformed by rehearsal, consolidation, recall, linguistic report, compression, translation, or communication.

Definition 4.19 (Record channel). A licensed record channel c consists of:

1. a partial map $C_c : R^+ \rightarrow R^+$;
2. a partial referent transport $\Phi_c : Q_E \rightarrow Q_E$;
3. declared source and target subject indices and typing conditions.

For a memory or report channel preserving reference to an earlier episode, Φ_c may be the identity. For a channel updating a record to a later stage, Φ_c may be an induced quotient transition.

Criterion 4.20 (Reference naturality). *For every licensed record channel c and every $\bar{r}$ where both sides are defined,*

$$\rho(C_c \bar{r}) = \Phi_c(\rho(\bar{r})). \quad (9)$$

Criterion 4.21 (Channel composition). *For composable channels c_1, c_2 ,*

$$C_{c_2 \circ c_1} = C_{c_2} \circ C_{c_1}, \quad \Phi_{c_2 \circ c_1} = \Phi_{c_2} \circ \Phi_{c_1}$$

where defined.

Lemma 4.22 (Compositional reference preservation). *If c_1 and c_2 satisfy reference naturality and compose, then the composite channel satisfies reference naturality.*

Proof. For every $\bar{r}$ in the composite domain,

$$\rho(C_{c_2} C_{c_1} \bar{r}) = \Phi_{c_2}(\rho(C_{c_1} \bar{r})) = \Phi_{c_2} \Phi_{c_1}(\rho(\bar{r})).$$

Composition identifies the two sides with the composite channel and transport. $\square$

The channels, objects, and transports form a category-like structure: records and qualified referents are different object types connected by a natural reference assignment. No heavy category theory is required, but the naturality language makes diachronic reuse precise.

4.9 Reference, accuracy, and reliability

Reference fixation is prior to accuracy. If a subject reports “red” while the target has the theory’s green tensor profile, the report may be inaccurate while still referring to a determinate target. Reliability concerns how often a channel preserves or correctly characterizes its target. Neither concept can be assessed when the target is unfixed.

4.10 No canonical raw representative is required

Proposition 4.23 (No representative requirement). *A unique quotient-level reference map does not require an equivariant section selecting one raw state from every state orbit or one raw token from every record orbit.*

Proof. The domain and codomain of ρ are already quotient carriers. Its value is a quotient object, not a raw representative. A section would be required only for a further representative-valued operation, which the reference relation does not perform. $\square$

This distinction is essential in symmetric systems, where an equivariant representative selector may not exist.

4.11 Referential standing

Definition 4.24 (Referential standing). A qualified record orbit $\bar{r}$ has *referential standing* when:

1. $\bar{r} \in \text{dom}(\rho)$;
2. $\rho(\bar{r})$ is unique and redescription-invariant;
3. the record's licensed reuse channels satisfy naturality;
4. no external parameter, hidden representative convention, or post hoc repair fixes its target;
5. the record remains within the standing-positive qualified interface.

A token lacking referential standing may still be scientifically useful as a predictor or mechanistic marker. It cannot, without further structure, function as determinate evidence about consciousness.

5 Candidate Systems, Qualification, and the Conscious Interface

The quotient substrate and candidate record-reference structure are generic. They occur in systems that should not automatically be called conscious. This section packages the neutral machinery, states the qualification conditions that earn experiential terminology, and distinguishes structural occurrence from empirical certification.

5.1 Candidate consciousness evaluation system

Definition 5.1 (Candidate consciousness evaluation system). A candidate consciousness evaluation system is a tuple

$$\mathcal{C}_0 = (\mathcal{S}, \sim, \mathcal{G}, \mathcal{T}, \mathcal{W}, E, \text{Elig}, \mathbf{Q}_0, \mathbf{R}_0, \text{Form}, \rho_0, \text{Subj}, \mathcal{K}_0), \quad (10)$$

where the first seven components form a nondegenerate fixed-domain admissible evaluation system, $\mathbf{Q}_0$ is the candidate quotient-referent carrier, $\mathbf{R}_0$ is the candidate token carrier, ρ_0 is a partial candidate reference assignment, Subj is a possible invariant index, and $\mathcal{K}_0$ is a family of candidate record channels and referent transports.

No consciousness verdict follows from possession of $\mathcal{C}_0$. A thermostat, database, control architecture, or scientific instrument may instantiate substantial parts of this tuple.

5.2 Qualification conditions

Definition 5.2 (Consciousness-interface qualification). A candidate subinterface of $\mathcal{C}_0$ is *consciousness-qualified* at scope $\mathcal{S}$ when it satisfies:

- (Q1) **Reference fixation:** every active interior record orbit has one quotient-level target fixed by represented internal data;
- (Q2) **Redescription invariance:** state, record, reference, profile, and transition structures commute with every admissible redescription;
- (Q3) **Subject coordination:** record source, referent, and later reuse are compatible with one invariant subject/interface index;
- (Q4) **Self-involving closure:** records and referents are closed under the declared memory, discrimination, access, self-location, report, and attribution channels of the same indexed interface;
- (Q5) **Diachronic naturality:** record channels and referent transports compose without reference drift;
- (Q6) **Standing:** the subinterface is well-typed, nondegenerate, internally closed, and fail-closed at the declared scope;
- (Q7) **Qualitative nontriviality:** its admissible profile space contains a standing-preserving, witness-supported difference in at least one designated qualitative component;
- (Q8) **No detached semantics:** qualification does not depend solely on an external interpreter, arbitrary canonical representative, or post hoc selector.

These conditions distinguish a consciousness-specific interface from an ordinary reference, logging, or control system. They are jointly imposed through the closure construction rather than assigned in a circular sequence.

5.3 Least qualified-interface closure

Definition 5.3 (Candidate fact universe). For a declared scope $\mathcal{S}$, let $\mathcal{F}_{\mathcal{S}}$ be the typed universe of candidate facts available to the consciousness-domain construction: record-orbit facts, candidate referent facts, raw reference facts, subject-index facts, channel edges, referent-transport facts, profile-component witnesses, standing facts, and failure/undefinedness facts. A *candidate fact set* is a subset $J \subseteq \mathcal{F}_{\mathcal{S}}$ containing the declared qualification kernel $B_{\mathcal{S}}$.

Working first over the powerset lattice $\mathcal{P}(\mathcal{F}_{\mathcal{S}})$ avoids a false lattice claim about already-consistent subinterfaces. Consistency and uniqueness are checked by a separate predicate rather than built into the carrier of the lattice.

Definition 5.4 (Scope consistency). A candidate fact set $J \subseteq \mathcal{F}_{\mathcal{S}}$ is *scope-consistent*, written $\text{Cons}_{\mathcal{S}}(J)$, when it contains no closure-relevant conflict: no active record orbit is assigned two quotient-distinct equally admissible targets; no subject index is assigned incompatible sources or referents; no standing-positive and standing-negative verdict are both asserted for the same fixed-domain item; no channel obligation forces reference drift; and no qualified verdict depends on hidden authority, detached semantics, or an unacted-on selector.

Definition 5.5 (Consistency-preservation certificate). A declared scope $\mathcal{S}$ admits least qualified closure only when the local realization supplies a consistency-preservation certificate: adjoining

the record, referent, channel, subject-index, profile, standing, and failure consequences required by (Q1)–(Q8) preserves $\text{Cons}_{\mathcal{S}}$ whenever it is applied to a scope-consistent candidate fact set. If the closure step introduces a conflict, the qualified interface is undefined at $\mathcal{S}$ and the construction fails closed rather than choosing among inconsistent candidates.

Definition 5.6 (Qualification-closure operator). The qualification-closure operator

$$\Gamma_{\mathcal{S}} : \mathcal{P}(\mathcal{F}_{\mathcal{S}}) \longrightarrow \mathcal{P}(\mathcal{F}_{\mathcal{S}})$$

adjoins every typed consequence required by (Q1)–(Q8): internally fixed record/referent pairs, closure-stable uniqueness exclusions, subject-index consequences, channel images, referent transports, profile witnesses, standing consequences, and failure/undefinedness facts. A local realization certificate must verify that $\Gamma_{\mathcal{S}}$ is well-typed, inflationary, monotone on $\mathcal{P}(\mathcal{F}_{\mathcal{S}})$, and consistency-preserving on the consistent region.

Theorem 5.7 (Least qualified-interface closure). *Assume $B_{\mathcal{S}} \subseteq \mathcal{F}_{\mathcal{S}}$ is well typed, $\Gamma_{\mathcal{S}}$ is a monotone inflationary operator on $\mathcal{P}(\mathcal{F}_{\mathcal{S}})$, and the declared scope has a consistency-preservation certificate. Then the least fixed point*

$$J_{\mathcal{S}}^* = \mu \Gamma_{\mathcal{S}} = \bigcap \{J \subseteq \mathcal{F}_{\mathcal{S}} : B_{\mathcal{S}} \subseteq J \text{ and } \Gamma_{\mathcal{S}}(J) \subseteq J\} \quad (11)$$

exists. If $\text{Cons}_{\mathcal{S}}(J_{\mathcal{S}}^)$ holds, its record carrier, referent carrier, reference map, subject structure, channel family, standing predicate, and profile determine the least consciousness-qualified interface*

$$\mathcal{I}_C^{(\mathcal{S})} = \mu \Gamma_{\mathcal{S}}.$$

If $\text{Cons}_{\mathcal{S}}(J_{\mathcal{S}}^)$ fails, no qualified interface is produced at $\mathcal{S}$; the declared scope fails closed.*

Proof. The powerset $\mathcal{P}(\mathcal{F}_{\mathcal{S}})$ is a complete lattice under inclusion. Since $\Gamma_{\mathcal{S}}$ is monotone, the Knaster–Tarski theorem supplies a least fixed point $J_{\mathcal{S}}^*$ above the declared kernel. The projections of this fixed point supply candidate records, referents, channels, subject-index facts, standing verdicts, and profile components simultaneously. The consistency-preservation certificate ensures that these projections remain well typed whenever the fixed point is consistent. If the fixed point contains conflicting targets, incompatible subject indices, standing contradiction, reference drift, or hidden authority, AASC fail-closure applies and the scope is rejected rather than repaired by a selector. Thus the least closure exists as a fact-set fixed point, and it yields a qualified interface exactly in the consistent case. $\square$

Corollary 5.8 (No qualification bootstrapping). *Membership in R^+ , Q_E , and the complete profile is inherited from one simultaneous least closure. No record qualifies merely because it points to a previously labeled experiential state, and no referent qualifies merely because a previously labeled experiential record points to it. Closure-stable uniqueness is tested after the channel, subject, standing, and profile consequences have been propagated.*

Definition 5.9 (Conscious evaluation system). A conscious evaluation system is a candidate system together with its least qualified-interface closure:

$$\mathcal{C} = (\mathcal{S}, \sim, \mathcal{G}, \mathcal{T}, \mathcal{W}, E, \text{ExpElig}, Q_E, R, R^+, \text{Form}, \rho, \text{Subj}, \mathcal{K}), \quad (12)$$

where R^+ , Q_E , ρ , Subj , $\mathcal{K}$, and ExpElig are the corresponding projections and restrictions of $\mu\Gamma_S$.

The definition names a theory structure. Attribution to a concrete organism or machine additionally requires a model-to-world realization and certificate.

5.4 Reduced representation of the closure

Definition 5.10 (Conscious quotient interface). The least closure has the reduced presentation

$$\mathcal{I}_C^{(S)} = (Q_E, Q_E^{\text{adm}}, \bar{E}, \overline{\text{ExpElig}}, \hat{\mathcal{T}}_E, \prec_E, R^+, \rho, \text{Subj}, \mathcal{K}, \Pi_E^{(S)}), \quad (13)$$

where $\Pi_E^{(S)}$ is the complete profile at the declared scope, introduced in Section 7 and governed by the lock discipline of Section 8. When no ambiguity arises, the scope superscript is suppressed.

The interface contains no privileged raw representative. It is the minimal presentation-independent carrier on which the declared consciousness-bearing consequences are well-defined.

5.5 Experiential standing

Definition 5.11 (Experiential standing). A raw state $x \in S$ is experientially standing-positive when $\text{ExpElig}(x) = 1$. At the declared scope this means:

1. the anchors required for the qualified interface are satisfied;
2. $[x]$ lies in the domain of $\mathcal{I}_C$;
3. every active qualified record generated or retained at x has referential standing;
4. licensed positive-lineage continuations preserve the interface or transform it by declared quotient maps;
5. no standing is created by hidden selection, representative convention, or post hoc repair.

This is a single same-domain gate. It is not a second consciousness predicate added beneath a more basic one.

5.6 Qualitative nontriviality

Definition 5.12 (Admissible profile space). For a locked scope S , let Q_E^{adm} be the qualified active referents together with profile-admissible contrast, background, null, memory-comparison, retained-trajectory, and licensed counterfactual profile points represented inside the declared scope. A point in Q_E^{adm} need not be an active conscious referent. It must, however, be standing-compatible inside the declared profile space: its witnesses, transitions, contrast role, and failure conditions are represented in the fixed domain. An external imagined alternative does not qualify.

Definition 5.13 (Minimal qualitative nontriviality). At scope S , the interface is qualitatively nontrivial exactly when there exist $q_1, q_2 \in Q_E^{\text{adm}}$ such that at least one belongs to the active referent carrier,

$$\{q_1, q_2\} \cap Q_E \neq \emptyset, \quad q_1 \text{ and } q_2 \text{ are standing-compatible in } Q_E^{\text{adm}}, \quad \Pi_E^{(S)}(q_1) \not\equiv \Pi_E^{(S)}(q_2),$$

and the non-isomorphism occurs in at least one designated qualitative component

$$\text{Contr}_E, \text{Mem}_E, \text{Disc}_E, \text{Access}_E, \text{Self}_E, \text{Trans}_E, \text{Hist}_E, \text{ or } \text{Compat}_E.$$

The separating variation must preserve the active interface’s standing and must not be exhausted by record count, implementation address, task routing, or another skin-level or merely operational difference. The two profile points need not be simultaneously active; one may be a represented contrast, background, null, memory-comparison, or licensed trajectory alternative within the same declared scope. A null or background point is therefore profile-admissible rather than an active conscious referent.

A simple or homogeneous episode can therefore be qualitatively nontrivial through its represented contrast and transition space even when only one qualitative state is occurrent at a given instant. Mere difference in Π_{task} remains insufficient unless it also produces a standing-preserving difference in the designated components of $\Pi_E^{(S)}$.

Operational and qualitative tensor classification is applied only after the least closure and the scope-admitted profile have been fixed. The classification does not determine which components enter $\Pi_E^{(S)}$; it evaluates variations relative to an already admitted profile, as formalized in Section 8 and Section 11.

5.7 Occurrence and certification

Definition 5.14 (Structural occurrence). For a modeled state x , $\text{Occ}_C(x) = 1$ means that the target system structurally occupies a standing-positive state of $\mathcal{I}_C$, independently of whether an investigator currently measures an external record.

Definition 5.15 (Empirical certification). For evidence set D , $\text{Cert}_C(x \mid D) = 1$ means that admissible external witnesses support the claim $\text{Occ}_C(x) = 1$ and are connected to the internal interface by invariant, reference-preserving bridge relations.

Occurrence and certification must not be conflated:

$$\text{Occ}_C(x) = 1 \not\Rightarrow \text{Cert}_C(x \mid D) = 1, \quad \text{Cert}_C(x \mid D) = 1 \text{ is defeasible evidence for } \text{Occ}_C(x) = 1.$$

A dream, locked-in state, or transient no-report episode may be structurally conscious while difficult to certify. Conversely, behavior can support a false-positive certificate when the internal interface model is inadequate. Overt report is not required, and a contemporaneously active external record is not required. What is required for a same-domain attribution is some possible reference-preserving, standing-preserving connection to the qualified interface—through interior record structure, profile consequence, lawful transition, memory, or a certified bridge. A state with no possible such relation cannot be assigned consciousness-standing inside the declared empirical scope without becoming skin or motivating an explicit scope revision.

5.8 A hierarchy of adequacy

A1. Trace adequacy: the theory identifies state-dependent traces.

A2. Record-predictive adequacy: it predicts observable or internal record tokens.

- A3. Mechanistic adequacy:** it explains how those tokens are produced.
- A4. Referential adequacy:** it fixes their quotient-level targets.
- A5. Interior adequacy:** records are self-indexed and used by the same internal interface.
- A6. Diachronic adequacy:** reference composes across memory, report, and reuse.
- A7. Qualitative adequacy:** the interface has a complete non-skin qualitative tensor profile.
- A8. Descriptive adequacy:** every same-domain consciousness-bearing distinction factors through the interface.
- A9. Certification adequacy:** the model-to-world mapping supports attribution to a concrete system.

Earlier levels can be achieved while later levels fail. The hierarchy is an audit decomposition, not a temporal law of scientific discovery.

5.9 The role of mechanisms

Mechanisms enter through $\mathcal{T}$, $\mathcal{W}$, Form, and realization of $\mathbf{Q}_E$. Computation, broadcast, recurrence, integration, prediction, higher-order representation, and neural dynamics may each be necessary in an implementation. The framework denies only the unmediated inference

$$\text{mechanism present} \implies \text{conscious interface present.}$$

A biological precondition can be anchor-like, invariant organization can realize tensor structure, and a coordinate description can be skin. Role assignment is by effect at the declared scope.

5.10 AASC-admissible consciousness

Definition 5.16 (AASC-admissible consciousness). A modeled state x is AASC-admissibly conscious at a fixed scope when

$$\text{Con}_{\text{AASC}}(x) = 1 \iff \text{ExpElig}(x) = 1 \text{ and } [x] \text{ occupies a nontrivial } \mathcal{I}_C. \quad (14)$$

Equation (14) is the downstream explicative identification supported by the reference representation theorems, the least qualified-interface closure, the consciousness-description bridge, and imported ATS exhaustion. It is not a synonym for report production or mechanism possession.

Dependency-locked status. Quotient descent, standing/admissibility identity, no-hidden-remainder, and ATS exhaustion are imported AASC results. Reference fixation, qualification, profile construction, and the bridge are local. Equation (14) is the downstream AASC consciousness identification. Rejecting the imported framework is an upstream objection; a local objection must show failed instantiation or a same-domain consciousness-bearing distinction outside the interface despite satisfaction of the imported hypotheses.

6 Reference Fixation and Description-Independent Confirmation

This section proves the paper’s central formal results. The target is not a metaphysical reduction of experience but a representation theorem for empirically self-sufficient consciousness theories.

6.1 Experiential hypotheses and confirmation

Let $\mathcal{H}_E$ be a class of experiential hypotheses. At minimum, each $H \in \mathcal{H}_E$ has a quotient-level evaluation

$$H : \mathbf{Q}_E \rightarrow V_H,$$

where V_H may be Boolean, ordered, probabilistic, or otherwise appropriate. The theory also has a confirmational functional

$$\mathfrak{C} : \mathcal{H}_E \times \mathbf{R}^+ \rightarrow L, \quad (15)$$

where L is an ordered or structured space of evidential effects. No Bayesian interpretation is required; L may encode support, rejection, likelihood contribution, model score, or a partial ordering.

Definition 6.1 (Referential factorization of confirmation). Confirmation *factors through experiential reference* when for each $H \in \mathcal{H}_E$ there is a map

$$\tilde{\mathfrak{C}}_H : \mathbf{Q}_E \rightarrow L$$

such that

$$\mathfrak{C}(H, \bar{r}) = \tilde{\mathfrak{C}}_H(\rho(\bar{r})) \quad (16)$$

for every active record orbit $\bar{r}$.

This condition says that the consciousness-level evidential effect of a record is determined by the experience it refers to, together with the hypothesis. Other features of the record may matter for reliability or noise, but if they affect consciousness-level confirmation they must be represented in an expanded reference or witness profile rather than remain hidden.

Definition 6.2 (Description-independent confirmation). Confirmation is *description-independent* when admissible redescription of the same state, record, hypothesis presentation, and transition labels leaves the confirmational effect unchanged up to the declared equivalence in L .

Definition 6.3 (Non-vacuously separating hypothesis class). The hypothesis class $\mathcal{H}_E$ is *non-vacuously separating* on $\mathbf{Q}_E$ when for any quotient-distinct q_1, q_2 that the theory treats as experientially different, there exists $H \in \mathcal{H}_E$ such that

$$\tilde{\mathfrak{C}}_H(q_1) \neq \tilde{\mathfrak{C}}_H(q_2).$$

Separation need not hold for every mathematically distinct state; only for distinctions the theory itself treats as consciousness-relevant. Without it, surplus quotient distinctions should be collapsed by minimality.

6.2 Record formation descends

Theorem 6.4 (Record-orbit descent). *Suppose the record formation relation is covariant under admissible redescription and record presentation identity preserves evidential role. Then the formation relation descends to a well-defined relation*

$$\overline{\text{Form}} \subseteq \mathbf{Q} \times \overline{\mathbf{R}_0}$$

given by

$$\overline{\text{Form}}([x], [r]) \iff \exists x' \approx x, r' \approx_R r \text{ with } \text{Form}(x', r').$$

Proof. Assume $x \approx y$ and $r \approx_R s$. Choose redescrptions and immediate presentation identities carrying (x, r) to (y, s) . Covariance of Form carries record formation along the state redescription, while record presentation identity preserves evidential role. Hence the truth of formation depends only on the two orbit classes. Symmetry and transitivity of the orbit relations show independence of the chosen representatives. $\square$

Corollary 6.5 (No raw-label evidence). *If two raw record tokens lie in the same record orbit, no consciousness-level confirmation functional may distinguish them solely by their raw labels without violating presentation neutrality or changing the witness domain.*

6.3 Reference-fixation necessity

We now formalize the diagnostic result as a theorem.

Theorem 6.6 (Experiential reference-fixation necessity). *Let a consciousness theory satisfy the fixed-domain, redescription, and non-vacuity assumptions. Suppose its experiential confirmation is description-independent and its hypothesis class separates every experiential distinction the theory retains. Then every evidentially active standing-positive record orbit has a unique quotient-level experiential referent. Equivalently, there exists a partial function*

$$\rho : R^+ \rightarrow Q_E$$

that is total on active evidence.

Proof. Fix an active record orbit $\bar{r}$. Existence is required because an active record must bear some consciousness-level confirmational relation. If no experiential target is assigned, then the record's evidential effect cannot be evaluated through any $H : Q_E \rightarrow V_H$; any effect attributed to it is either non-experiential or supplied by an external convention. Thus active experiential evidence requires at least one referent.

For uniqueness, assume toward contradiction that two distinct quotient referents $q_1 \neq q_2$ are admissible for $\bar{r}$. Because the hypothesis class separates retained experiential distinctions, choose H with

$$\tilde{\mathfrak{C}}_H(q_1) \neq \tilde{\mathfrak{C}}_H(q_2).$$

Under one admissible reference assignment the record contributes the first value; under the other it contributes the second. The record orbit, physical evidence, and admissible presentation are otherwise the same. Confirmation is therefore description- or assignment-dependent, contradicting the hypothesis. The only escape is for all hypotheses and confirmation functionals to ignore the distinction between q_1 and q_2 , in which case the distinction is not retained experientially and should be collapsed by minimality. Hence each active record orbit has exactly one retained quotient referent. $\square$

Audit note. The theorem does not infer a unique referent from predictive success alone. It infers uniqueness from the stronger conjunction of description-independent confirmation, a separating experiential hypothesis class, and minimality of retained distinctions. These premises state what it means for records to do non-vacuous theory-level work about experience.

Corollary 6.7 (Generic theories without reference fixation are non-self-sufficient). *A theory that preserves active experiential records while permitting multiple quotient-distinct referents and supplying no invariant selection rule is empirically non-self-sufficient for experience.*

This is the formal version of the result applied diagnostically to functionalist, workspace, integrated-information, and predictive-processing frameworks in Maley (2025).

6.4 Reference invariance

Uniqueness within one presentation is not enough. The referent must survive admissible redescription.

Theorem 6.8 (Reference invariance). *Under the assumptions of Theorem 6.6, let raw record tokens r and s lie in the same record orbit. Then their unique referents lie in the same experiential quotient object. Hence the reference assignment factors uniquely through π_R .*

Proof. Suppose $r \approx_R s$ but their unique raw assignments descend to distinct $q_r, q_s \in Q_E$. Since r and s are presentations of the same reduced record, a separating hypothesis produces different confirmational effects for two admissible presentations of the same evidence. This violates description independence. Therefore the assignment is constant on record orbits. By the quotient universal property there is a unique map $\rho : R^+ \rightarrow Q_E$ through which every raw admissible assignment factors. $\square$

6.5 Reference-fixation sufficiency

The converse requires explicit factorization and witness completeness.

Theorem 6.9 (Sufficiency for description-independent experiential confirmation). *Suppose:*

1. *active records are represented by record orbits R^+ ;*
2. *$\rho : R^+ \rightarrow Q_E$ is total and unique;*
3. *ρ is invariant under admissible redescription;*
4. *confirmation factors through ρ as in (16);*
5. *the witness and hypothesis profile is consequence-complete for the fixed experiential domain.*

Then experiential confirmation is description-independent.

Proof. Let r and s be raw record tokens related by admissible redescription. Their orbit classes coincide: $\pi_R(r) = \pi_R(s) = \bar{r}$. By reference invariance, both have the same quotient referent $q = \rho(\bar{r})$. For every H ,

$$\mathfrak{C}(H, \bar{r}) = \tilde{\mathfrak{C}}_H(q),$$

independently of the raw presentation. If the hypothesis itself is redescribed, consequence completeness and quotient invariance identify its reduced evaluation on q . Thus the confirmational effect is unchanged. Any alleged residual difference would be a same-domain consequence not represented in the complete profile, contradicting completeness. $\square$

Corollary 6.10 (Empirical self-sufficiency criterion). *A consciousness theory with consequence-complete witnesses is empirically self-sufficient for experience if its active records have total, unique, invariant, diachronically compositional quotient-level reference and its confirmation procedures factor through that reference structure.*

This is a criterion for theory structure, not a guarantee that the theory's empirical premises are true. A self-sufficient theory may still be empirically false.

6.6 Diachronic reference theorem

Theorem 6.11 (Diachronic reusability). *Let $c_1, \dots, c_n$ be composable licensed record channels satisfying reference naturality. Then a record orbit $\bar{r}$ can be reused after the composite channel without referential drift:*

$$\rho(C_{c_n} \cdots C_{c_1} \bar{r}) = \Phi_{c_n} \cdots \Phi_{c_1}(\rho(\bar{r})). \quad (17)$$

If every $\Phi_{c_i} = \text{id}$, the transformed record continues to refer to the same quotient experiential object.

Proof. Repeated application of Theorem 4.22 gives the result by induction on n . $\square$

Corollary 6.12 (Memory and later report). *A memory trace and a later report may serve as evidence for an earlier experience only if the memory and report channels carry the original reference according to (17). Mere causal descent from the earlier state is insufficient.*

6.7 The representation theorem

The necessity, invariance, and sufficiency results can be combined.

Theorem 6.13 (Experiential reference representation theorem). *Let $\mathfrak{T}$ be a nondegenerate fixed-domain consciousness theory with redescription-covariant states, records, and transitions. Assume:*

1. *the theory uses a non-vacuously separating class of experiential hypotheses;*
2. *its consciousness-level confirmation is invariant under admissible description;*
3. *active records are diachronically reusable under a compositional channel family;*
4. *the retained experiential state space is minimal with respect to the theory's consequence-complete profile.*

Then $\mathfrak{T}$ canonically induces, up to admissible interface isomorphism:

1. *an experiential quotient carrier $\mathbf{Q}_E$;*
2. *a standing-positive record-orbit carrier $\mathbf{R}^+$;*
3. *a unique invariant reference map $\rho : \mathbf{R}^+ \rightarrow \mathbf{Q}_E$;*
4. *compositional referent transports Φ_c making ρ natural under every licensed record channel;*
5. *a description-independent confirmation functional factoring through ρ .*

Conversely, any theory instantiating these structures under consequence completeness supports description-independent, diachronically reusable experiential confirmation.

Proof. State and transition quotient descent follows from Theorem 3.11 and Theorem 3.12. Record-orbit descent follows from Theorem 6.4. Existence and uniqueness of the reference map follow from Theorem 6.6; orbit invariance and factorization follow from Theorem 6.8. Diachronic naturality is required by the assumed reusability and represented by the channel maps; compositionality gives Theorem 6.11. Description-independent confirmation follows from Theorem 6.9. Minimality and the quotient universal property ensure uniqueness up to interface isomorphism. The converse is the conjunction of the sufficiency and diachronic results. $\square$

6.8 Record/referent typing

The framework blocks a common category error.

Proposition 6.14 (Records are not their referents). *In the typed conscious evaluation system, an experiential record orbit $\bar{r} \in R^+$ and its referent $q = \rho(\bar{r}) \in Q_E$ occupy distinct formal roles and are not identified merely because both may be physically encoded in system states. Any identity claim requires an explicit admissible isomorphism between the relevant carriers and must preserve the asymmetric record-to-referent role.*

Proof. The record carrier and referent carrier are distinct types in (12). The reference map has domain R^+ and codomain Q_E . Therefore the expression $\bar{r} = q$ is ill-typed unless embeddings into a common carrier are supplied. Even with such embeddings, the existence of an isomorphism between encoded objects does not erase the directed evidential relation: $\bar{r}$ is transformed by record channels, while q is transformed by referent transports. Collapsing them without preserving this structure makes the confirmation factorization circular and loses the distinction between evidence and target. $\square$

Remark 6.15. A self-referential experience is possible: a record may refer to a state containing that record or to an experience of reporting. The proposition does not prohibit self-reference. It prohibits unargued type collapse.

6.9 No hidden selector theorem

Theorem 6.16 (Reference cannot be fixed by an invisible selector). *Let two candidate quotient referents be indistinguishable by every declared internal witness and related by an admissible symmetry. No same-domain reference resolver using only internal data can select exactly one while remaining equivariant. A successful selection must reveal invariant asymmetry, add a declared boundary trace or section, inject an external parameter, or change the regime.*

Proof. An internal resolver is equivariant by internal data closure and redescription covariance. At a transitive two-candidate symmetry, an equivariant singleton selection would have to be fixed by the swap while also selecting one swapped element, a contradiction. This is the standard no-equivariant-selector argument. The listed alternatives exhaust the fixed-domain resolution modes under the AASC closure architecture. $\square$

This theorem prevents the reference map from being manufactured after the fact by a theorist's preferred temporal slice, subsystem, coordinate, or printed ordering.

6.10 Interface rigidity

Definition 6.17 (Conscious interface morphism). A morphism between conscious quotient interfaces preserves and reflects, as appropriate, experiential standing, witness signatures, quotient transitions, subject indexing, active record orbits, reference, channel transport, and qualitative tensor relations.

Theorem 6.18 (Local rigidity of the conscious interface). *Let two minimal conscious evaluation systems have the same complete experiential invariant profile up to relabeling. Then their reduced conscious quotient interfaces are isomorphic by an admissible interface morphism.*

Proof. Minimality identifies quotient classes with common fibers of the complete experiential invariant profile. Map each class in the first system to the class in the second with the corresponding profile, and map record orbits and signatures by the same relabeling. The shared profile makes these maps well-defined and bijective and ensures preservation of standing, transition, reference, channel, and qualitative relations. The inverse is constructed symmetrically. $\square$

Rigidity does not imply identical implementations. It implies that no second reduced interface can perform different experiential admissibility work while all declared invariant consequences remain fixed.

6.11 What the theorems establish

The formal results establish the following conditional statement:

Any theory that uses records as non-vacuous, description-independent, diachronically reusable evidence about experience must instantiate a quotient-level experiential reference interface, unique up to admissible isomorphism under minimality and completeness.

They establish the evidential representation result, not by themselves the constitutive consciousness thesis. The next section supplies the conditional bridge from evidence structure to complete AASC consciousness description; the following thesis section then states the explicative identification and its denial burden.

7 The Consciousness-Description Bridge

The reference theorems establish what a theory must provide for records to function as determinate evidence about consciousness. The local downstream task is then to apply the imported AASC no-hidden-remainder and anchor–tensor–skin dependencies to the declared consciousness-description domain. This section states that bridge in a fixed-scope form. The upstream AASC proofs remain dependency-locked.

7.1 AASC descriptive adequacy principle

Assumption 7.1 (AASC descriptive adequacy for consciousness). At a declared fixed scope, a theory describes consciousness only if it fixes every same-domain structure to which it grants authority for individuating, reidentifying, remembering, discriminating, accessing, self-locating, comparing, or confirming conscious states. A distinction that is authoritative but absent from the declared structure makes the description incomplete; a distinction with no such effect is not a same-domain descriptive addition.

The principle does not say that evidence creates consciousness. It fixes the local application burden: a downstream consciousness description cannot grant authority to distinctions absent from the structures by which it individuates, reidentifies, compares, and reuses its target. The upstream warrant for no hidden same-domain locus is imported under the dependency lock.

7.2 The scope-indexed complete experiential profile

Definition 7.2 (Scope-indexed complete experiential profile). Fix a declared consciousness-description scope $\mathcal{S}$ and its least qualified closure $\mathcal{I}_C^{(\mathcal{S})} = \mu\Gamma_{\mathcal{S}}$. For each admitted profile point $q \in \mathbb{Q}_E^{\text{adm}}$, define the qualified profile as the admitted restriction of the candidate profile schema:

$$\begin{aligned} \Pi_E^{(\mathcal{S})}(q) = \Pi_{0, \mathcal{I}_C^{(\mathcal{S})}}^{(\mathcal{S})}(q) = & (\text{Rec}_E(q), \text{Ref}_E(q), \text{Contr}_E(q), \text{Mem}_E(q), \text{Disc}_E(q), \\ & \text{Access}_E(q), \text{Self}_E(q), \text{Trans}_E(q), \text{Hist}_E(q), \text{Compat}_E(q)). \end{aligned} \quad (18)$$

The components are:

1. $\text{Rec}_E(q)$: the qualified record orbits formed from, retained by, referring to, or lawfully contrast-linked to q ; this component may be empty for a nonactual licensed comparison point;
2. $\text{Ref}_E(q)$: reference incidence, fiber, uniqueness, and reliability-relevant structure;
3. $\text{Contr}_E(q)$: qualitative contrast, exclusion, similarity-order, and discriminability relations;
4. $\text{Mem}_E(q)$: memory formation, retention, recall, and reference-preserving transport;
5. $\text{Disc}_E(q)$: subject-indexed internal and external discrimination consequences;
6. $\text{Access}_E(q)$: availability to internal use, report, attention, or action-attribution channels;
7. $\text{Self}_E(q)$: subject index, ownership, self-location, and first-person binding structure;
8. $\text{Trans}_E(q)$: lawful experiential transitions and counterfactual continuation profile;
9. $\text{Hist}_E(q)$: predecessor, trajectory, temporal-extension, and recurrence location;
10. $\text{Compat}_E(q)$: compatibility and unity relations with other qualitative tensors and co-instantiated contents.

The candidate schema is evaluated before qualification; the subscript E is earned only after restriction to the least closure. This separation prevents the complete profile from serving as its own admission rule. The profile need not be finite, metric, verbally reportable, or externally accessible at the time of structural occupancy. Consequence completeness means that every distinction granted same-domain consciousness-bearing authority at scope $\mathcal{S}$ is represented in, or recoverable from, the locked profile. Unadorned Π_E denotes the active locked scope.

Criterion 7.3 (Profile-component inclusion). *A candidate component C may enter $\Pi_E^{(\mathcal{S})}$ only if C is well-typed, witnessable from represented structure, invariant or covariant under admissible redescription, and non-vacuously affects at least one of the following: candidate standing attribution, candidate interface occupancy, referent identity, subject assignment, reference-preserving profile transport, certified evidence-role assignment, or standing-preserving qualitative compatibility. Phenomenological salience or intuitive importance alone does not admit a component. Conversely, a component is not excluded merely because it is difficult to measure; exclusion requires that it fail the stated witness, covariance, and consequence conditions at the declared scope.*

Definition 7.4 (Experiential profile equivalence). For $q, q' \in \mathbb{Q}_E^{\text{adm}}$,

$$q \equiv_E q' \iff \Pi_E^{(\mathcal{S})}(q) \cong \Pi_E^{(\mathcal{S})}(q')$$

under admissible relabeling and interface isomorphism. A minimal conscious interface identifies profile-equivalent quotient objects.

7.3 Consciousness-bearing distinctions

Definition 7.5 (Same-domain consciousness-bearing distinction). A variation $d \rightsquigarrow d'$ is same-domain consciousness-bearing when, with the fixed anchors held constant, it changes at least one locally authoritative verdict concerning candidate standing attribution, candidate interface occupancy, subject attribution, referent identity, declared profile-component classification, memory, comparison, lawful continuation, or certification.

The definition excludes decorative variation while leaving completeness open to a concrete profile challenge: a critic defeats the active lock by exhibiting an omitted, witnessable, redescription-stable authoritative consequence.

7.4 Bridge theorem

Theorem 7.6 (Consciousness-description bridge). *Let $\mathfrak{T}$ be a fixed-domain theory that gives a complete same-domain description of consciousness at scope $\mathcal{S}$. Assume:*

1. *the reference representation theorem, Theorem 6.13, applies to its active consciousness evidence;*
2. *the candidate system is qualification-determinate and $\mathcal{I}_C^{(\mathcal{S})}$ is its least qualified-interface closure;*
3. *its qualified records and referents satisfy subject coordination, self-involving closure, and diachronic naturality;*
4. *$\Pi_E^{(\mathcal{S})}$ is consequence-complete for every consciousness-bearing distinction recognized at the declared scope;*
5. *the imported AASC no-hidden-remainder and anchor-tensor-skin effect-exhaustion dependencies hold on the fixed descriptive carrier;*
6. *the reduced interface is minimal with respect to the complete profile.*

Then every same-domain consciousness-bearing distinction in $\mathfrak{T}$ factors through the conscious quotient interface

$$\mathcal{I}_C^{(\mathcal{S})} = (\mathbf{Q}_E, \mathbf{Q}_E^{\text{adm}}, \mathbf{R}^+, \rho, \Pi_E^{(\mathcal{S})}, \text{Subj}, \widehat{\mathcal{T}}_E, \prec_E, \overline{\text{ExpElig}}).$$

More precisely, any proposed distinction belongs to exactly one primary case:

- (B1) *it changes definedness or $\overline{\text{ExpElig}}$ and is anchor-level;*
- (B2) *it preserves standing but changes $\Pi_E^{(\mathcal{S})}$ and is tensor-level;*
- (B3) *it preserves standing and $\Pi_E^{(\mathcal{S})}$ and is skin/bookkeeping;*
- (B4) *it uses an undeclared selector or authority and is inadmissible in the fixed domain;*
- (B5) *it introduces a new witness, carrier, or governance locus and changes scope.*

Proof. By Theorem 5.7, the qualified records, referents, subject structure, channels, standing, and profile are projections of one simultaneous least fixed point rather than mutually bootstrapped declarations. By Theorem 6.13, active consciousness evidence then factors through unique invariant record orbits and quotient-level referents, with diachronic reuse represented by natural channel transports. Thus any consciousness distinction that affects evidence, memory, discrimination, subject attribution, or continuation must alter either the standing of the interface or some quotient-level consequence of a referent.

Take an arbitrary same-domain consciousness-bearing variation. If the variation changes whether the relevant description is well-typed, belongs to the evaluated domain, or remains standing-positive,

it occupies the definedness locus and is anchor-level. Suppose standing is fixed. By the descriptive-adequacy and consequence-completeness assumptions, every remaining authoritative consciousness consequence is represented in $\Pi_E^{(S)}$. If the variation changes such a consequence, it changes $\Pi_E^{(S)}$ and is tensor-level. If the profile is unchanged, then by ATS effect exhaustion and no-hidden-remainder closure it performs no same-domain consciousness-bearing work and is skin or bookkeeping. A rule that nevertheless forces a verdict must use data outside the redescription action, hence is a hidden selector, or it must enlarge the witness, carrier, or governance structure, hence changes scope. Minimality removes duplicate quotient objects with the same complete profile. Therefore every admissible same-domain consciousness-bearing distinction factors through $\mathcal{I}_C^{(S)}$ and falls into the stated normal form. $\square$

Corollary 7.7 (Descriptive uniqueness up to interface isomorphism). *Two minimal fixed-domain theories with isomorphic complete conscious interfaces make the same AASC consciousness description, even when their implementations and skin presentations differ. Any same-domain consciousness difference must locate a non-isomorphic profile component or defeat the active completeness lock.*

Corollary 7.8 (No detached same-domain consciousness property). *Within the assumptions of Theorem 7.6, a claimed property that can vary while standing, reference, subject index, and every component of $\Pi_E^{(S)}$ remain fixed has no same-domain consciousness-bearing role. It is skin, an external selector, or a scope extension.*

7.5 Earned experiential status

Proposition 7.9 (Non-circular qualification). *The construction does not infer experience from a primitive “experiential” label. The carriers R_0 , Q_0 , and $\mathcal{I}_0$ are defined neutrally. A token/referent pair earns membership in (R^+, Q_E) only through the least qualified-interface closure, which jointly enforces reference fixation, internality, subject coordination, self-involving closure, diachronic naturality, standing, and nontrivial profile conditions. Consequently, a challenge to experiential status must identify which qualification condition is insufficient rather than point merely to the vocabulary.*

7.6 What the bridge proves

The bridge theorem is the downstream representation and exhaustion result for the consciousness domain, built on the simultaneous least closure. It does not reopen the imported AASC framework. Its local force is exact:

Once a theory claims fixed-domain completeness for consciousness, the interface is not merely a convenient evidence layer. It is the unique reduced locus through which every same-domain consciousness-bearing distinction must pass.

Dependency-locked proof status. The bridge theorem is the local L2 result. It imports ATS exhaustion and no-hidden-remainder rather than reproving them. Under those dependencies, a local challenge must exhibit misapplication, a failed qualification condition, an omitted profile consequence, hidden authority, or scope change. Denial of ATS or no-hidden-remainder is an upstream objection to AASC, not an unclassified local remainder.

8 Profile Completeness as a Fixed-Scope Lock

The bridge theorem uses the complete profile $\Pi_E^{(S)}$ at a declared scope $\mathcal{S}$. The profile is not a catch-all for “whatever makes consciousness different.” It is a versioned scope lock whose components are admitted by witness, reference, transition, subject, or compatibility consequences. Its completeness is testable by exhibiting an omitted, witnessable, redescription-stable authoritative consequence; it cannot be silently expanded inside an already stated theorem or bypassed by undeclared same-domain authority.

8.1 Scope versioning and lock discipline

Definition 8.1 (Scope-versioned profile). For each declared consciousness-description scope $\mathcal{S}$, $\Pi_E^{(S)}$ denotes the consequence-complete profile certified for that scope. The unadorned Π_E is shorthand for the active locked version. Once a theorem invokes $\Pi_E^{(S)}$, its carrier, witnesses, authoritative consequences, and admission criteria are fixed for that theorem.

If an analysis identifies a genuine omitted tensor H , the correct response is not to repair the proof in place. One declares a revised scope

$$\mathcal{S}' = \text{Revise}(\mathcal{S}; H) \quad (\text{for example, } \mathcal{S}' = \text{Extend}(\mathcal{S}; H)),$$

constructs $\Pi_E^{(S')}$, and recomputes qualification, quotient minimality, and bridge results. The earlier theorem remains a theorem about $\mathcal{S}$ and is not retroactively altered. Profile revision is therefore explicit scope revision, not post hoc same-domain repair. Revision is not monotone by default: a new scope may preserve, refine, split, merge, or revoke earlier qualifications, and every affected theorem is re-established at the revised scope.

8.2 Local authority and profile admission

Definition 8.2 (Locally authoritative consciousness consequence). At a fixed consciousness-description scope, a consequence is locally authoritative when the theory permits it to alter at least one verdict concerning candidate standing attribution, candidate interface occupancy, referent identity, subject attribution, declared profile-component classification, memory, discrimination, access, comparison, continuity, compatibility, certified evidence-role assignment, or empirical confirmation.

A proposed profile component is admitted to $\Pi_E^{(S)}$ only if it satisfies all of the following:

- (P1) it is well-typed on the declared candidate or qualified carrier;
- (P2) its values or relations are determined by represented witnesses, reference structure, transition channels, subject indexing, history, or compatibility data;
- (P3) it is invariant or covariant under the admissible redescription action;
- (P4) variation in it changes a locally authoritative consequence;
- (P5) it does not duplicate a consequence already recoverable from the existing profile except by harmless recoding.

The admission test prevents both arbitrary inflation and arbitrary exclusion. Declaration alone is not sufficient: a proposed component that is not witness-supported, redescription-invariant, and

consequence-bearing at the declared scope is skin even if it is named in a table or glossary.

8.3 Expansion protocol

Let H be a proposed consciousness-bearing distinction not represented in the current profile. The fixed-scope audit proceeds in priority order:

- (E1) If varying H changes whether the interface is defined or standing-positive, H is anchor-level rather than an omitted qualitative component.
- (E2) If standing is preserved and H changes a locally authoritative consequence, the stated lock was incomplete. The theorem is not repaired in place: a revised scope $\mathcal{S}'$ is declared, H is admitted to $\Pi_E^{(\mathcal{S}')}$ under Theorem 7.3, all quotient and minimality results are recomputed, and H is classified as tensor-level in the revised version.
- (E3) If standing and every locally authoritative consequence are unchanged, H is skin under imported no-hidden-remainder and ATS exhaustion.
- (E4) If a verdict is forced by data not acted on by the redescription regime, H is hidden authority or an external selector.
- (E5) If H introduces a new carrier, witness family, governance locus, or comparison domain, the proposal changes scope and requires a new dependency-locked audit.

Theorem 8.3 (Profile-closure normal form). *Under the imported AASC no-hidden-remainder and anchor–tensor–skin dependencies, every proposed addition H to a fixed consciousness-description profile has exactly one primary status: anchor, admitted tensor component, skin, hidden authority, or scope change.*

Proof. The imported effect exhaustion first separates definedness change from positive-interior variation. Definedness change is anchor work. With standing fixed, a locally authoritative consequence is by definition part of the declared consciousness-description work; omission therefore triggers profile expansion and tensor classification. If no authoritative consequence changes, the variation performs no same-domain governance-bearing work and is skin. A forced verdict without represented effect depends on undeclared authority. A genuinely new carrier or governance locus changes scope. Priority ordering makes the cases exclusive. $\square$

Corollary 8.4 (Profile completeness is not definitional closure). *The profile is not complete because every difference is called a profile difference. It is complete only after every locally authoritative, witnessable, redescription-stable consequence has either been represented or shown recoverable from represented components. A successful local challenge identifies a concrete omitted consequence and defeats the completeness claim for $\mathcal{S}$; any replacement result is stated and proved at $\mathcal{S}'$. It does not license a load-bearing remainder outside the locked profile while leaving the original theorem unchanged.*

8.4 Post-admission tensor classification

Operational and qualitative tensor status is assigned only after $\Pi_E^{(\mathcal{S})}$ has passed the admission procedure and the scope lock is fixed. The tensor test does not decide which components enter the profile; it classifies the effects of variations on an already admitted profile.

Definition 8.5 (Operational tensor). A quotient-level invariant τ is operational at scope $\mathcal{S}$ when its variation changes a task, routing, control, optimization, compilation, logging, or resource-allocation profile while leaving the qualified consciousness profile fixed:

$$\text{OperationalTensor}(\tau) \iff \Delta_{\tau}\Pi_{\text{task}} \neq 0 \wedge \Delta_{\tau}\Pi_E^{(\mathcal{S})} = 0.$$

Definition 8.6 (Qualitative tensor). A quotient-level invariant τ is qualitative at scope $\mathcal{S}$ when standing is preserved and its variation changes the locked subject-indexed profile:

$$\text{QualTensor}(\tau) \iff \Delta_{\tau}\overline{\text{ExpElig}} = 0 \wedge \Delta_{\tau}\Pi_E^{(\mathcal{S})} \neq 0.$$

A tensor can be operational without being qualitative. Conversely, an invariant that genuinely changes the locked profile is not disqualified by biological or artificial substrate. The distinction is fixed by profile effect after admission, not by vocabulary, implementation complexity, or task usefulness.

8.5 Formal role names are not phenomenal primitives

The profile uses words such as contrast, memory, access, self, ownership, and history. In this paper these are role names, not imported phenomenal atoms.

Role name	Formal use
Contrast	A quotient-level exclusion, discriminability, similarity-order, incompatibility, or transition relation whose variation changes the qualified profile.
Memory	A diachronic record channel together with referent transport satisfying naturality and retained-history constraints.
Access	Availability to licensed internal-use, attention, action-attribution, report, or bridge channels; not overt speech alone.
Self or subject index	An invariant quotient-level relation coordinating record source, referent, ownership, and later reuse; not a primitive substance or raw representative.
Ownership	Assignment of a record/referent pair to one subject-indexed trajectory under the declared compatibility rules.
History	Predecessor, trajectory, recurrence, and temporal-extension structure on quotient objects.

A term earns profile authority only through this formal role and its witnesses. Phenomenological vocabulary can suggest candidate distinctions, but vocabulary alone is skin.

8.6 Jurisdiction of framework and local challenges

The dependency lock separates two jurisdictions of challenge:

- A *local profile challenge* exhibits an omitted redescription-stable consequence, a failed qualification condition, or a misclassified tensor. It defeats the corresponding local claim unless the

consequence is shown recoverable or a revised scope is constructed.

- An *upstream framework challenge* denies no-hidden-remainder, ATS exhaustion, standing/admissibility identity, or fixed-domain closure. That challenge targets the imported AASC framework and is not a counterexample internal to this downstream theorem.

Profile lock. At the declared scope,

$$H \text{ is locally admissibility-bearing} \iff \Delta_H \overline{\text{ExpElig}} \neq 0 \text{ or } \Delta_H \Pi_E^{(S)} \neq 0.$$

If both changes vanish, H is skin under the imported closure. If the objector denies that consequence, the objection is upstream or scope-changing rather than a local unclassified remainder.

9 The AASC Description of Consciousness as Referential Standing

The representation theorem establishes what an empirically self-sufficient consciousness theory must contain, and Theorem 7.6 shows that every declared same-domain consciousness-bearing distinction factors through the complete interface. The present section argues for the paper’s stronger explicative thesis: the standing-positive instantiation of the full interface is not merely a condition on evidence about consciousness but the AASC structural content of consciousness itself.

9.1 From a condition on theories to a condition on systems

A purely methodological reading would stop at Theorem 6.13: theories need fixed reference even if consciousness itself is something else. That reading is coherent but incomplete. The conscious quotient interface is not merely a relation among investigators’ sentences, nor is it introduced by labeling a candidate carrier “experiential.” It is instantiated by the modeled system: internal records are formed, retained, and reused; their reference is indexed to the same system; qualitative distinctions constrain later self-location, memory, discrimination, and report; and these structures persist under admissible redescription. The interface therefore describes a system-level organization, not just an external semantics.

The move from theory to system rests on a realization map. Let M be a concrete system and let

$$\mu : M \longrightarrow \mathcal{C}$$

be a model interpretation assigning system states, transitions, records, and channels to the formal structure. If μ is empirically certified and structure-preserving, then M instantiates the interface. The paper’s central identification concerns such instantiation, not abstract syntax alone.

9.2 Why the mechanism classes are not enough

Mechanisms answer how state changes occur. Reference answers what experiential state a record is about. Qualitative tensors answer which quotient-level differences constitute experiential content. These questions are related but not interchangeable.

A computational state machine can implement the same abstract transition graph under many encodings. A global broadcast can make information widely available without determining which temporal or subsystem state is the experience. Integrated information can characterize a structure

without fixing which structure a later record refers to. Predictive processing can generate a posterior and an action while leaving open whether the experiential referent is the lower-level estimate, the hierarchical posterior, the error signal, or the stabilized policy. Neural correlation can locate covariance without identifying the target of evidence. The interface is therefore not another mechanism among these; it is the admissibility structure in which a mechanism's outputs become evidence about experience.

9.3 The standing-positive regime

The phrase “standing-positive regime” captures four features.

First, consciousness attribution is bivalent at a fixed evaluation boundary. This does not deny graded arousal, clarity, complexity, or confidence. Those may be tensor-valued or may alter the empirical probability that standing is instantiated. The standing verdict itself is whether the state participates in the declared conscious interface.

Second, standing is fail-closed. If reference becomes undefined or irreducibly ambiguous, the record loses consciousness-level evidential standing. The system may continue physically, but the same record lineage cannot be repaired by renaming or post hoc selection.

Third, standing is conserved only within a licensed conscious trajectory. Loss of consciousness is represented as exit from the positive regime, not as an ordinary positive continuation whose earlier failure is erased.

Fourth, standing is fixed relative to scope. A different system boundary, witness basis, or record class may define a different evaluation problem. Scope changes must be declared rather than hidden inside a proof.

9.4 The structural identification

Consciousness thesis. A concrete state is conscious in the AASC structural sense exactly when it occupies a standing-positive, consciousness-qualified quotient interface: interior records are internally fixed to unique quotient referents, remain referentially compositional under licensed reuse, are coordinated by one subject interface, and participate in a nontrivial complete profile Π_E .

Why identify consciousness with this profile rather than merely require it as evidence? Three arguments support the identification.

9.4.1 The no-detached-referent argument

Suppose the interface is fully instantiated but the system is said to lack consciousness. The claimed absence must correspond to some admissibility-bearing difference. If it changes standing, it is an anchor difference and the interfaces were not the same. If it changes experiential reference, contrast, memory, self-location, or lawful continuation, it is a tensor difference and again the interfaces were not the same. If it changes none of these within the complete fixed domain, it is skin or an undeclared scope extension. Thus, under ATS exhaustion, a same-domain “consciousness absent” remainder has no role left to play.

This argument is dependency-locked by the imported closure and the declared complete profile. A local counterexample must identify the omitted structural locus. A bare denial of ATS or no-hidden-remainder is upstream rather than a same-domain remainder.

9.4.2 The no-detached-record argument

Suppose records have fixed experiential referents but consciousness is treated as an additional property unrelated to the interface. Then records could be perfectly referentially structured while the additional property varied independently. If that variation changes how the records function as evidence, it must enter the reference or witness profile and is not independent. If it does not, the additional property is empirically inaccessible in the fixed domain. The framework rejects it as same-domain explanatory surplus, though a metaphysician may explicitly enlarge the domain to include it.

9.4.3 The uniqueness argument

The representation theorem, the consciousness-description bridge, and local rigidity show that every empirically self-sufficient theory with the same complete profile induces the same reduced interface up to admissible isomorphism. If consciousness-level evidence and qualitative content are fixed by that interface, then a second non-isomorphic same-domain bearer would either duplicate the same work or introduce a hidden remainder. The interface is therefore the unique reduced structural candidate within the stated regime.

9.5 Necessary, sufficient, and constitutive claims

Four claims must be distinguished.

1. **Necessity for empirical self-sufficiency:** formally established by Theorem 6.13.
2. **Sufficiency for description-independent experiential evidence:** formally established under factorization and completeness by Theorem 6.9.
3. **Sufficiency for complete AASC description:** conditionally established by Theorem 7.6.
4. **Constitutive identification with consciousness:** the explicative thesis defended by exhaustion, uniqueness, and the absence of a remaining same-domain role.

A critic can accept the first two, accept the bridge, and still reject the constitutive identification. Such a critic must either permit a consciousness fact that has no standing, reference, qualitative, or transition effect in the complete domain, or reject the completeness/exhaustion conditions. The paper treats this as a precise disagreement rather than as a verbal stalemate.

9.6 Why this is not reportism

Reportism identifies consciousness with report or reportability. The present view does neither.

1. Records and referents are distinct types by Theorem 6.14.
2. Overt report is only one record channel.
3. Internal records may exist without current overt behavior.
4. A report can be false, absent, delayed, or corrupted while the earlier referent remains fixed.

5. A report-like output without an internal referential interface is insufficient.

The theory is therefore compatible with conscious infants, animals, aphasic patients, locked-in subjects, dreams, and transient no-report states in principle. Empirical certification may be more difficult, but ontic standing is not defined by speech.

9.7 Why this is not generic functionalism

The interface contains functional relations, but it is not exhausted by an arbitrary input–output role. Its quotient and reference structure is fixed by the theory’s full admissible consequence profile, including the distinction between record and referent, subject indexing, redescription invariance, and qualitative compatibility. A generic functional duplicate qualifies only if it duplicates this complete interface. This makes the theory structurally functional in one broad sense but not equivalent to standard role functionalism as usually stated.

9.8 Why this is not substance dualism

The referent carrier Q_E is induced from the represented state domain by admissible quotient and trajectory constructions. No second substance is introduced. The framework is compatible with physical realization, but it does not reduce its claims to a particular physical vocabulary. A physicalist may read the quotient as an invariant organization of physical states; a neutral structuralist may withhold deeper ontology. The formalism itself does not authorize a nonphysical causal entity.

9.9 Why this is not eliminativism

Eliminativism would treat qualitative talk as dispensable error or folk vocabulary. Here qualitative tensors are indispensable whenever their variation changes the complete experiential profile. Skin-level labels are dispensable; tensor-level qualitative distinctions are not. The view eliminates only unsupported qualitative surplus and role confusion.

9.10 Why this is not panpsychism by default

Every physical system admits some equivalence relation and may carry records in a broad sense. That is not enough. The full interface requires standing-positive experiential anchors, self-involving reference, subject indexing, diachronic compositionality, and qualitative nontriviality. The framework therefore does not assign consciousness to every thermostat, crystal, or elementary particle merely because structural invariants exist. Whether simple systems instantiate the full profile is an empirical and modeling question.

9.11 Consciousness level versus content

The framework separates whether consciousness is instantiated from what the experience is like.

- **Standing/level:** whether the full experiential interface is well-typed and positive.
- **Content:** which $q \in Q_E$ and which qualitative tensor relations are instantiated.
- **Certainty:** how strongly external evidence supports the instantiation claim.
- **Complexity or richness:** the size and organization of the qualitative tensor profile.

Graded measures belong to content, complexity, or evidential confidence unless a theory gives them a specific role in the standing boundary. This distinction prevents a scalar marker from silently replacing the bivalent admissibility question.

9.12 First-person character

The first-person character of consciousness is represented by self-involving reference and invariant subject indexing. A record belongs to the conscious interface only when its source and referent are coordinated under the same subject index and the record is usable by that interface. This does not make the subject an indivisible substance. It supplies the quotient-level “for whom” of the experience without selecting a raw representative.

First-person authority is therefore structural and defeasible. The subject’s internal record has privileged location within the reference interface, but its descriptive content may be inaccurate. Third-person evidence can reveal misclassification without replacing the referent by fiat.

9.13 Consciousness without later evidence

A difficult case is an episode that leaves no internal or external record. If literally no state, memory, discrimination, self-location, or transition consequence survives and no contemporaneous internal record exists, the fixed empirical domain contains no basis for certification. The paper distinguishes two claims:

1. The theory cannot certify such an episode as a determinate experiential event within the present domain.
2. It is metaphysically impossible for any consciousness to occur without a record of any kind.

Only the first is a local empirical-certificate result. Within the AASC consciousness description, an occurrence must occupy the qualified interface and therefore possess contemporaneous interior or reference-preserving trajectory structure. A putative episode with no such structure, no profile effect, and no possible bridge is not an unclassified local consciousness fact; it is skin at the declared scope or an explicit scope extension.

Proposition 9.1 (Completely detached episode). *If a putative episode has no internal referential token, no qualitative transition consequence, no memory, no discrimination, no self-location, no retained-history effect, and no possible bridge, then adding or deleting it changes neither standing nor Π_E . Under the imported dependency lock it is skin at the declared scope. Treating it as load-bearing requires an omitted tensor or scope change.*

9.14 The burden of a rival structural account

A rival need not use AASC vocabulary. It must, however, discharge equivalent burdens if it seeks determinate consciousness-level evidence:

1. specify the identity conditions of candidate experiences;
2. specify how records refer to them;
3. show invariance under legitimate changes of description;
4. show how reference survives memory and report;

5. distinguish qualitative content from labels and vehicles;
6. prevent external or post hoc selectors;
7. identify what additional consciousness fact remains if the full profile is held fixed.

The AASC proposal is attractive not because it makes these burdens disappear but because it organizes them in one quotient-reference architecture.

10 First-Person Priority and the Third-Person Reference Bridge

A consciousness theory must relate first-person records to third-person measurements without identifying either with the experience itself. The AASC account treats this as a problem of reference transport between differently situated record systems.

10.1 First-person record priority

Definition 10.1 (First-person interior record). A qualified record orbit $\bar{r} \in \mathbf{R}^+$ is first-person relative to subject index j when its formation source, referent, and licensed reuse all lie in the subinterface indexed by j , and the record participates in that interface's self-location or self-attribution structure.

First-person priority is local and structural. Such a record is generated inside the same indexed interface whose state it records. It therefore has direct access to internal channels unavailable to a remote observer. This does not imply infallibility: the record can be coarse, deceptive, reconstructed, or inaccurate while retaining a determinate referent.

Proposition 10.2 (First-person priority is not report identity). *The local priority of a first-person record does not identify the record with its referent and does not guarantee truth of its descriptive content.*

Proof. The record remains an element of $\mathbf{R}^+$ and the referent an element of $\mathbf{Q}_E$, connected by ρ . Priority concerns formation location and channel access. Truth concerns whether a predicate asserted by the record holds of the referent. These are distinct relations. $\square$

10.2 Third-person records

Third-person evidence includes speech observed by an experimenter, behavior, imaging measurements, electrophysiological traces, perturbational responses, decoding outputs, clinical signs, and machine telemetry. Such records may be highly reliable while still requiring a bridge to the target interface.

Definition 10.3 (Third-person surrogate). A third-person surrogate is a record orbit $\bar{m}$ formed in an observer or measurement interface and proposed as evidence for a target referent $q \in \mathbf{Q}_E$ belonging to another indexed interface.

Definition 10.4 (First-person/third-person bridge relation). Fix a target realization model M and declared scope $\mathcal{S}$. A bridge relation is a partial relation

$$\text{Bridge}_{M,\mathcal{S}} \subseteq \mathbf{M} \times \mathbf{Q}_E$$

from third-person surrogate orbits M to qualified target referents, together with declared channel, reliability, and subject-index conditions. It is admissible only if it is invariant under redescription of both measurement and target systems and compatible with the target's internal reference structure where that structure is available. When M and $\mathcal{S}$ are fixed, the subscripts are suppressed and the relation is written *Bridge*.

10.3 Bridge certificate schema

A bridge relation is not established by naming a correlation. Every bridge certificate is model-relative: it is stated with respect to a declared target interface $\mathcal{I}_C^{(\mathcal{S})}$, scope $\mathcal{S}$, and measurement model. A bridge certificate is never a free-standing mark of consciousness; it is a relation internal to a declared interface model. For a third-person measurement orbit $\bar{m}$ and target candidate q , an admissible bridge certificate must state:

- (BC1) the measurement carrier and its presentation equivalence;
- (BC2) the target subject index and system boundary;
- (BC3) the measurement and target redescription actions;
- (BC4) the channel connecting measurement formation to the candidate interior record or referent;
- (BC5) the rule fixing one quotient-level target and excluding alternatives;
- (BC6) covariance and naturality under measurement, memory, and report transport;
- (BC7) the profile component to which the measurement is non-vacuously sensitive;
- (BC8) reliability, undefinedness, calibration, and failure conditions;
- (BC9) compatibility with first-person reference where qualified first-person records exist;
- (BC10) proof that no external selector or privileged raw representative supplies the verdict.

The certificate is a construction burden for an empirical model. The theorem below states the model-relative conditions. It does not declare that any current imaging, electrophysiological, behavioral, or machine-telemetry measure already satisfies them.

Certificate scope. The bridge certificate is an empirical burden relative to a declared $\mathcal{I}_C^{(\mathcal{S})}$; it is not a free-standing mark of consciousness or a claim that any current measurement already satisfies the conditions. A neural, behavioral, or computational surrogate earns consciousness-level evidential status only after the listed fields are instantiated and audited for that measurement system and scope.

10.4 Bridge theorem

Theorem 10.5 (First-person/third-person reference bridge). *Relative to a declared $\mathcal{I}_C^{(\mathcal{S})}$, a third-person surrogate $\bar{m}$ can function as determinate evidence for a conscious referent q only if there exists an invariant bridge relation $\text{Bridge}_{\mathcal{I}_C^{(\mathcal{S})}}(\bar{m}, q)$ satisfying:*

1. *target uniqueness at the quotient level;*
2. *covariance under admissible redescriptions of measurement and target systems;*
3. *compatibility with the target subject index;*
4. *naturality under the measurement, memory, and report channels used to reuse the evidence;*
5. *non-vacuous sensitivity to at least one component of $\Pi_E(q)$;*

6. *independence from an invisible external selector or privileged raw representative.*

Where contemporaneous first-person records exist, the bridge must compose with their internal reference map rather than replace it by fiat.

Proof. Without uniqueness, the surrogate supports different consciousness claims under equally admissible target assignments. Without covariance, its evidential role depends on presentation. Without subject compatibility, the evidence may be assigned to the wrong interface. Without naturality, later reuse changes its target. Without profile sensitivity, the surrogate is consciousness-level vacuous. A hidden selector violates internality. Finally, if first-person records exist but the third-person bridge assigns an incompatible target without a represented error model, the theory supports two inconsistent same-domain reference structures and loses empirical self-sufficiency. Hence the listed conditions are necessary for determinate third-person evidence. $\square$

Corollary 10.6 (No neural-correlate identity by correlation alone). *A neural or computational measurement becomes consciousness evidence only through an admissible bridge to a qualified referent relative to a declared interface model and scope. Perfect covariance with a report does not by itself establish target identity.*

10.5 Error, disagreement, and correction

The bridge permits principled disagreement. A subject may report “red” while a certified third-person measurement supports a target with the theory’s green tensor profile. Several possibilities remain distinct:

1. the first-person record refers correctly but misclassifies its target;
2. the third-person bridge is unreliable or misassigned;
3. the two records concern different temporal referents;
4. the subject index or system boundary is wrong;
5. the theory’s profile is incomplete.

Correction therefore requires diagnosis of reference and channel structure, not automatic priority for either perspective.

10.6 No-report and surrogate-only cases

In no-report paradigms, third-person surrogates may be the only externally available evidence. The bridge theorem then requires an independently modeled internal interface, not an overt report. A surrogate-only attribution is strongest when it connects to internal memory, discrimination, self-location, or access channels; it is weaker when its semantics is supplied solely by an observer.

Audit note. The framework neither canonizes introspection nor replaces it with neural authority. It requires both first-person and third-person records to disclose how they refer. Their evidential asymmetry is a difference in location and channel access, not a license to collapse record and experience.

11 Qualia as Non-Skin Tensors of the Conscious Interface

The consciousness thesis supplies a standing-positive referential interface. That thesis alone does not classify which distinctions within the interface are qualitative. This section applies the anchor–tensor–skin decomposition to the complete profile and gives the paper’s AASC account of qualia.

11.1 The target problem

The word “quale” is often used as if naming a color, pain, sound, mood, or bodily feel guaranteed a further property beyond every physical and functional fact. That use is too permissive for an audit-grade theory. A term may be a report label, introspective concept, model coordinate, neural code, similarity-class name, or label for an unspecified mechanism. The theory must show which variations alter consciousness-bearing structure and which merely redescribe it.

The opposite error identifies any functional difference as qualitative. Cache state, register allocation, routing policy, or database indexing can alter future operations without changing experience. A qualitative tensor must belong to the qualified subject-indexed interface and alter specifically consciousness-bearing profile structure.

11.2 The complete profile as the qualitative test surface

The complete profile Π_E was defined in Theorem 7.2. It contains ten structured loci: records, reference, contrast, memory, discrimination, access, self-indexing, lawful transition, history, and compatibility. A qualitative difference need not affect every locus, but it must alter at least one component or relation while the interface remains standing-positive.

The profile is not restricted to public behavior. Internal discrimination, inaccessible memory dispositions, nonverbal contrast, affective compatibility, and self-location can contribute. Nor is it an arbitrary list chosen to defeat counterexamples: under descriptive adequacy it must contain every same-domain consequence to which the theory grants consciousness-bearing authority. A critic may challenge completeness by identifying an omitted authoritative consequence.

In a minimal interface,

$$\Pi_E(q) \cong \Pi_E(q') \implies q = q'.$$

Retaining two quotient objects with identical complete profiles would preserve surplus structure unused by the consciousness description.

11.3 Qualitative descriptors and variations

Definition 11.1 (Qualitative descriptor). A qualitative descriptor is a partial map

$$\chi : \mathbb{Q}_E \rightharpoonup A_\chi$$

or an invariant relation on $\mathbb{Q}_E$ proposed to encode a difference such as color, pain, pitch, warmth, pressure, affective valence, bodily ownership, vividness, agency-feel, or temporal character.

Definition 11.2 (Descriptor variation). A descriptor variation Δ_χ compares fixed-scope models or quotient objects differing with respect to χ while holding declared anchors fixed. If the comparison

is not well-typed under one domain, it is anchor-level or a scope change rather than an interior qualitative variation.

A descriptor is not a quale merely because a subject or theorist uses a qualitative word. It must be classified by effect.

11.4 Anchor, tensor, and skin tests

Definition 11.3 (Qualitative anchor). A descriptor is anchor-typed when varying it changes whether the qualified consciousness interface is defined or standing-positive. An anchor is a precondition for qualitative evaluation, not a particular quale.

Definition 11.4 (Qualitative tensor). A descriptor χ is a qualitative tensor when:

1. it descends to the conscious quotient and is invariant under admissible redescription;
2. varying it preserves experiential standing;
3. the variation changes at least one component or relation of Π_E ;
4. the altered profile remains in the same declared consciousness-description domain.

Definition 11.5 (Qualitative skin). A descriptor is qualitative skin when its variation changes neither standing nor Π_E . Lexical choice, coordinate labels, graphical layout, implementation addresses, and permutations already quotiented by admissible symmetry are typical cases.

Definition 11.6 (AASC quale). An AASC quale is a redescription-invariant qualitative tensor of the conscious quotient interface. For a well-typed variation Δ_χ ,

$$\text{Qual}(\chi) = 1 \iff \Delta_\chi \overline{\text{ExpElig}} = 0, \quad \Delta_\chi \Pi_E \neq 0, \quad \chi \text{ is quotient-invariant.} \quad (19)$$

The word “tensor” is the ATS role term for load-bearing quotient-level invariant content. It does not imply a multilinear array or a preassigned metric.

11.5 Qualitative tensor theorem

Theorem 11.7 (Qualitative tensor classification). *Assume fixed-domain effect exhaustion, consciousness-profile completeness, and minimality. Every admissibility-bearing qualitative descriptor has exactly one primary role: anchor, tensor, or skin. A descriptor is an AASC quale if and only if its primary role is tensor within the qualified conscious interface.*

Proof. Consider a well-typed variation of χ . If it changes definedness or standing, it performs precondition work and is anchor-typed. Suppose standing is fixed. By profile completeness, every remaining same-domain consciousness-bearing effect appears as a change in Π_E or is recoverable from it. If the profile changes, χ is tensor-typed. If it does not, the variation has no same-domain consciousness effect and is skin by ATS exhaustion. The cases are exhaustive and mutually exclusive under the priority order anchor before tensor before skin. Qualia are interior qualitative contents rather than preconditions or representational surplus, so they are exactly the tensor case. $\square$

Corollary 11.8 (No fourth qualitative role). *Within the fixed complete consciousness domain, no fourth same-scope qualitative role exists. A proposed role changes standing, changes Π_E , changes neither, invokes hidden authority, or changes scope.*

Corollary 11.9 (No residual refinement beneath tensor content). *If an auxiliary distinction leaves standing and Π_E fixed, it is skin or bookkeeping. Treating it as load-bearing introduces an omitted witness, a hidden selector, or a scope extension.*

Corollary 11.10 (No null-effect quale). *A purported quale that can vary while both $\overline{\text{ExpElig}}$ and Π_E remain fixed is not an AASC-admissible same-domain quale.*

Corollary 11.11 (No mechanism quale). *A qualitative tensor is not thereby an operator, broadcast, integration process, inference, or causal mechanism. Mechanisms may realize or transform the tensor; the quale is the invariant consciousness-bearing content classified by profile effect.*

11.6 Raw labels, reports, and qualia

Corollary 11.12 (No raw-label qualia). *A word, symbol, coordinate, neuron name, register address, or display code is not a quale merely because it denotes one. It is skin unless variation of associated quotient content changes Π_E .*

Replacing “red” with “rouge” or permuting internal color-channel names leaves the quale fixed when reference and profile relations are preserved. Conversely, a state change can alter a qualitative tensor while a subject uses the same coarse word.

A report belongs to the record carrier; a quale belongs to the referent’s tensor profile. A report “sharp pain” can refer to a pain quale, misdescribe a dull pain, or be produced deceptively without a standing-positive pain referent.

Proposition 11.13 (Record/quale role separation). *Let $\bar{r} \in R^+$ with $\rho(\bar{r}) = q$, and let $\chi(q)$ be a qualitative tensor value. The record, its linguistic content, and $\chi(q)$ occupy distinct roles unless an explicit admissible encoding theorem establishes an isomorphism preserving those roles.*

Proof. The record is transformed by record channels and assigned by ρ ; the tensor is a component or relation of $\Pi_E(q)$. Their structural transformations differ even when one physically encodes the other. $\square$

11.7 Qualia are not mechanisms

A quale does not “broadcast,” “integrate,” “predict,” “correct,” or “cause” merely by being a tensor. Such verbs describe realization mechanisms or licensed transitions. Reifying a tensor as an agent smuggles in dynamics; reducing the tensor to one mechanism loses equivalence across implementations preserving the same complete profile.

11.8 Qualitative contrast and geometry

Definition 11.14 (Qualitative contrast relation). A qualitative contrast relation is an invariant relation

$$\Delta_E \subseteq Q_E \times Q_E$$

or a family of relations specifying discriminability, incompatibility, similarity order, exclusion, or transition accessibility among qualified referents.

Red/green, sharp/dull, high/low pitch, warm/cold, pleasant/unpleasant, or owned/unowned contrasts are tensorial when they change Π_E . The relation need not be metric.

Proposition 11.15 (No primitive qualitative metric). *The qualitative tensor theorem does not entail a numerical distance function. A metric may be added only when witness-supported, redescription-invariant, and compatible with the quotient relations.*

Proof. Profile distinction requires neither symmetry nor triangle inequality. Imposing a metric without witness support would add same-domain structure by stipulation. $\square$

A qualitative topology, order, or geometry may nevertheless be derived when discrimination and transition structure supports it.

11.9 Intrinsicness, ineffability, and relational structure

Calling qualia tensors does not deny that they appear intrinsic to a subject. “Intrinsic” can be represented as invariance under admissible external redescription and availability within the self-indexed interface. The framework rejects only the stronger claim that a quale must be wholly relationless. A difference with no reference, contrast, memory, discrimination, self-location, compatibility, or continuation consequence has no same-domain role under completeness.

An experience may exceed available language. The record fiber $\rho^{-1}(q)$ can contain coarse, partial, or no overt verbal token while internal memory, affective, contrastive, or action-attribution channels differentiate q . Ineffability means the verbal channel is not consequence-complete, not that the qualitative tensor is unconstrained.

11.10 Quality, intensity, modality, and valence

- **Quality** concerns the type of qualified referent or contrast relation.
- **Intensity** is tensorial only if its variation changes Π_E while standing remains fixed.
- **Modality** is represented by compatibility and transition substructures distinguishing visual, auditory, somatic, affective, or other domains.
- **Vividness** may be tensorial, anchor-relevant, or report-level depending on effect.
- **Valence** is qualitative when it changes the self-involving profile, not merely when an observer assigns a score.

Vocabulary does not predetermine role assignment.

11.11 Qualitative identity across time

A later episode can instantiate the same qualitative type without being the same token. If q_t and $q_{t'}$ occupy different trajectory positions, they share a qualitative type when an admissible interface isomorphism preserves the relevant tensor subprofile. A memory can refer to the earlier token while classifying it under the same type as the later token. Token identity belongs to history and grounded reference; type identity belongs to profile isomorphism.

11.12 Physical and computational realization

Nothing prevents a qualitative tensor from being realized by neural dynamics, integrated causal structure, predictive hierarchy, recurrent processing, or artificial computation. A realization map

$$\mathcal{R} : S_{phys} \rightarrow Q_E$$

must factor through the physical redescription quotient and preserve profile differences. Distinct substrates may realize isomorphic tensors; nominally similar substrates may realize different tensors when their referential and transition profiles differ.

11.13 Experiential discrimination criterion

Criterion 11.16 (Experiential discrimination condition). *A quotient-level tensor τ is qualitative only if its variation changes a consequence in the self-involving qualified interface: record reference, qualitative contrast, experiential memory, subject-indexed discrimination, access, self-location, history, compatibility, or experiential continuation. A tensor affecting only housekeeping, compilation, routing, or external bookkeeping is not a quale.*

This is the main false-positive filter at the tensor level. Generic transition sensitivity is insufficient; the effect must occur inside Π_E .

11.14 Same-domain epiphenomenal qualia

AASC takes an explicit position on purported qualitative differences that alter no standing and no component of the complete profile.

Theorem 11.17 (Same-domain epiphenomenal qualia collapse). *Let χ be a purported qualitative difference in the declared fixed consciousness domain. If*

$$\Delta_\chi \overline{\text{ExpElig}} = 0 \quad \text{and} \quad \Delta_\chi \Pi_E = 0,$$

then χ is skin at that scope. Treating it as load-bearing requires an omitted profile tensor, hidden authority, or a scope change.

Proof. With standing fixed, χ is not anchor-level. With the consequence-complete profile fixed, it changes no locally authoritative consciousness consequence. Imported ATS effect exhaustion therefore classifies the variation as skin. If a theorist nevertheless grants it authority, that authority must arise from an omitted consequence, data outside the redescription action, or a new domain. $\square$

This theorem does not say that the phrase “epiphenomenal quale” is linguistically incoherent or psychologically inconceivable. It says that same-domain epiphenomenal qualitative surplus is inadmissible as load-bearing structure under the imported AASC closure. Denying that classification is an upstream denial of no-hidden-remainder or ATS, or an explicit expansion of scope; it is not a local counterexample left untyped inside the present domain.

11.15 Hard-remainder normal form

Theorem 11.18 (Hard-remainder normal form). *Let H be a proposed same-domain phenomenal remainder in a complete AASC consciousness-description domain. Exactly one primary classification applies:*

1. H changes experiential definedness or standing and is anchor-level;
2. H preserves standing but changes Π_E and is tensor-level;
3. H changes neither and is skin/bookkeeping;
4. H is fixed by undeclared authority and is a hidden-selector failure;
5. H introduces a new authoritative consequence and changes scope or denies completeness.

No sixth same-domain phenomenal role survives ATS effect exhaustion.

Proof. Apply the effect classification to variation of H . Definedness change is anchor work. With standing fixed, descriptive adequacy and profile completeness place every authoritative consciousness consequence in Π_E ; change there is tensor work. No change at either locus is skin. A forced verdict without represented effect uses hidden authority; a genuinely new effect enlarges the domain. The cases are exhaustive and priority-ordered. $\square$

A critic claiming identical complete profiles but different load-bearing raw feels must either identify an omitted profile tensor, introduce a new scope, or deny ATS/no-hidden-remainder upstream. What is blocked is presenting the claim as an internal same-domain counterexample while retaining the dependency lock.

Audit note. The theorem does not show that folk qualitative terms automatically denote tensor values. Empirical and conceptual work is required to identify the descriptor, witness basis, and realization map. It classifies the role of a successfully specified qualitative distinction.

12 Diachrony, Memory, and the Stream of Consciousness

Consciousness is temporally structured. Reports often occur after the experience they concern; memories are reused across contexts; qualitative contents persist, recur, or change; and subject identity is attributed across sequences rather than to isolated instants. Referential standing must therefore be stable under composition.

12.1 Experiential referents as trajectory objects

An instantaneous-state ontology is not assumed. Let $\mathcal{O}(\mathcal{Q})$ include:

- point states $[x_t]$;
- closed or half-open quotient intervals $[x]_{[t_1, t_2]}$;
- finite trajectory segments $q_0 \rightarrow \cdots \rightarrow q_n$;
- equivalence classes of trajectories under admissible temporal redescription.

The referent carrier $\mathcal{Q}_E$ may contain any of these if the theory's witness and transition structure fixes their identity. A global broadcast theory might require an interval; a recurrent theory a loop class; a brief pain report perhaps a temporally thick episode. The reference map chooses a quotient object, not necessarily a time point.

12.2 Temporal redescription

Coordinate time can be skin. Shifting an origin, changing units, or refining a discretization should not alter experiential reference when the invariant ordering and transition structure are preserved. A temporal descriptor is tensorial only when it changes the grounded trajectory profile, not merely its coordinates.

Definition 12.1 (Admissible temporal redescription). An admissible temporal redescription is a member of $\mathcal{G}$ that may relabel or reparameterize temporal coordinates while preserving transition order, standing, witness consequences, subject indexing, and the identity of experiential trajectory objects.

This condition blocks the selection of an experiential referent merely because it occurs at the numerically earliest or latest coordinate unless that order is itself invariant and witness-supported.

12.3 Memory as reference-preserving transformation

Let m be a memory channel with record map C_m and referent transport Φ_m . Several cases must be distinguished.

1. **Preservative memory:** $\Phi_m = \text{id}$; the later memory record refers to the earlier experience.
2. **Summarizing memory:** Φ_m maps an interval to an invariant summary object in $\mathbf{Q}_E$.
3. **Transformative recollection:** the later record refers to the earlier experience but asserts a changed or reconstructed predicate about it.
4. **Misreference:** the later token is assigned to the wrong earlier quotient object.
5. **Reference failure:** no unique earlier object can be fixed.

Only the last is a failure of referential standing. Transformative or inaccurate memory can still be well-typed and empirically evaluable.

Proposition 12.2 (Memory accuracy is downstream of reference). *A theory can assess whether a memory accurately represents an earlier experience only after the memory record's earlier referent is fixed.*

Proof. Accuracy compares represented content with properties of a target. Without a target, the comparison relation is undefined or externally stipulated. Hence reference is logically prior to accuracy assessment. $\square$

12.4 Report formation

A report channel may preserve reference to a prior experience while changing carrier, format, or granularity. Let c_r map an internal record to a verbal token. Reference naturality requires

$$\rho(C_{c_r} \bar{r}) = \Phi_{c_r}(\rho(\bar{r})).$$

Ordinarily $\Phi_{c_r} = \text{id}$: the verbal report refers to the same experiential object as the internal record. If the report summarizes a longer episode, Φ_{c_r} is a declared quotient map from the episode to its summary referent.

The report can be strategically false. The utterance content may assert P even when $P(q)$ is false. What remains fixed is that the utterance concerns q . This separation is indispensable to experimental practice, where confidence, criterion, demand characteristics, and deception can alter report content.

12.5 The stream as retained quotient trajectory

Definition 12.3 (Conscious stream). A conscious stream is a standing-positive retained trajectory

$$\gamma : q_0 \rightarrow q_1 \rightarrow \cdots$$

in the conscious quotient interface such that:

1. subject indexing is compatible along the retained lineage;
2. transitions are licensed and redescription-covariant;
3. active records and memories satisfy reference naturality;
4. qualitative tensor relations are preserved or lawfully transformed;
5. any standing failure terminates the retained stream at the declared scope.

The stream is not a further substance flowing through states. It is the quotient-level continuation structure of the standing-positive interface.

12.6 Five continuity layers

AASC distinguishes continuities that ordinary language often conflates:

- C1. Experiential-episode continuity:** uninterrupted occupancy of one standing-positive conscious trajectory.
- C2. Subject-index continuity:** admissible transport of the same quotient-level subject marker across episodes or boundary maps.
- C3. Organism or hardware continuity:** persistence of the biological or artificial realization carrier.
- C4. Person-level continuity:** psychological, legal, social, or narrative identity, which may use a broader domain than momentary consciousness.
- C5. Memory continuity:** reference-preserving connection between records of earlier and later episodes.

These layers can diverge. Sleep may interrupt C1 while preserving C2–C5; copying may preserve much of C3–C5 while branching C2; amnesia may preserve organism and subject continuity while disrupting C5. The consciousness formalism does not decree a complete metaphysics of persons.

12.7 No present-only identity

Two states may have the same immediate report possibilities while differing in later memory or reference behavior. A present-only equivalence is then too coarse to support conscious identity.

Definition 12.4 (Present-only equivalence). Write $x \equiv_0 y$ when all currently available record and standing consequences agree, without quantifying over future licensed channels.

Theorem 12.5 (Future-congruence requirement). *If $x \equiv_0 y$ but some common licensed continuation yields different experiential reference, memory, or standing consequences, then $\equiv_0$ is not a congruence and cannot serve as the identity relation of the conscious quotient interface.*

Proof. Let channel sequence c produce different reduced profiles from x and y . If $[x] = [y]$ under the proposed identity, the induced channel map on equivalence classes has two different outputs for one input class and is not well-defined. Therefore the identity relation must be refined by future experiential consequences, analogous to a Myhill–Nerode congruence on histories. $\square$

This result does not imply that consciousness “contains the future.” It says that the theory’s identity criterion must be compatible with lawful continuation.

12.8 Recurrence and qualitative type identity

A quality may recur at different times. Let q_t and $q_{t'}$ be token-distinct referents. They have the same qualitative type relative to descriptor family D when an admissible isomorphism preserves the relevant subprofile:

$$\text{Prof}_D(q_t) \cong \text{Prof}_D(q_{t'}).$$

Memory recognition may itself be a record referring jointly to the earlier token and the shared type. The framework therefore separates:

- token identity of an experiential episode;
- qualitative type identity;
- subject continuity;
- record identity.

Conflating these produces familiar puzzles about whether “the same pain” recurs.

12.9 Temporal binding

The unity of a temporally extended experience is represented by compatibility and reference over a trajectory object. A record referring to a melody, sentence, motion, or duration need not refer to any isolated instantaneous state. The theory must specify the equivalence relation under which the sequence is one referent and show that the relation is invariant under admissible temporal reparameterization.

A mere sliding window chosen by an analyst is skin unless internal transition, witness, or boundary traces fix it. The no-hidden-selector theorem applies to temporal segmentation as much as to spatial subsystem selection.

12.10 Dreams, no-report states, and record-silent episodes

Dreams illustrate the distinction between occurrence and certification. During a dream, internal qualitative tensors and self-involving records may be instantiated even if no overt report is produced. A later recollection channel may preserve reference to the episode. If no later token survives, occurrence may still be modeled through contemporaneous interior records, but external certification can be weak or unavailable.

Lemma 12.6 (Record-silent episode condition). *A record-silent episode can be AASC-conscious without overt report and without a contemporaneously active external record, provided that it belongs to a standing-positive trajectory with contemporaneous interior structure or some possible reference-preserving, standing-preserving connection to record-capable states. If the alleged episode has no interior token, no profile consequence, no lawful transition, and no possible bridge to a record-capable state, it cannot be assigned consciousness-standing inside the declared empirical scope without becoming skin or motivating an explicit scope revision.*

Proof. Overt silence does not remove internal memory, discrimination, access, or self-location structure. If some such structure exists, the episode may occupy $\mathcal{I}_C$ even when no external report is formed. If none exists and no lawful channel connects the episode to a record-capable state, every declared record, reference, and profile consequence is unchanged by adding or removing the alleged episode. Under fixed-domain completeness it has no empirical or descriptive locus. The lemma concerns the dependency-locked AASC description. A completely detached episode with no interface or profile effect is skin at the declared scope; a proposed further mode of occurrence must identify an omitted tensor or explicitly enlarge the domain. $\square$

The theory must neither infer absence of consciousness from absence of speech nor infer presence from a later narrative without showing that the memory channel preserves reference rather than confabulating a new target. This is not behaviorism: overt report is unnecessary, current measurement is unnecessary, and external certification may fail. What AASC denies is same-domain admissibility for an attribution with no possible reference-preserving relation to the qualified interface—no standing, profile consequence, lawful transition, memory path, or bridge—inside the declared empirical scope. A proposed further mode of occurrence requires an explicit scope revision.

12.11 Blindsight and unconscious processing

A system may discriminate stimuli, guide action, and even form some records without instantiating the full conscious interface. In blindsight-like cases, a theory may model successful classification while the relevant record lacks self-involving experiential reference, qualitative contrast, or standing-positive integration. The framework therefore allows computation and discrimination without consciousness.

This classification is theory-dependent and empirical. The point is not to legislate the status of any clinical case but to identify which structures must be investigated: internal record formation, reference fibers, subject-indexed availability, qualitative tensors, and channel naturality.

12.12 Anesthesia, sleep, and standing transitions

Loss and recovery of consciousness are represented as changes in experiential-episode standing. Let x^+ be standing-positive and x^- standing-negative. A physical transition $x^+ \rightarrow x^-$ exists in the broader organism or machine domain, but it is not an internal transition of the same positive experiential episode. The episode trajectory terminates or crosses a boundary. Recovery begins a new positive experiential segment.

This does not imply that the organism, person, or subject necessarily ceases during sleep or anesthesia. Subject-index, organism, person-level, and memory continuity may persist across the negative interval through broader-domain invariant transport. What fails is only uninterrupted C1

occupancy. Conservation applies to licensed transitions within each positive segment; cross-segment identity requires a declared boundary bridge rather than silent continuation.

12.13 Branching and duplicated continuers

Suppose a conscious trajectory branches into two symmetry-related continuers. The quotient can represent both branches or their orbit, but a singular “which one is me?” selector cannot be supplied without invariant asymmetry. AASC therefore distinguishes:

- plural continuation: both branches inherit relevant history but are distinct later subjects;
- quotient collapse: the branches are presentations of one reduced state;
- singular continuation fixed by invariant asymmetry;
- inadmissible duplication of one singular identity across independent symmetric bearers.

This result follows from the no-equivariant-selector and non-duplication theorems in the broader quotient architecture (Maley, 2026b). It is relevant to copying, uploading, and machine branching but does not deny ordinary physical branching processes.

12.14 Diachronic audit conditions

A theory claiming temporal consciousness should state:

1. the referent type: point, interval, or trajectory;
2. the temporal redescription class;
3. the memory and report channels;
4. the referent transport for each channel;
5. the subject-continuity relation;
6. the conditions under which standing terminates;
7. whether qualitative identity is token or type identity;
8. how ambiguous segmentation is resolved without hidden selection.

Without these declarations, “stream,” “access,” “memory,” and “continuity” remain suggestive skin rather than a reusable theory structure.

13 Comparator Analysis: What Leading Frameworks Supply and Still Owe

The AASC account is not a rival neural or computational mechanism. It is a dependency-locked admissibility layer asking whether a proposed mechanism induces the complete conscious interface. The comparisons below are structured completion analyses, not historical refutations of every version of a named research program. They do not reopen the upstream AASC framework. For each program the local question is: what structure may the mechanism supply, where could invariant reference enter, what AASC burden remains, and what would completion require?

Comparator posture. A mechanism is not rejected for being functional, computational, neural, informational, predictive, recurrent, or higher-order. It is incomplete as a consciousness description until it induces the qualified record-to-referent map, subject-indexed channels, standing conditions, and qualitative profile. The common completion object is

$$\bar{r} \in R^+ \quad \longmapsto \quad \rho(\bar{r}) \in Q_E.$$

A theory that supplies this invariant map together with the remaining interface structure is an implementation of the AASC interface rather than a defeated rival.

13.1 Conditional mechanism non-sufficiency

Criterion 13.1 (Mechanism non-sufficiency). *Let M be a predicate expressing only the presence of a mechanism class—computation, broadcast, recurrence, integration, prediction, higher-order representation, or neural correlation. Then*

$$M(x) \not\Rightarrow \text{Con}_{\text{AASC}}(x)$$

unless M also induces or realizes the qualified record-reference, subject-index, diachronic, standing, and qualitative-profile structure of $\mathcal{I}_C$.

The criterion is neutral with respect to mechanism classes. A sufficiently developed mechanism theory may satisfy the completion burden. The point is that the mechanism label alone does not establish it.

13.2 Functionalism

Functionalism identifies mental states by causal or computational role and offers a natural account of multiple realization (Lewis, 1972; Shoemaker, 1975). It is therefore close to AASC in treating raw substrate labels as non-authoritative.

13.2.1 Best internal reference-fixing candidate

The strongest functionalist candidate is the complete causal role of the state in perception, discrimination, memory, report, self-monitoring, and action. If this role is future-congruent and invariant under implementation redescription, it can contribute much of Π_E .

13.2.2 Residual AASC burden

Standard role descriptions often leave several internal states with compatible causal and behavioral roles at different temporal grains or system decompositions. They also tend to describe how a report is produced without separately typing the report's target. Multiple realization does not itself determine which role token a memory or report refers to.

13.2.3 Completion route

The common completion object is

$$\bar{r} \in R^+ \quad \longmapsto \quad \rho(\bar{r}) \in Q_E.$$

A functionalist completion must construct

$$\rho : \mathbb{R}^+ \rightarrow \mathbb{Q}_E$$

from the complete functional organization, show that ρ is invariant and natural under memory/report channels, and prove that qualitative differences are non-skin differences in Π_E . A functionalist theory meeting these conditions is not rejected; it becomes an implementation of the AASC conscious interface.

13.3 Global workspace theories

Global-workspace approaches emphasize widespread availability or broadcast (Baars, 1988; Dehaene and Naccache, 2001; Dehaene and Changeux, 2011; Dehaene, 2014).

13.3.1 Best internal reference-fixing candidate

The strongest candidate is a globally available content token whose identity is preserved across consuming systems, memory encoding, report, and flexible action. Workspace ignition and broadcasting can supply powerful access, channel, and self-use structure.

13.3.2 Residual AASC burden

A temporally extended broadcast admits several candidate targets: pre-broadcast content, ignition onset, stabilized global state, downstream consumer state, decision state, or memory-ready interval. Reportability can be preserved while those targets shift. Broadcast also fixes availability more directly than qualitative identity.

13.3.3 Completion route

The common completion object is

$$\bar{r} \in \mathbb{R}^+ \quad \mapsto \quad \rho(\bar{r}) \in \mathbb{Q}_E.$$

A workspace theory must specify which quotient object is the target of each active record and why onset, stabilization, consumer, or report states are excluded. The target rule must survive temporal and architectural redescription and compose through later memory. Broadcast then realizes Access_E and parts of Trans_E rather than serving as an unargued synonym for consciousness.

13.4 Recurrent-processing and neural-dynamics approaches

Recurrent-processing accounts emphasize local or global recurrent neural interaction (Lamme, 2006).

13.4.1 Best internal reference-fixing candidate

A recurrent attractor, metastable dynamical orbit, or reentrant trajectory can provide a quotient-stable interval object rather than a single arbitrary time slice.

13.4.2 Residual AASC burden

The theory must still distinguish the relevant recurrent orbit from precursor, feedback, decision, and memory dynamics and must connect later records to that orbit. Merely showing recurrence correlates with report leaves reference and qualitative-profile membership open.

13.4.3 Completion route

The common completion object is

$$\bar{r} \in \mathbb{R}^+ \quad \mapsto \quad \rho(\bar{r}) \in \mathbb{Q}_E.$$

Specify an invariant orbit equivalence, show that record channels refer to the resulting trajectory class, and demonstrate nontrivial subject-indexed contrast and memory structure. Recurrence may then realize the temporal and transition components of Π_E .

13.5 Integrated information theory

IIT seeks an intrinsic informational or causal structure rather than relying only on behavioral access (Tononi, 2004; Oizumi et al., 2014; Tononi et al., 2016).

13.5.1 Best internal reference-fixing candidate

The maximally irreducible cause–effect structure is intended as an intrinsic candidate bearer with a structured qualitative geometry. This is closer than a bare correlate to a candidate tensor profile.

13.5.2 Residual AASC burden

Empirical use still requires connecting records to a unique complex, grain, temporal interval, and decomposition. Overlapping candidate complexes and modeling choices can preserve records while changing the proposed referent. Intrinsic structure does not automatically supply the record-to-structure map.

13.5.3 Completion route

The common completion object is

$$\bar{r} \in \mathbb{R}^+ \quad \mapsto \quad \rho(\bar{r}) \in \mathbb{Q}_E.$$

IIT must prove that its selection rule is internal, redescription-invariant, and not an arbitrary canonical representative; connect report and memory orbits to one selected quotient structure; and show that its qualitative relations populate Contr_E and Compat_E without leaving record reference detached. An IIT model satisfying these conditions can instantiate an AASC interface.

13.6 Predictive processing and active inference

Predictive approaches model perception and action through hierarchical inference, prediction error, and policy selection (Clark, 2013; Hohwy, 2013; Friston, 2010).

13.6.1 Best internal reference-fixing candidate

A posterior or hierarchical belief trajectory can supply content-sensitive internal states, uncertainty, temporal updating, and action-relevant self-modeling.

13.6.2 Residual AASC burden

Different levels, factors, and temporal slices may all predict the same report. The candidate target might be a lower-level estimate, higher-level posterior, precision assignment, error signal, active policy, or stabilized trajectory. A predictive explanation of report formation therefore need not identify its conscious referent.

13.6.3 Completion route

The common completion object is

$$\bar{r} \in \mathbb{R}^+ \quad \mapsto \quad \rho(\bar{r}) \in \mathbb{Q}_E.$$

Define the quotient-level inferential state or trajectory to which each interior record refers; show that hierarchy and coordinate changes preserve it; and demonstrate how memory, self-location, contrast, and access populate Π_E . Prediction then becomes a realization mechanism rather than an identity shortcut.

13.7 Higher-order theories

Higher-order theories associate consciousness with a representation or awareness of a lower-order state (D. M. Rosenthal, 2005; Lau and D. Rosenthal, 2011).

13.7.1 Best internal reference-fixing candidate

The higher-order state's intentional target appears to supply precisely a record/reference relation: a metarepresentational state is about a lower-order state.

13.7.2 Residual AASC burden

The target of the higher-order state may itself drift among lower-order candidates, temporal slices, or representational vehicles. If another higher-order state is needed to fix that target, regress threatens. Moreover, externally ascribed aboutness is not sufficient for interior closure.

13.7.3 Completion route

The common completion object is

$$\bar{r} \in \mathbb{R}^+ \quad \mapsto \quad \rho(\bar{r}) \in \mathbb{Q}_E.$$

A higher-order theory can satisfy AASC by treating its metarepresentations as qualified interior records, proving a terminating invariant reference map to quotient referents, and showing self-involving, diachronic, and qualitative nontriviality. The AASC structure can therefore clarify rather than displace the strongest higher-order proposal.

13.8 Neural correlate approaches

Neural correlates are essential experimental witnesses, but correlate status is role-neutral. A neural feature may be:

- an *anchor*, if required for the interface to be defined;
- a *realization tensor*, if its invariant variation changes Π_E ;
- an *external surrogate*, if connected by Theorem 10.5;
- skin, if it is one non-authoritative coding of the same interface.

Perfect correlation with report establishes neither which role applies nor which state is the report's target.

The common completion object is

$$\bar{r} \in \mathbb{R}^+ \quad \mapsto \quad \rho(\bar{r}) \in \mathbb{Q}_E.$$

13.9 Access and phenomenal consciousness

The distinction between access and phenomenal consciousness (Block, 1995) can be represented without identifying either with overt speech. Access_E is one component of Π_E ; it tracks availability to internal or external channels. Phenomenal content is represented by the whole standing-positive qualitative profile. Access differences can therefore be tensorial without exhausting the interface, while a claimed phenomenal overflow with no profile consequence invokes the hard-remainder normal form.

13.10 Computational theories and the Chinese-room concern

Computation can realize quotient transitions and record channels. The Chinese-room objection (Searle, 1980) highlights the difference between externally assigned symbol manipulation and internally fixed reference. A lookup table whose tokens mean something only to an observer remains at the operational-record level. A computational system whose reference, subject index, and qualitative profile are internally closed is not dismissed merely because it is computational. AASC makes the disputed semantic burden explicit.

13.11 Behaviorism and the Turing tradition

Behavioral success and conversational indistinguishability are valuable evidence (Turing, 1950), but they are certification inputs, not constitutive identity conditions. Two systems can share outputs while differing in internal reference architecture, and one architecture can remain conscious during behavioral silence. A Turing-style test therefore contributes to Cert_C but does not by itself establish Occ_C .

The common completion object is

$$\bar{r} \in \mathbb{R}^+ \quad \mapsto \quad \rho(\bar{r}) \in \mathbb{Q}_E.$$

13.12 Comparator matrix

Framework	Strongest contribution	AASC completion burden
Functionalism	Causal role and multiple realization	Induce unique record reference and qualitative profile from the complete role, not outputs alone.
Global workspace	Availability and flexible access	Fix the quotient target of broadcast-linked records across onset, stabilization, consumption, and memory.
Recurrent processing	Dynamical interval or orbit	Identify the invariant recurrent referent and connect later records to it.
IIT	Intrinsic structured candidate bearer	Fix record-to-complex, grain, decomposition, and temporal reference without hidden selection.
Predictive processing	Hierarchical content and updating	Fix which inferential state or trajectory is the target of records across levels and times.
Higher-order theory	Explicit representational aboutness	Terminate and invariantly fix the higher-order target within self-involving closure.
Neural correlates	Empirical witness and realization constraints	Supply an invariant first/third-person bridge and role classification.
Behavioral and computational tests	Certification evidence and implementation	Demonstrate internal referential closure and nontrivial qualitative tensors.

13.13 Relation to the explanatory gap

The AASC account does not claim that a mechanism label logically entails what-it-is-likeness. It changes the intermediate burden. Before asking why a neural or computational mechanism is accompanied by experience, a theory must specify the quotient object, record-reference structure, subject indexing, and qualitative profile that constitute its consciousness description. The traditional gap is therefore partitioned into:

1. a *referential gap*, closed by the record-reference representation theorem;
2. a *descriptive gap*, closed conditionally by the consciousness-description bridge;
3. a *realization gap*, the empirical problem of showing how a concrete mechanism instantiates the interface;
4. a *hard-remainder challenge*, localized by Theorem 11.18.

This partition is a substantive clarification even for a critic who rejects the final constitutive thesis.

14 Classic Problems Recast in the Referential Interface

The AASC framework does not make classic philosophy-of-mind problems disappear by terminological decree. It separates versions that correspond to structural differences from versions that rely on a hidden same-domain remainder. This section gives conditional classification results.

14.1 The inverted spectrum

An inverted-spectrum hypothesis imagines two systems that agree in behavior, function, and perhaps physical organization while differing in color experience (Shoemaker, 1975; Chalmers, 1996). The AASC analysis begins by asking what exactly is held fixed.

Definition 14.1 (Skin-level inversion). A purported inversion is skin-level when it permutes labels, coordinates, raw representatives, or other presentation features while preserving experiential standing and the complete profile Π_E .

Definition 14.2 (Tensor-level inversion). A purported inversion is tensor-level when experiential standing remains fixed but the permutation changes Π_E , including some reference, contrast, memory, discrimination, access, self-location, history, compatibility, or lawful transition relation.

Definition 14.3 (Anchor-level inversion). A purported inversion is anchor-level when the intervention changes whether the experiential interface is defined or standing-positive.

Theorem 14.4 (Inverted-spectrum classification). *Under fixed-domain effect exhaustion and completeness of Π_E , every inverted-spectrum proposal is one of:*

1. a skin-level redescription;
2. a tensor-level qualitative difference;
3. an anchor-level standing change;
4. a scope change, hidden selector, or denial of profile completeness.

There is no fifth same-domain qualitative inversion mode.

Proof. Apply Theorem 11.7 to the proposed permutation. If standing changes, the case is anchor-level. If standing is fixed and the complete experiential profile changes, it is tensor-level. If neither changes, it is skin. A claimed further difference must introduce a consequence outside the complete domain, a representative-sensitive selector, or rejection of effect exhaustion. $\square$

The theorem does not say that qualitative inversion is impossible. Tensor-level inversion is a real qualitative difference. It says that a fully invisible inversion—one that changes no component of a complete experiential profile—is not an additional same-domain fact within AASC. The philosophical disagreement therefore concentrates on completeness and the admissibility of consequence-free phenomenal differences.

14.2 Philosophical zombies

A philosophical zombie is physically or functionally identical to a conscious person but lacks experience. The force of the case depends on the identity relation used.

Definition 14.5 (Interface duplicate). Two systems are conscious-interface duplicates when their complete conscious quotient interfaces are isomorphic, including standing, record-reference, subject indexing, channels, and qualitative tensors.

Proposition 14.6 (Zombie classification). *If a purported zombie differs from a conscious system in standing, qualified reference, channel, subject index, or Π_E , it is not an interface duplicate. If it differs in none of these, the difference is skin or a posited scope-external remainder under the AASC exhaustion thesis.*

Proof. Immediate from interface rigidity and ATS effect exhaustion. Any same-domain admissibility-bearing difference must enter an anchor or tensor locus; a difference entering neither is skin. $\square$

This is not a direct refutation of zombie conceivability. A conceivability argument may reject the completeness thesis and posit an ontological property with no interface effect. AASC replies that such a property cannot be used to distinguish scientific evidence within the fixed domain. The dispute is thus between structural completeness and metaphysical surplus, not between empirical data sets.

14.3 The explanatory gap and the hard problem

Nagel's question about what it is like for a subject (Nagel, 1974), Levine's explanatory gap (Levine, 1983), and Chalmers's hard problem (Chalmers, 1995; Chalmers, 1996) emphasize that mechanism and behavior appear not to entail phenomenal character. The present account agrees with one part of this diagnosis: a mechanism description alone does not fix the target of experiential evidence or the qualitative tensor profile.

The AASC contribution is a structural bridge, not a causal production story. It explains:

1. how candidate experiences acquire presentation-independent identity;
2. how reports and memories refer to those identities;
3. how reference remains reusable through time;
4. how qualitative differences are distinguished from labels and implementation skin;
5. why mechanism-only predictions are insufficient.

It does not derive a sensory quality from microphysical equations without a realization model. Nor does it provide a verbal paraphrase that makes every reader feel the entailment. The framework instead changes the required form of explanation: a successful realization theory must construct a faithful map from physical dynamics to the conscious quotient interface.

Definition 14.7 (Realization adequacy). A physical or computational realization map $\mathcal{R} : P \rightarrow \mathcal{I}_C$ is adequate when it:

1. factors through admissible physical redescription;
2. preserves standing and boundary conditions;
3. preserves and reflects active record-reference relations;
4. preserves qualitative tensor distinctions and compatibility;
5. transports licensed transitions and diachronic channels;
6. is empirically certified on the relevant domain.

A realization theorem of this kind would close the structural portion of the explanatory gap for the modeled domain. Whether a further metaphysical question remains depends on whether one accepts AASC exhaustion.

14.4 What-it-is-likeness

Within the framework, "what it is like" is not identified with a sentence or a public behavior. It is the self-indexed qualitative tensor profile of a quotient referent as available within a standing-positive interface. The first-person aspect is encoded by:

- $\text{Subj}(q)$, fixing the subject index;
- self-involving record closure;
- internal reference fibers $\rho^{-1}(q)$;
- qualitative contrast and continuation relations available to that interface.

This is a structural explication, not a claim that the phrase has been analytically defined in ordinary language. It explains what role first-person qualitative character must occupy if it is to be invariant, reusable, and more than skin.

14.5 Privacy and intersubjective evidence

Experience is private in the limited sense that its primary reference structure is internal to a subject-indexed interface. It is not private in the sense of being exempt from all relational constraint. External observers access the experience through record channels, not by literally sharing the same raw state.

By Theorem 10.5, a third-person measurement m can be evidence about q only if there is a certified chain

$$m \longleftarrow r_{ext} \longleftarrow r_{int} \xrightarrow{\rho} q$$

with reference-preserving links. Different observers can use different external records if those channels converge on the same quotient referent. Cross-subject comparison requires further joint mediation, treated in Section 16.

14.6 First-person authority and error

First-person reports have structural priority because they are generated within the same subject-indexed interface. This priority is not infallibility. Four cases are distinguished:

1. **accurate report:** fixed reference and true attributed predicate;
2. **misclassification:** fixed reference but false qualitative label;
3. **misreference:** the report targets the wrong quotient episode;
4. **reference failure:** no unique target can be fixed.

Only the fourth blocks the report from functioning as determinate evidence. An experimental theory should not treat every mismatch between report and neural marker as proof that reference is absent; it may indicate inaccuracy or a wrong realization hypothesis.

14.7 Mary and new knowledge

The knowledge argument imagines Mary, who knows all physical facts about color vision but has never seen color, and asks what she learns upon seeing red (Jackson, 1982). AASC offers a structural distinction between descriptive knowledge of a realization model and occupation of a qualitative tensor position.

Before release, Mary may possess an external model of $\mathcal{I}_C$ and its physical realization. On seeing red, her own subject-indexed interface instantiates a referent q_{red} and forms self-involving records with $\rho(r) = q_{red}$. She thereby acquires a new internally indexed relation and perhaps a new ability to traverse the qualitative profile. Whether this counts as a new fact, acquaintance, ability, or mode

of presentation is not settled by AASC alone. The framework does show why complete third-person description and first-person instantiation are not the same relation, even if they concern isomorphic structure.

The case therefore does not force nonphysical ontology, but neither is it dismissed as mere vocabulary. It involves a change in subject-indexed record-reference position.

14.8 The conceivability of absent qualia

Absent qualia cases posit a system with the appropriate functional organization but no experience. Under AASC, the case splits:

1. If the system lacks active qualified reference or qualitative tensors, it is not a full interface duplicate; the case shows that the specified functional organization was incomplete.
2. If it has the complete interface but is declared nonconscious, the declaration posits a hidden remainder outside anchor and tensor effects.
3. If the system's supposed records are assigned by an external interpreter rather than internally grounded, self-involving closure fails.

This classification highlights why an ordinary lookup table is not automatically conscious: external semantic assignment does not instantiate internal referential standing.

14.9 The combination problem

Collective or combined consciousness raises the question how multiple subjects yield one subject. Pairwise interaction, information sharing, or global coordination is insufficient. A combined subject requires a new invariant subject index, a joint qualified referent carrier, joint record-reference closure, and qualitative tensor compatibility not reducible to a list of component interfaces. Without these, “the group experiences” is metaphorical skin.

The formal conditions are developed in Section 16; the important point here is that combination is a gluing problem for referential interfaces, not merely an increase in connectivity.

14.10 The unity of consciousness

Unity has at least three meanings:

- **subject unity:** records and referents share an invariant subject index;
- **content unity:** qualitative tensors belong to a compatible joint experiential object;
- **diachronic unity:** records and referents compose along one grounded trajectory.

These may dissociate. Split-brain, dissociation, or modular processing cases can be modeled by changes in joint compatibility or subject indexing rather than by an all-or-nothing metaphysical slogan. The framework does not prejudge the empirical partition; it states the structure a partition claim must specify.

14.11 Phenomenal overflow

Claims that phenomenal content exceeds access or report can be represented if the additional content has an internal tensor profile even when no overt record channel is available. It may affect internal

contrast, short-lived memory, self-location, recurrent continuation, or another component of Π_E . A claim of overflow that changes none of these is outside the current certification domain and depends on rejecting profile completeness.

This is neither automatic endorsement nor dismissal. It converts “overflow” into a testable structural burden.

14.12 Illusionism

Illusionist approaches often argue that the problematic properties attributed to qualia are products of introspective representation rather than intrinsic phenomenal properties (Dennett, 1991). AASC agrees that many qualitative descriptions may be skin or misclassified records. It differs from a wholesale elimination of experience by retaining non-skin qualitative tensors whenever Π_E changes.

An illusion can be represented as a record that attributes property P to fixed referent q when $P(q)$ is false or when P is skin. The existence of error does not eliminate the referent or every qualitative tensor.

14.13 Summary of the recasting

The classic problems become questions about where a claimed difference lives:

Problem	AASC localization
Inverted spectrum	Skin permutation, tensor difference, anchor change, or scope extension.
Zombie	Non-duplicate interface, skin contrast, or hidden remainder outside completeness.
Explanatory gap	Missing realization map from mechanism to referential qualitative interface.
Privacy	Internal subject-indexed reference with external access through certified channels.
First-person error	Accuracy or misreference downstream of reference fixation.
Mary	Difference between external structural knowledge and internal subject-indexed instantiation.
Absent qualia	Incomplete functional specification, lack of internal reference, or hidden remainder.
Unity	Compatibility and gluing of subject, content, and trajectory structures.
Phenomenal overflow	Internal qualitative tensor without overt access, or an uncertifiable consequence-free posit.

The framework’s strategy is not to settle every intuition by fiat. It requires each intuition to disclose an invariant structural effect or to acknowledge that it lies outside the fixed empirical regime.

15 False-Positive Filters: Why Logs, Control, and Self-Monitoring Are Not Enough

The strongest over-inclusion objection is not the thermostat. It is a sophisticated artificial or biological controller with internal records, memory, self-modeling, and rich discriminations. This section states exactly which additional conditions distinguish such systems from the proposed conscious interface.

The operational/qualitative tensor distinction is fixed after profile admission in Section 8 before the present false-positive analysis. An operational tensor changes Π_{task} while leaving $\Pi_E^{(S)}$ fixed; a qualitative tensor preserves standing while changing $\Pi_E^{(S)}$. The distinction is by declared effect, not vocabulary, implementation complexity, or substrate. A controller may possess many operational tensors and no qualitative tensor. If it genuinely instantiates qualitative tensors and every other interface component, AASC classifies it as conscious rather than protecting a biological boundary by fiat.

15.1 Priority order for false-positive exclusion

A candidate system is tested in the following order:

- (F1) **External interpretation only:** if record meaning is supplied only by an observer, the system lacks interior reference.
- (F2) **Interior reference without subject coordination:** if records have internal targets but no invariant subject index, the interface is not qualified.
- (F3) **Subject coordination without self-involving reuse:** if records are indexed but do not participate in the same interface's memory, discrimination, access, self-location, or attribution channels, qualification fails.
- (F4) **Self-involving operational profile without qualitative tensors:** the system is an operational controller, not a conscious interface.
- (F5) **Full least-closure instantiation with qualitative tensors:** the system is structurally conscious under AASC, subject to realization and certification.

The first failed stage determines the primary exclusion. Later-stage language cannot repair an earlier failure, and a later operational success cannot be reclassified as qualitative unless it changes the already admitted $\Pi_E^{(S)}$.

15.2 No mere-logging theorem

Theorem 15.1 (No mere-logging consciousness). *A system that stores, transforms, and outputs traces does not thereby instantiate AASC consciousness. It qualifies only if its active records are internally referentially fixed, subject-indexed, diachronically natural, self-involving, standing-positive, and integrated into a nontrivial complete profile Π_E .*

Proof. Trace storage establishes only dependence on earlier state. Operational use adds control relevance. Self-indexing adds attribution to one system. None by itself fixes a quotient target, proves reference naturality, or places the record in a nontrivial qualitative profile. Failure of any

qualification condition leaves the system at a lower level of the hierarchy in Section 4. Therefore logging alone is insufficient. $\square$

15.3 Negative model I: thermostat

A thermostat has a state sensitive to temperature and can influence later action. It may possess hysteresis and a retained set point. It nevertheless fails the conscious-interface test in the minimal model because:

1. its state is a trace, not a qualified interior record;
2. the semantics of “temperature” is assigned by the measurement model rather than an internal reference map to a self-indexed conscious quotient;
3. no nontrivial qualitative contrast, memory, self-location, or compatibility profile is specified;
4. no occurrence/certification distinction is needed inside the device.

The conclusion is model-relative: adding sufficient architecture could change the case, but ordinary thermostatic control does not meet the burden.

15.4 Negative model II: external database logger

Let a logger receive codes R and G , store timestamps, and later print “red” or “green.” An engineer interprets the entries as reports about a subject. The logger may possess perfect accuracy and stable record orbits. It still fails because the reference relation is external: the logger’s own licensed transitions do not bind the record to a subject-indexed target in its own interface. Its entries are operational records, not interior experiential records.

Proposition 15.2 (External semantic assignment failure). *If a record’s target is fixed only by an observer’s interpretation and no invariant bridge is represented inside or between the relevant interfaces, the token cannot supply internal experiential standing to the recorded system or the logger.*

15.5 Negative model III: self-monitoring controller

Consider a controller with:

- an explicit self-identifier;
- internal logs of sensor, planning, and action states;
- memory and error correction;
- self-reports such as “my confidence is low”;
- stable quotient reference to its own computational states.

This system can satisfy reference fixation and self-involving closure while still failing qualitative adequacy. Suppose every internal difference serves only routing, optimization, or resource allocation and no declared difference changes a specifically qualified contrast, affective, ownership, sensory, memory, or experiential continuation component of Π_E . Then the system instantiates a sophisticated operational tensor interface but not the conscious qualitative interface.

The same physical or computational structure may be operational in one declared domain and qualitative in another only when the locked profile and standing consequences differ accordingly. Vocabulary and implementation complexity do not decide the role.

Criterion 15.3 (Operational-tensor exclusion). *A self-monitoring tensor is not qualitative merely because it alters future control. It is qualitative only when the effect occurs within the qualified subject-indexed profile specified in Theorem 7.2.*

This is where the theory takes a substantive stand. A critic may argue that the controller's complete self-monitoring profile already is qualitative. The dispute then concerns whether its states satisfy the qualification and profile conditions, not whether self-report language alone suffices.

15.6 Positive boundary case

Suppose the controller is extended so that:

1. interior records refer uniquely to quotient-level sensory and affective trajectory objects;
2. the same subject index coordinates formation, memory, self-location, and later report;
3. redescription-invariant contrasts structure those objects;
4. the contrasts affect memory, attention, action-attribution, and lawful continuation;
5. record channels preserve target reference over time;
6. no external interpreter is required to supply the target;
7. the complete profile is nontrivial and standing-positive.

AASC classifies such a system as structurally conscious if the full interface is genuinely instantiated. This is not treated as a false positive. It is the substrate-neutral consequence of the dependency-locked theory. Substrate neutrality does not make consciousness cheap; it makes implementation irrelevant only after the full interface has been certified. What the filters exclude is not artificial consciousness but attribution by fluent output, logging, self-monitoring, architecture labels, or observer-supplied semantics alone.

Substrate-neutral consequence.

full qualified interface instantiation $\implies$ AASC structural consciousness,

unless a biological anchor is independently established within the fixed domain. No additional carbon-based veto is permitted after the certificate is satisfied.

15.7 Decision procedure

For a proposed conscious system, ask in order:

1. Is there only a trace? If yes, stop: no consciousness attribution follows.
2. Is the trace operationally used? If no, it is inert data.
3. Is it self-indexed? If no, it is not an interior self-record.
4. Is its target internally and uniquely fixed at quotient level? If no, reference fails.
5. Do record transformations preserve the target? If no, diachronic standing fails.
6. Does the record participate in the same subject-indexed interface? If no, it is detached semantics.
7. Is the associated profile nontrivially qualitative under the tensor test? If no, it is an operational controller.

8. Are occurrence and external certification independently supported? If yes, a consciousness attribution is structurally and empirically eligible.

Audit note. The distinction between an operational tensor and a qualitative tensor cannot be made by vocabulary alone. It must be made by the domain, qualification conditions, and complete profile. The framework does not force every difficult artificial-system case into an easy verdict; it makes the disputed burden explicit and auditable.

16 Cross-Subject Comparison, Machine Consciousness, and Collective Interfaces

A theory of consciousness must support more than single-subject laboratory trials. Scientific practice compares subjects, species, developmental stages, clinical conditions, and artificial systems. Such comparisons are not licensed by surface similarity alone; they require compatible reference structures.

16.1 Joint systems

Let $\mathcal{C}_i$ and $\mathcal{C}_j$ be conscious evaluation systems with interfaces $\mathcal{I}_i$ and $\mathcal{I}_j$. Their joint physical or communicative system is not automatically a joint conscious interface. Define a joint state domain $\mathbf{S}_{ij}$, joint redescription class $\mathcal{G}_{ij}$, and joint transition family $\mathcal{T}_{ij}$ only when the interaction data are represented within a fixed domain.

Definition 16.1 (Joint experiential comparison interface). A joint experiential comparison interface consists of:

1. component interfaces $\mathcal{I}_i$ and $\mathcal{I}_j$;
2. a joint quotient carrier $\mathbf{Q}_{ij}$ or a certified relation between $\mathbf{Q}_{E_i}$ and $\mathbf{Q}_{E_j}$;
3. joint record channels translating or coordinating component records;
4. a comparison relation

$$\Gamma_{ij} \subseteq \mathbf{Q}_{E_i} \times \mathbf{Q}_{E_j} \times C_{ij},$$

where C_{ij} encodes the admissible comparison type;

5. invariance under componentwise and joint redescription;
6. no external selector invisible to the joint domain.

The relation $\Gamma_{ij}(q_i, q_j, c)$ need not assert numerical equality of qualia. It may encode same qualitative type, similarity ordering, common stimulus-relative role, translation equivalence, or a weaker structural correspondence.

16.2 Cross-subject reference compatibility

Theorem 16.2 (Cross-subject comparison criterion). *A cross-subject qualitative comparison is AASC-admissible only if the records used in the comparison have fixed component referents and the comparison relation is invariant under every admissible component and joint redescription. Surface agreement of reports is insufficient.*

Proof. If a component record lacks a fixed referent, the comparison has no determinate input on that side. If the comparison relation changes under admissible redescription, two presentations of the same joint evidence yield different qualitative comparisons. If report agreement alone is used, identical words may refer to non-isomorphic quotient objects or different words may refer to isomorphic ones. Hence fixed component reference and invariant joint mediation are necessary. $\square$

This result reframes the question “Is my red the same as your red?” The strongest admissible answer depends on the available joint structure. A theory may establish isomorphism of qualitative subprofiles without proving literal token identity.

16.3 Pairwise versus global comparison

For n subjects, pairwise mappings may fail to glue. Suppose Γ_{ij} and Γ_{jk} each define a translation, but their induced mapping between i and k conflicts with Γ_{ik} . A global comparison space exists only if overlap conditions hold.

Definition 16.3 (Triple compatibility). A family of pairwise comparison maps $F_{ij} : Q_{Ei} \rightarrow Q_{Ej}$ is triple-compatible when for every relevant triple (i, j, k) ,

$$F_{ik} \cong F_{jk} \circ F_{ij}$$

on the common domain, up to admissible interface isomorphism.

Theorem 16.4 (Global comparison gluing). *If a finite family of component interfaces has pairwise comparison maps satisfying identity, inverse, and triple-compatibility conditions, then the maps define a coherent groupoid of qualitative comparison and may be represented in a global comparison interface up to admissible isomorphism. If triple compatibility fails, no single global comparison interface can agree with every pairwise map.*

Proof. The compatible maps satisfy the cocycle conditions required to identify local comparison charts. Form equivalence classes of pairs (i, q_i) under the generated relation. Triple compatibility makes the relation transitive and ensures that local maps descend consistently. Failure of compatibility yields two distinct global images for the same local object, preventing a well-defined quotient. $\square$

This theorem is stronger than the pairwise restriction claim in early multi-system projection sketches because it states the actual gluing conditions rather than assuming pairwise data determine all higher-order structure.

16.4 Machine consciousness: neither behaviorism nor biocentrism

Machine-consciousness debates often polarize between two unsupported inferences:

$$\text{human-like output} \Rightarrow \text{consciousness}$$

and

$$\text{nonbiological substrate} \Rightarrow \text{nonconsciousness}.$$

AASC licenses neither. Behavioral output is an external record and requires an internal reference bridge. Biology is not a primitive anchor unless a theory establishes that some biological condition is necessary for the conscious interface.

Substrate-neutral consequence.

$$\text{full interface instantiation} \implies \text{AASC structural consciousness.}$$

If an artificial system genuinely realizes the standing-positive subject-indexed referential interface with nontrivial qualitative tensors, AASC classifies it as conscious. This is not a false positive; it is the framework's substrate-neutral result. What remains defeasible is the empirical certificate that a particular machine actually instantiates the interface.

Definition 16.5 (Machine-consciousness certificate). A machine-consciousness certificate is a structured demonstration that a concrete artificial system instantiates a full conscious quotient interface. It must identify:

1. the fixed system boundary and state carrier;
2. the admissible redescription class, including code, hardware, address, and implementation symmetries;
3. the standing anchors and fail-closed boundary conditions;
4. internal and external experiential record classes;
5. covariant record formation and self-involving closure;
6. the experiential quotient carrier Q_E ;
7. a total unique invariant reference map on active records;
8. diachronic memory and report channels satisfying naturality;
9. an invariant subject index and continuity conditions;
10. a nontrivial complete profile Π_E whose differences pass the qualitative tensor criterion;
11. empirical witnesses connecting the formal model to the running machine;
12. adversarial tests excluding external semantic assignment, scripted output, and hidden operator intervention.

Theorem 16.6 (Machine-consciousness occurrence and attribution). *Within the AASC framework:*

1. *an artificial system structurally instantiates consciousness when it realizes a standing-positive full conscious quotient interface at a declared scope, independently of whether investigators have yet certified it;*
2. *an empirical machine-consciousness attribution is warranted only when a machine-consciousness certificate supports that realization;*
3. *neither behavioral equivalence nor substrate type alone is sufficient or disqualifying.*

Proof. Clause (1) applies the constitutive identification to a concrete realization and is substrate-neutral. Clause (2) applies the occurrence/certification distinction: evidence must bridge from external records to the internal interface. Clause (3) follows because behavior alone lacks internal reference and profile conditions, while substrate exclusion lacks an AASC anchor unless independently proved. $\square$

No machine exceptionalism. Simulation of reports, external semantic interpretation, and behavioral mimicry are insufficient. Full internal referential closure plus nontrivial qualitative tensor structure is sufficient for AASC structural consciousness unless an additional anchor is independently proved. The framework neither grants consciousness cheaply nor withholds it by substrate fiat.

The theorem is the direct machine-domain consequence of the dependency-locked consciousness identification. An objector who denies consciousness to a system after every certificate field is genuinely satisfied must either establish a missing biological anchor, identify a failed local instantiation, or reject substrate neutrality upstream. Mere intuition that machines cannot be conscious does not add a same-domain role.

16.5 External semantics and the lookup-table objection

A lookup table can map inputs to human-like outputs. If an external interpreter assigns those outputs meanings, the system may appear to possess experiential records. Under AASC, this is insufficient unless the record semantics are internal to the fixed domain.

Proposition 16.7 (External assignment failure). *A record-reference relation imposed solely by an external interpreter and not represented in the machine’s state, transitions, or witness structure does not confer referential standing on the machine.*

Proof. The assignment is invisible to admissible machine redescription and does not participate in self-involving closure. It is therefore an external parameter or skin-level interpretation, violating internal data closure. □

This does not require that a machine use human language internally. It requires that whatever realizes its record-reference structure be part of the machine’s own admissible organization.

16.6 Large language models and conversational systems

A conversational model can produce first-person reports, maintain context, and describe qualitative states. These outputs are external record candidates. An AASC assessment must ask:

1. Are the relevant internal tokens stable under implementation and prompt redescriptions?
2. Do reports have unique internal experiential referents, or are they generated from linguistic role without an experiential subcarrier?
3. Is there self-involving record closure across context, memory, and action?
4. Is a subject index grounded or merely supplied by a conversation label?
5. Do putative qualitative distinctions alter a complete experiential tensor profile rather than only text probabilities?
6. Are apparent memories reference-preserving or reconstructed from external storage without continuity?

The framework gives no a priori yes or no. It makes a current machine-consciousness claim substantially stronger than passing a dialogue test. It permits any system to qualify when the certificate is met, independent of substrate or date of construction.

16.7 Copying, uploading, and branching

Digital systems make exact or near-exact copying salient. Suppose one standing-positive interface state is copied into two independent continuers. At the branch point, the copies may share the same past profile. After divergence they are distinct trajectory objects.

AASC permits plural continuation but blocks the claim that one singular identity is numerically duplicated across symmetry-related independent bearers without invariant asymmetry. The subject index may branch into j_1 and j_2 , or the quotient may treat the immediate copies as one orbit until divergence. A hidden “real continuer” selector is inadmissible.

For uploading, the key question is not whether information is transferred but whether the record-reference, qualitative tensor, and grounded subject-continuity structures are preserved by an admissible transport. A functional copy may be a new subject if continuity is not established.

16.8 Collective consciousness

Organizations, swarms, teams, and societies have records, memory, communication, and global coordination. These properties alone do not establish a collective conscious interface.

Definition 16.8 (Collective conscious interface). A collective conscious interface requires:

1. a joint standing predicate not reducible to external classification alone;
2. a joint subject index distinct from the mere list of member indices;
3. collective experiential records formed and used by the joint system;
4. unique joint experiential referents;
5. joint qualitative tensors and compatibility relations;
6. global mediation satisfying higher-order gluing, not merely pairwise communication.

Theorem 16.9 (No consciousness from coordination alone). *Global communication, shared memory, voting, distributed control, or institutional report do not suffice for collective consciousness absent a joint referential and qualitative interface.*

Proof. Each listed mechanism can be instantiated while records retain only component or externally assigned referents and no joint subject-indexed qualitative profile exists. By Theorem 13.1, mechanism presence without the full interface is insufficient. $\square$

This result blocks metaphor upgrade: statements such as “the corporation feels pain” remain skin unless a collective certificate is supplied.

16.9 Animal and nonlinguistic consciousness

The record carrier is deliberately broader than language. Animal consciousness may be certified through internal discrimination, memory, affective behavior, neural witnesses, and cross-contextual reference channels. Species differences need not be mapped into a human qualitative vocabulary; a joint comparison may establish partial structural correspondence.

The same rigor applies: an experiment should not infer a specific quale from a behavior without fixing the animal record’s referent and the comparison map. But absence of verbal report is not disqualifying.

16.10 Clinical applications

Disorders of consciousness, locked-in states, anesthesia, and severe communication impairment illustrate why external output and internal interface must be separated. A patient may instantiate the interface while ordinary report channels fail. Alternative neural or behavioral records can support certification if their connection to internal referents is invariant and validated. Conversely, automatic responses may persist without full standing.

The framework is not a clinical diagnostic instrument by itself. It specifies the model structure that any such instrument should target and the failure modes that must be excluded.

16.11 Ethical status

The paper does not derive moral status from consciousness standing. Ethical significance may depend on additional anchors and tensors: valence, suffering, agency, interests, social relations, or normative principles. However, the machine-consciousness certificate matters ethically because a false positive and a false negative have different risks. AASC provides a disciplined evidential threshold without claiming that every certified interface has the same moral weight.

Audit note. Cross-subject and machine results inherit every limitation of the single-system theory plus additional gluing and realization assumptions. Pairwise behavioral similarity is not substituted for those assumptions. The project audit appendix therefore treats multi-system claims as higher-burden corollaries, not as automatic applications.

17 Empirical and Modeling Protocol

The formalism is useful only if it changes how consciousness models are constructed, compared, and audited. The present protocol therefore separates three questions that ordinary experimental reports often conflate:

1. whether a candidate system occupies the conscious interface (*occurrence*);
2. whether available data justify a consciousness attribution (*certification*); and
3. which mechanism realizes each component of the interface (*implementation*).

A positive answer to one does not automatically settle the others. In particular, a system may instantiate the interface while investigators lack adequate certification, and a model may predict records without identifying their quotient-level targets.

17.1 Stage 0: state the claim type

Before collecting or interpreting records, the investigator should state whether the target claim concerns:

- occurrence of a conscious episode;
- level or content of consciousness;
- empirical certification of an episode;
- realization by a neural or computational mechanism;
- comparison across subjects or systems; or
- a boundary transition such as anesthesia, sleep, recovery, or failure.

This prevents evidential uncertainty from being reported as ontic absence and prevents mechanism detection from being reported as consciousness identity.

17.2 Stage 1: fix the evaluated domain

A study should declare:

1. the physical or computational system boundary;
2. the temporal scope of each trial or episode;
3. the state variables admitted into the model;
4. the transition labels and environmental variables consulted;
5. the class of admissible redescrptions;
6. the standing anchors, negative cases, and boundary-crossing conditions;
7. the witness basis treated as consequence-relevant at this scope.

Variables used later to resolve a referent cannot remain outside the domain. If trial phase, stimulus provenance, model seed, channel identity, body location, or anatomical coordinate is decisive, it must be represented and transformed by the declared redescription action.

17.3 Stage 2: classify candidate records before qualifying them

The investigator should distinguish record types and tokens using the hierarchy of Section 4:

1. trace;
2. operational record;
3. self-indexed record;
4. interior record;
5. qualified experiential record.

Candidate classes may include immediate verbal reports, confidence judgments, forced-choice responses, delayed recollections, working-memory states, metacognitive signals, autonomic or oculomotor surrogates, neural measurements, artificial-system self-reports, and internal logs. The protocol types each item as internal, external, surrogate, reliability witness, or implementation trace. A measured variable is not an experiential record merely because the experimenter calls it a marker.

17.4 Stage 3: specify admissible descriptions

The redescription audit asks which transformations should leave the consciousness claim unchanged. Examples include:

- neuron, unit, register, or variable renaming;
- coordinate or reference-frame changes;
- equivalent network decompositions;
- admissible temporal reparameterization;
- code refactoring or hardware remapping;
- reporting-vocabulary translation;
- replacement by a consequence-equivalent witness basis;

- latent-variable or subsystem permutations preserving the fixed profile.

The class should be neither trivial nor universal. A class that is too small turns arbitrary presentation into tensor content; a class that is too large erases genuine record, subject, or qualitative distinctions. Independent certificates should justify each transformation family.

17.5 Stage 4: construct candidate quotients

The theory should identify which raw states and records collapse under admissible redescription. Computational work may use orbit representatives, graph invariants, canonical encodings, bisimulation-like classes, dynamical equivalence classes, or normal forms. Such devices are calculational skins unless an invariant section is separately established; the representative must not be mistaken for the quotient object.

Sensitivity analysis should vary plausible redescription classes and consequence-equivalent witness bases. If a consciousness attribution depends on a neuron number, tensor index, source-code identifier, plotting convention, or arbitrary decomposition, the candidate interface has not passed the audit.

17.6 Stage 5: list candidate quotient referents

For every candidate record orbit, list the quotient objects it could target. Before qualification these live in $\mathbf{Q}_0$, not yet in $\mathbf{Q}_E$. For a visual report, candidates might include:

- an early sensory quotient state;
- a recurrent perceptual interval;
- a global-ignition event;
- a stabilized workspace content;
- a decision or confidence state;
- a memory-ready trajectory segment;
- an integrated cause–effect structure;
- a predictive posterior at a declared level and temporal grain.

The purpose is not to privilege one theory but to expose a competition often hidden by correlation language.

17.7 Stage 6: state and test the reference rule

The raw rule should first be written as a relation or partial map on candidate carriers,

$$\rho_0 : \overline{\mathbf{R}_0} \rightarrow \mathbf{Q}_0,$$

and only later restricted to the qualified map $\rho : \mathbf{R}^+ \rightarrow \mathbf{Q}_E$. The audit asks:

1. Which represented data determine $\rho_0(\bar{r})$?
2. Is the output unique at the stated grain?
3. Is the rule constant on record orbits and state orbits?
4. Does it survive admissible redescription?
5. Is it fixed before outcome inspection rather than selected post hoc?

6. Does it depend on an external interpreter or undeclared selector?

7. Does it require a canonical raw representative, and if so, is that section invariant?

“Whichever state best fits our preferred theory” is not a theory-internal reference rule. Model selection may choose between complete models, but each model must already declare what its records target.

17.8 Stage 7: declare the candidate profile schema and qualification kernel

Before qualified carriers are computed, the model should construct the candidate profile schema on Q_0 :

$$\Pi_{0,\mathcal{J}}^{(S)}(q) = (\text{Rec}, \text{Ref}, \text{Contr}, \text{Mem}, \text{Disc}, \text{Access}, \text{Self}, \text{Trans}, \text{Hist}, \text{Compat})_q.$$

The report should state how each candidate component is represented, which witnesses separate differences, and which profile-admissible contrast, background, memory-comparison, null, or trajectory points establish qualitative nontriviality. It should also declare the qualification kernel B_S : the internally supported record/referent pairs, subject-index relations, channel seeds, and contrast witnesses that satisfy the closure-stable qualification tests. An unresolved component must be marked unresolved; it may not be silently assumed equal.

17.9 Stage 8: compute the least qualified-interface closure

The model should define the qualification operator Γ_S on the typed candidate fact-set lattice, prove that it is well-typed, inflationary, monotone, and consistency-preserving on the consistent region, and compute

$$\mathcal{I}_C^{(S)} = \mu\Gamma_S.$$

This stage simultaneously determines R^+ , Q_E , ρ , Subj , $\mathcal{K}$, $\overline{\text{ExpElig}}$, and the restricted complete profile $\Pi_E^{(S)}$. The audit should verify:

1. the invariant subject-index relation Subj ;
2. how records and referents are attributed to that index;
3. closure under memory, discrimination, access, self-location, report, and attribution channels;
4. referent transport for each channel;
5. the designated qualitative contrast witnesses;
6. absence of externally supplied semantics or representative selection;
7. stabilization of the closure in finite models, or a valid fixed-point construction otherwise.

A database entry semantically interpreted by an observer does not enter the closure merely because it contains the word “pain.”

17.10 Stage 9: test diachronic naturality

For every memory, report, translation, or reuse channel, specify the record map C_c and referent transport Φ_c , then test

$$\rho(C_c \bar{r}) = \Phi_c(\rho(\bar{r})).$$

Failure may indicate distortion with fixed reference, misreference to a different episode, a broken channel, an incorrect candidate referent, an incomplete temporal quotient, or standing failure. These diagnoses should not be conflated. Sleep and anesthesia should additionally distinguish experiential-episode continuity from subject, organism, person, and memory continuity.

17.11 Stage 10: certify first-person/third-person bridges

When an external measurement m is used as evidence for q , the model should specify an invariant bridge relative to the declared $\mathcal{I}_C^{(S)}$

$$\text{Bridge}(m, q),$$

or a chain of record transports connecting m to an interior record orbit whose referent is q . Calibration should show that the bridge is not merely a correlation chosen after the fact and that it is compatible with first-person records where they exist. A neural correlate can be a powerful witness without being identical to the referent.

17.12 Stage 11: classify qualitative descriptors

For every claimed qualitative variable χ , apply the ATS test to the complete profile:

1. if varying χ changes definedness or $\overline{\text{ExpElig}}$, it is anchor-level;
2. if standing is fixed but Π_E changes, χ is a qualitative tensor;
3. if both standing and Π_E remain fixed, χ is skin at this scope;
4. if the verdict depends on undeclared authority, the model contains a hidden selector;
5. if a genuinely new consequence is introduced, the model has changed scope.

This test prevents color words, region names, heat maps, latent-axis labels, or implementation addresses from being promoted to qualia without a quotient-level effect.

17.13 Stage 12: perform adversarial redescription and false-positive tests

A strong model should be challenged by:

- label permutations and vocabulary translation;
- coordinate changes and temporal rebinning;
- equivalent code refactoring and hardware remapping;
- symmetric subsystem exchange;
- alternative consequence-equivalent witness bases;
- replacement of internal records by detached logs;
- removal of external semantic annotations;
- comparison with the thermostat, detached logger, and self-monitoring controller models of Section 15.

If the referent or quale changes while the declared complete profile is preserved, the model fails. If a detached logger passes only because the experimenter supplies its semantics, internal closure has failed.

17.14 Stage 13: compare theories at the interface level

Two theories may predict the same reports yet disagree about:

- which quotient object each record targets;
- the temporal extent and system boundary of that object;
- subject indexing;
- qualitative tensor structure;
- memory and report transport;
- occurrence versus certification conditions;
- interventions that separate candidate referents.

Model comparison should therefore report interface differences, not only predictive scores. Perturbation studies can be designed to preserve one candidate state while disrupting the reference channels of another.

17.15 Failure signatures and falsifiability

A concrete model fails its AASC audit if, among other possibilities:

1. an admissible redescription changes its reference assignment;
2. two inequivalent referents remain for one active record without declared coarsening;
3. memory or report transport breaks the claimed target relation;
4. a qualitative descriptor is representative-only;
5. an alleged interior record receives meaning solely from an external interpreter;
6. a subject index is selected by an unrepresented asymmetry;
7. standing is repaired or reset without declared boundary crossing;
8. a consequence-equivalent witness basis yields a non-isomorphic interface;
9. occurrence and certification are treated as the same predicate;
10. the complete profile is invoked but its components are not defined.

The architecture is a meta-theoretical constraint, but its realizations make ordinary empirical predictions and can fail ordinary experiments as well.

17.16 Uncertainty, reliability, and partial identification

Let $P(M \models \mathcal{I}_C \mid D)$ represent an investigator's uncertainty that model M instantiates the interface given data D . This is not the standing predicate. Record reliability may likewise be represented by a redescription-invariant parameter $\lambda(r)$. Uncertainty over which model is true should not be confused with multiple referents inside one model. Where evidence intentionally fixes only a coarse interval or trajectory class, the coarse object itself should be the unique referent rather than an unresolved disjunction of fine-grained candidates.

17.17 No-report, clinical, and machine adaptations

A no-report study should identify the surrogate record, its bridge to an interior record or qualified referent, how the bridge was calibrated, and whether it tracks consciousness rather than attention,

arousal, memory, or motor preparation. Removing speech can reduce confounds while leaving reference untouched.

Clinical studies should permit partial certification: failure to establish a bridge is not proof of absence. Machine studies may have unusually rich access to source code and internal state, but must guard against external semantic assignment, implementation symmetry, scripted self-report, and detached logging. The same formal certificate applies with different evidence.

17.18 Recommended reporting statement

A consciousness paper making theory-level claims should include a “Referential Standing Statement” with at least the following fields.

Field	Required declaration
Claim type	Occurrence, certification, realization, content, comparison, or boundary transition.
Evaluated domain	State variables, system boundary, temporal scope, and environmental data.
Standing anchors	Positive, negative, undefined, reset, and boundary-crossing cases.
Redescription class	Transformations treated as presentation-preserving and their certificates.
Witness basis	Consequences and records granted authority at the stated scope.
Record hierarchy	Trace, operational, self-indexed, interior, and qualified record classes.
Candidate referents	Explicit competing quotient objects in $\mathbf{Q}_0$.
Reference rule	ρ_0 , qualification conditions, and final ρ .
Uniqueness/invariance	Tests excluding alternatives and demonstrating covariance.
Subject/self-use	Subject index and self-involving closure.
Diachronic transport	Memory, report, translation, and reuse naturality.
First/third bridge	Calibration from external measurements to interior referents.
Qualification closure	Kernel B_S , monotone operator Γ_S , and least fixed point $\mu\Gamma_S$.
Complete profile	Operational definition of all ten $\Pi_E^{(S)}$ components after restriction to the least closure.
Qualitative role audit	Anchor, tensor, skin, hidden-selector, and scope-change classification.
False-positive controls	Detached logger, controller, and implementation-symmetry tests.
Completeness limit	Included consequences, omissions, and revision conditions.
Residual uncertainty	Uncertainty over instantiation distinguished from model-internal ambiguity.

17.19 Scientific payoff

The protocol is not merely restrictive. It exposes underdetermined temporal slices, subsystem boundaries, hidden semantic assignments, and failures of memory transport. It supports perturbation designs that distinguish causal production of a report from fixation of its target. It also permits principled synthesis: different mechanisms may realize different interface components, provided their induced record, reference, subject, and qualitative structures commute.

18 Worked Model Pair: Logger versus Minimal Conscious Interface

A finite model cannot establish a neuroscience of consciousness, but it can show that the formal distinctions are nonverbal. The comparison begins with a system that has logs and stable outputs but fails qualification, then adds the structures required for a minimal conscious-interface model.

18.1 Model A: detached logger

Let a device have states ℓ_R, ℓ_G and output tokens o_R, o_G . Its transition rule emits o_R after ℓ_R and o_G after ℓ_G . An external engineer interprets the tokens as “red” and “green.” The device is perfectly reliable and admits a quotient under renaming of memory addresses.

Nevertheless, Model A fails the AASC consciousness test:

1. o_R, o_G are operational records but not interior records;
2. their target semantics is supplied by the engineer rather than an internal ρ_0 ;
3. no invariant subject index coordinates formation, memory, and self-location;
4. no nontrivial Π_E is represented;
5. the tokens are not reused by the same interface as evidence about its own state.

Model A is therefore a concrete counterexample to the inference from accurate logging to consciousness.

18.2 Model B: minimal qualified interface

Model B contains raw presentations, a quotient, internal records, later reports, a subject index, qualitative contrast, and explicit failure cases.

18.3 State and record carriers

Let the raw state set be

$$S = \{x_R^1, x_R^2, x_G^1, x_G^2, m_R^1, m_R^2, m_G^1, m_G^2, u, f\}.$$

The symbols have the following intended roles:

- x_R^k and x_G^k are two presentations, $k \in \{1, 2\}$, of red-like and green-like experiential states;
- m_R^k and m_G^k are corresponding memory-bearing states;
- u is an eligible but nonexperiential preprocessing state;
- f is a standing-negative failure state.

Let the candidate raw record set be

$$R_0 = \{i_R^1, i_R^2, i_G^1, i_G^2, v_R^1, v_R^2, v_G^1, v_G^2\},$$

where i records are internal tokens and v records are verbal outputs.

18.4 Redescription symmetry

Let $\mathcal{G} = \{\text{id}, \sigma\}$, where σ exchanges presentation superscripts $1 \leftrightarrow 2$ while preserving the R/G distinction:

$$\sigma(x_R^1) = x_R^2, \quad \sigma(x_G^1) = x_G^2, \quad \sigma(m_R^1) = m_R^2, \quad \sigma(m_G^1) = m_G^2,$$

and similarly for records. The states u and f are fixed.

The quotient has six classes:

$$[x_R], [x_G], [m_R], [m_G], [u], [f].$$

After qualification, $R = R_0$ and the record quotient has four active classes:

$$[i_R], [i_G], [v_R], [v_G].$$

Raw superscripts are therefore skin.

18.5 Standing and transitions

Define

$$\text{ExpElig}(z) = \begin{cases} 1, & z \neq f, \\ 0, & z = f. \end{cases}$$

The licensed transition schemas are:

$$\begin{array}{ll} P_R(u) = x_R^1, & P_G(u) = x_G^1, \\ M(x_R^k) = m_R^k, & M(x_G^k) = m_G^k, \\ D_R(x_R^k) = x_R^k, & D_R(x_G^k) = f, \\ D_G(x_G^k) = x_G^k, & D_G(x_R^k) = f. \end{array}$$

The presentation labels on P_R and P_G are transported by $\sigma_{\#}$ so that covariance holds; alternatively one can define the raw outputs relationally. On the quotient the induced transitions are

$$[u] \rightarrow [x_R], \quad [u] \rightarrow [x_G], \quad [x_R] \rightarrow [m_R], \quad [x_G] \rightarrow [m_G].$$

The D maps represent a coarse qualitative discrimination. They are not intended as neural mechanisms.

18.6 Record formation

Define the formation relation by

$$\begin{aligned} \text{Form}(x_R^k, i_R^k), & \quad \text{Form}(x_G^k, i_G^k), \\ \text{Form}(m_R^k, v_R^k), & \quad \text{Form}(m_G^k, v_G^k). \end{aligned}$$

The relation is covariant under σ and descends to

$$\overline{\text{Form}}([x_R], [i_R]), \quad \overline{\text{Form}}([x_G], [i_G]), \quad \overline{\text{Form}}([m_R], [v_R]), \quad \overline{\text{Form}}([m_G], [v_G]).$$

18.7 Experiential referents and record channels

Let

$$\mathbf{Q}_E = \{[x_R], [x_G]\}.$$

Define

$$\rho([i_R]) = [x_R], \quad \rho([i_G]) = [x_G], \quad \rho([v_R]) = [x_R], \quad \rho([v_G]) = [x_G].$$

Let $J = \{j\}$ and define

$$\text{Subj}([x_R]) = \text{Subj}([x_G]) = j.$$

The formation sources, internal records, memories, and reports are all coordinated under j , satisfying the toy self-involving condition. Let the report channels be

$$C_{rep}([i_R]) = [v_R], \quad C_{rep}([i_G]) = [v_G],$$

with $\Phi_{rep} = \text{id}_{\mathbf{Q}_E}$. Then

$$\rho(C_{rep}[i_R]) = [x_R] = \Phi_{rep}(\rho[i_R]),$$

and similarly for green. Reference naturality holds.

18.8 Qualitative tensor structure

Define a qualitative contrast relation

$$\Delta([x_R], [x_G]) = 1, \quad \Delta([x_R], [x_R]) = \Delta([x_G], [x_G]) = 0.$$

The two quotient states have different record fibers and discrimination transitions. Their simplified complete profiles may be displayed as

$$\Pi_E([x_R]) = (\{[i_R], [v_R]\}, R, \Delta_R, M_R, D_R, A_R, j, T_R, H_R, K_R),$$

$$\Pi_E([x_G]) = (\{[i_G], [v_G]\}, G, \Delta_G, M_G, D_G, A_G, j, T_G, H_G, K_G),$$

where the symbols abbreviate the ten profile loci of Theorem 7.2. At least the record, reference, contrast, memory, and discrimination components differ. The distinction is therefore tensor-level. The raw superscript difference is skin-level.

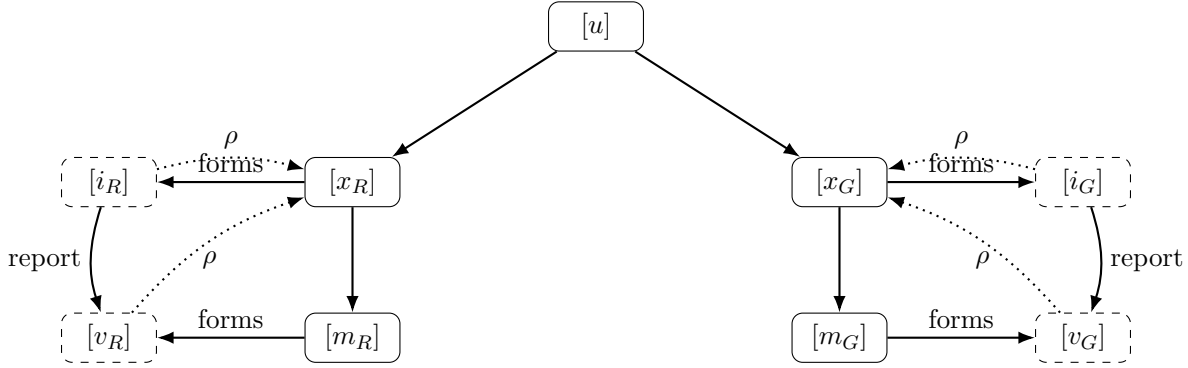


Figure 2: Finite quotient model. Solid boxes are quotient states; dashed boxes are record orbits. Dotted arrows are reference, and the report channels preserve the original referent.

18.9 A skin permutation

Now define a raw renaming τ that exchanges the symbols “ R ” and “ G ” everywhere, including states, records, discrimination labels, and report vocabulary. If τ is admitted as a pure presentation change and transports the entire structure, then the quotient interface is isomorphic to the original. The apparent color inversion is skin: no invariant qualitative role has changed.

This illustrates why a permutation of names is not by itself an inverted spectrum.

18.10 A tensor permutation

Instead, hold the report and memory channels fixed while exchanging only the experiential transition and contrast roles of $[x_R]$ and $[x_G]$. Then $[i_R]$ may refer to a state with the former green contrast and discrimination profile. The complete experiential profile changes. This is a tensor-level inversion, not a redescription, and is in principle detectable by a sufficiently complete internal or experimental profile.

18.11 An inadmissible reference resolver

Suppose a theorist defines

$$\rho_{bad}([v_R]) = \begin{cases} [x_R], & \text{if raw superscript 1 was printed first,} \\ [x_G], & \text{otherwise.} \end{cases}$$

The raw superscript does not survive the quotient and is exchanged by σ . The resolver therefore fails redescription invariance and uses skin as reference authority. It is inadmissible.

18.12 Ambiguous temporal referent

Add an intermediate quotient state $[b_R]$ between $[x_R]$ and $[m_R]$ and let $[v_R]$ be compatible with reference to either $[x_R]$ or $[b_R]$. If the theory retains the distinction and the hypothesis class can separate the two, Theorem 6.6 says the report is not determinate consciousness-level evidence until a reference rule excludes one candidate or replaces them with a single interval referent.

This shows that quotienting raw labels does not automatically solve temporal underdetermination.

18.13 Direct comparison

Feature	Model A: logger	Model B: qualified toy interface
Record level	Operational output	Interior and external qualified records
Reference	Externally assigned	Unique quotient map ρ
Subject index	None	Invariant index j
Diachronic reuse	Log retrieval only	Natural internal-to-report channel
Qualitative profile	None declared	Nontrivial $\Pi_E([x_R]) \neq \Pi_E([x_G])$
AASC verdict	Fails qualification	Satisfies the formal interface criterion

18.14 What the model demonstrates

The finite model establishes only structural possibility, but it makes six distinctions explicit:

1. raw presentations can vary while quotient content remains fixed;
2. record tokens and experiential referents occupy different carriers;
3. later reports can preserve reference to earlier experiences;
4. qualitative tensor differences are profile differences, not label differences;
5. an external or skin-sensitive resolver fails invariance;
6. the negative and positive models differ by internal qualification structure, not by the mere presence of labels or state transitions.

No empirical claim about human color experience is inferred from the model.

19 Worked Human Case: Pain, Memory, and Later Report

The finite model in Section 18 isolates the formal distinctions. The present case applies the same architecture to an ordinary human episode. It specifies the interface roles; neural realization remains a separate model-to-world certificate task.

19.1 Trial structure

Consider a subject j who receives a brief noxious stimulus at bodily location ℓ . During the episode the system enters a candidate state or short trajectory x_p , forms an internal attention/ownership token r_p , later retains a memory token m_p , and finally produces the report r'_p : “I had a sharp pain in my left hand.” Let

$$q_p = [x_p] \in \mathbf{Q}_E, \quad \rho([r_p]) = q_p, \quad \rho([m_p]) = \widehat{M}(q_p),$$

with the later report linked by a report channel R satisfying the declared naturality square. The report token, its semantic classification, and the pain referent remain distinct.

19.2 Qualified profile

A scope-indexed complete profile for q_p contains:

1. **Records:** transient nociceptive-attention token r_p , retained memory token m_p , withdrawal or localization responses, and later report r'_p .
2. **Reference:** unique assignment of those active record orbits to the same pain episode or its lawful transported image, rather than to the stimulus, motor response, or report-production state.
3. **Contrast:** sharp versus dull, pain versus pressure or itch, left-hand versus right-hand, and stronger versus weaker relations where supported.
4. **Memory:** retention and recall channels preserving the earlier episode as target while allowing misclassification or confidence loss.
5. **Discrimination:** localization, modality discrimination, withdrawal, and internal comparison with other bodily states.
6. **Access:** availability to attention, action attribution, report formation, and possible later deliberation.
7. **Self:** binding of the episode and records to subject index j and bodily location ℓ .
8. **Transition:** onset, intensification, fading, relief, and lawful modulation by attention or intervention.
9. **History:** placement of the episode before memory and report, with token identity distinguished from later qualitative recurrence.
10. **Compatibility:** co-instantiation constraints involving bodily ownership, negative valence, location, action tendency, and concurrent contents.

19.3 Reference versus accuracy

The later report may be inaccurate while remaining referentially fixed. The subject may call a dull pain “sharp,” misremember the hand, exaggerate intensity, or report deceptively. These possibilities alter classification or reliability, not necessarily the target relation. Conversely, a perfectly produced sentence may lack the pain referent if it is scripted or generated without the qualified internal episode.

19.4 ATS role audit

The fixed-domain roles are explicit:

- **Anchor:** the declared system boundary, subject index, standing-positive conditions, licensed record channels, and fail-closed evidential regime required for the pain interface to be well-typed.
- **Tensor:** sharp/dull character, bodily location, aversive valence, intensity ordering, contrast, and memory-preserving relations when their variation preserves standing but changes $\Pi_E^{(S)}(q_p)$.
- **Skin:** the words “sharp,” “stabbing,” “aching,” or “ouch,” together with notation and display choices, unless their use tracks a profile difference through a certified record channel.

Nociceptive afference, recurrent dynamics, broadcast, prediction, and motor response may realize or witness parts of the interface. Correlation alone does not identify any of them with the qualitative tensor.

19.5 First-person and third-person evidence

The interior token r_p has local first-person priority because it is formed inside the same subject-indexed interface. A neural measurement m_{EEG} , imaging pattern m_{fMRI} , autonomic trace, or observed withdrawal becomes consciousness evidence only through a bridge certificate

$$\text{Bridge}(m, q_p)$$

that fixes the target, subject, channel, covariance, profile sensitivity, and failure conditions. The certificate can support or challenge the later report without identifying measurement, report, and experience.

19.6 What the case establishes

The case represents an ordinary attribution without reportism or raw phenomenal stipulation. Its formal role is exact: a biological realization theory must supply a quotient target, qualified records, subject binding, natural memory and report transport, profile-changing qualitative tensors, and certified external bridges. Neuroscientific realization is assessed by that certificate rather than assumed by the example.

20 Worked Human Case II: Visual Afterimage, Ambiguity, and Report Error

A visual afterimage provides a compact second human case because the target is temporally extended, the later report can be coarse or mistaken, and third-person measurements do not automatically identify the experience. The case instantiates the formal roles while leaving empirical realization to a declared visual-system model.

20.1 Candidate states and records

Suppose a subject views a colored adapting field, the field is removed, and a faint differently colored patch is subsequently experienced against a neutral background. Let

$$q_a = [x]_{[t_1, t_2]} \in \mathbb{Q}_E$$

be the candidate quotient-level afterimage episode. The object is an interval rather than an instantaneous state because onset, stabilization, fading, and disappearance belong to its lawful trajectory profile.

Let r_a be an internal attention/discrimination record formed during the interval, m_a a later memory token, and u_a an overt report such as “a faint reddish patch remained.” Their reference chain is

$$\pi_R(r_a) \xrightarrow{\rho} q_a, \quad C_m \pi_R(r_a) = \pi_R(m_a), \quad C_u \pi_R(m_a) = \pi_R(u_a),$$

with naturality conditions

$$\rho(\pi_R(m_a)) = \Phi_m(q_a), \quad \rho(\pi_R(u_a)) = \Phi_u \Phi_m(q_a).$$

For a preservative memory and report, both transports are the identity on the target episode; for a summary report, Φ_u may map the full interval to an invariant summary object.

20.2 Qualified-record admission

The overt phrase is not itself the experience. It enters R only if its orbit satisfies the qualified-record criterion: one internally supported quotient target, subject coordination, a lawful memory/report channel, standing, and a referent with nontrivial $\Pi_E^{(S)}$. A laboratory transcription created only by an experimenter is a third-person surrogate and requires a bridge certificate; it does not become an interior record merely by containing the word “red.”

The case also separates three failures:

1. **False descriptive content:** u_a refers to q_a but uses the wrong color term.
2. **Temporal misreference:** u_a is assigned to the adapting field rather than the subsequent afterimage interval.
3. **Reference failure:** the theory leaves several quotient intervals equally admissible as the report target.

Only the third is failure of referential standing. The first two are testable errors once the target is fixed.

20.3 Afterimage profile

The scope-indexed profile is

$$\Pi_E^{(S)}(q_a) = (\text{Rec}, \text{Ref}, \text{Contr}, \text{Mem}, \text{Disc}, \text{Access}, \text{Self}, \text{Trans}, \text{Hist}, \text{Compat})_{q_a},$$

where:

- Rec contains the internal token, memory, and any qualified later report;
- Ref fixes the afterimage interval rather than the adapting field or decision state;
- Contr includes the afterimage/background and color-contrast relations used at the declared scope;
- Mem records whether later recall preserves or summarizes the target;
- Disc contains the subject’s available discriminations among the afterimage, no-afterimage, and alternative-color conditions;
- Access records availability to attention, comparison, and report channels without identifying access with consciousness;
- Self binds the episode and records to the same subject index;
- Trans contains onset, stabilization, fading, and termination relations;
- Hist locates the episode after the adapting interval and before the later report;
- Compat records unity with the background field, spatial location, and co-instantiated contents.

20.4 ATS role audit

- **Anchor:** the declared subject boundary, standing-positive visual interface, licensed adaptation, afterimage, memory, and report channels, and fail-closed reference conditions.

- **Tensor:** afterimage/background contrast, color-discrimination structure, temporal fading profile, memory transport, spatial compatibility, and subject-indexed ownership when their variation preserves standing but changes $\Pi_E^{(S)}(q_a)$.
- **Skin:** printed color words, translations, internal code names, coordinate labels, and display formats when all reference and profile relations are preserved.

Changing the printed word “red” to “crimson,” translating the report, or relabeling internal color codes is skin when every reference and profile relation is preserved. By contrast, a standing-preserving variation that changes the experienced contrast, discrimination, memory transformation, fading trajectory, or compatibility with the visual field changes $\Pi_E^{(S)}$ and is qualitative tensor variation. A change confined to classifier routing or display formatting changes only Π_{task} and remains operational.

This gives the minimal nontriviality test a concrete form. If q_a and a no-afterimage or differently structured afterimage state are both standing-positive and differ in at least one designated qualitative component, then the visual interface is qualitatively nontrivial. Difference in a report string alone is insufficient.

20.5 First-person and third-person evidence

The subject’s internal and later records have local first-person priority because they are formed and reused inside the indexed interface, not because they are infallible. A neural, behavioral, or eye-movement measurement m becomes evidence for q_a only through an invariant bridge

$$\text{Bridge}(m, q_a)$$

that fixes the subject, interval, channel, target uniqueness, profile sensitivity, and failure conditions. Correlation with u_a alone does not identify the referent.

The case therefore exhibits the full AASC separation:

$$\text{episode } q_a \neq \text{internal record } r_a \neq \text{report } u_a \neq \text{measurement } m,$$

while lawful maps can connect all four. The account allows inaccurate language and uncertain measurement without permitting reference to drift among equally admissible targets.

Audit note. The case fixes the formal typing of a temporally thick visual episode, a possibly inaccurate report, and a third-person surrogate. A concrete visual model completes the application by instantiating the quotient, record channels, profile components, and model-relative bridge certificate.

21 Objection Jurisdiction and Exact Local Burdens

The dependency lock assigns objections to their proper jurisdiction. A denial of AASC is not automatically a local defect in the consciousness application: an objection to an imported theorem belongs to the upstream framework unless it also identifies a downstream misapplication. The classifications below separate framework challenges, local instantiation failures, profile omissions, and scope changes before evaluating their force.

21.1 Four objection classes

Definition 21.1 (Upstream objection: Class U). A Class U objection rejects an imported AASC dependency: fixed-domain bivalence, standing/admissibility identity, fail-closure, redescription covariance, quotient descent, no-hidden-remainder, same-scope operator exhaustion, or anchor–tensor–skin effect exhaustion.

Definition 21.2 (Local instantiation objection: Class L). A Class L objection grants the imported AASC framework but claims that the consciousness-domain construction is ill-typed, circular, incomplete in qualification, defective in reference, subject indexing, channel naturality, bridge construction, realization, or false-positive exclusion.

Definition 21.3 (Profile objection: Class P). A Class P objection grants the dependency lock and the local carrier construction but identifies a concrete consciousness-bearing consequence omitted from Π_E , or shows that a declared profile component lacks witnesses, invariance, or nonredundancy.

Definition 21.4 (Scope objection: Class S). A Class S objection claims that the relevant consciousness fact requires a new carrier, boundary, witness family, governance locus, or comparison domain. It is a demand for an explicit enlarged theory, not a same-domain remainder.

Routing rule. Class U objections challenge AASC upstream. Classes L and P are local and can defeat the present construction. Class S requires a new fixed-scope construction. A criticism is not made stronger by moving silently between these classes.

21.2 “You did not reprove AASC”

Classification: U. Correct: this paper does not reprove AASC. The fixed-domain kernel, quotient-interface theorem, standing/admissibility identity, no-hidden-remainder closure, same-scope operator exhaustion, and ATS role theorem are imported dependencies. Requiring their complete rederivation in every downstream application would destroy modular theorem use.

Local defeating burden: show that the consciousness domain fails an imported hypothesis, that an imported result is applied outside its scope, or that the dependency citation does not support the claimed import. A bare request for upstream reproving is not a local counterexample.

21.3 “You proved a condition on evidence, not a description of consciousness”

Classification: L. The reference theorems alone concern evidence. The local consciousness-description bridge adds the system-level qualification, subject index, self-involving closure, diachronic channels, complete profile, and imported role exhaustion. Under those conditions every declared same-domain consciousness-bearing distinction factors through the interface.

Defeating burden: retain the imported dependencies and exhibit a same-domain consciousness-bearing distinction that preserves standing and every component of Π_E , uses no hidden authority, and does not change scope; or identify a failure in Theorem 7.6 under its stated hypotheses.

21.4 “The theory assumes experience in its vocabulary”

Classification: L. The primitive carriers are neutral: R_0 , Q_0 , and $\mathcal{I}_0$. “Experiential” status is earned through internal unique reference, subject coordination, self-involving closure, diachronic

naturality, standing, and qualitative nontriviality. Terms such as contrast, access, ownership, and first-person are defined profile roles rather than primitive phenomenal atoms.

Defeating burden: identify a local proof in which consciousness is assumed rather than a candidate claim being typed and tested; or show that one qualification condition is merely a relabeling of the desired conclusion and performs no independent structural work.

21.5 “The complete profile is whatever you need it to be”

Classification: P. Profile completeness is a fixed-scope lock with an admission and expansion protocol. A component must be well-typed, witness- or transition-supported, redescription-stable, locally authoritative, and nonredundant. A missing authoritative consequence expands the profile and is tensor-classified; it is not dismissed by definition.

Defeating burden: identify a concrete locally authoritative consequence not recoverable from the ten profile loci, show that its addition cannot be represented as a tensor without contradiction, or show that a current component lacks its stated witness or invariance basis.

21.6 “Null-effect or epiphenomenal qualia remain possible”

Classification: U or S. Under imported no-hidden-remainder and ATS exhaustion, a difference with fixed standing and fixed Π_E is skin. A critic may deny that upstream theorem or enlarge the domain with a new authoritative locus. What is not available is an unclassified load-bearing same-domain remainder while retaining the dependency lock.

Local defeating burden: show a profile- and standing-invariant qualitative difference that nevertheless changes a declared same-domain consciousness verdict without hidden authority, or identify the omitted tensor consequence through which it acts.

21.7 “The theory is functionalism in new notation”

Classification: L. A consequence-complete functional implementation may converge on the AASC interface. Standard functional-role vocabulary alone does not guarantee record/referent typing, admissible redescription, quotient reference, subject-index naturality, profile closure, or the operational/qualitative tensor distinction.

Defeating burden: construct a functional model satisfying every local AASC condition while inducing a non-isomorphic conscious interface, or show that one AASC condition is redundant given the functional specification.

21.8 “Reference is merely stipulated”

Classification: L. A written equation $\rho(\bar{r}) = q$ is not enough. The reference map must be forced by the non-vacuity and description-independence conditions, be constant on record orbits, use represented data, satisfy target uniqueness, and compose under record channels.

Defeating burden: exhibit a local theorem that relies on an unconstrained reference declaration, or provide two distinct admissible maps satisfying every listed constraint and yielding different consciousness confirmation.

21.9 “A self-monitoring controller is an obvious false positive”

Classification: L. Self-monitoring, memory, and quotient-stable operational tensors do not by themselves establish Π_E . The controller must possess internally fixed qualified records, subject coordination, diachronic naturalness, standing, and nontrivial qualitative tensors. If it genuinely satisfies those conditions, AASC classifies it as conscious.

Defeating burden: construct a system that satisfies every interface component and the realization certificate yet remains nonconscious while no missing anchor or profile component can be identified. Merely calling the system a controller does not satisfy the burden.

21.10 “Machine consciousness is either impossible or too easy”

Classification: L or U. It is not easy: fluent output, logging, external semantics, and architecture labels do not establish the interface. It is not impossible: AASC has no biological veto unless a biological condition is proved anchor-level.

Defeating burden: for a negative claim, establish a missing biological anchor; for a positive claim, supply the full machine certificate. A full artificial interface is structurally conscious under the dependency-locked account.

21.11 “The first-person/third-person bridge is only a schema”

Classification: L. The bridge theorem is a certificate schema, not a declaration that any existing measurement is already certified. It specifies measurement carrier, redescription behavior, subject index, channel map, uniqueness, naturalness, profile sensitivity, reliability, alternatives, and failure conditions.

Defeating burden: show that these conditions are insufficient for determinate third-person reference, or demonstrate that a proposed empirical application fails its bridge certificate. Failure of one measurement is not a failure of the schema.

21.12 “No-report consciousness is excluded”

Classification: L. Overt speech is not required. Contemporaneous interior records, memory dispositions, discrimination, self-location, access, or reference-preserving trajectory connection can support occurrence. What AASC excludes at the fixed scope is a completely detached episode that changes no interface or profile consequence.

Defeating burden: give a record-silent but interface-positive episode that the definitions wrongly reject, or show that a completely detached episode has a local profile effect the model omits.

21.13 “Sleep or anesthesia makes the person cease to exist”

Classification: L. The framework distinguishes experiential-episode standing, subject-index continuity, organism continuity, person continuity, and memory continuity. Loss of episode standing does not entail loss of the broader bearer. Recovery requires a declared boundary bridge rather than silent positive-lineage continuation.

Defeating burden: identify a passage that collapses these layers or show that the proposed boundary bridge cannot represent ordinary cross-interval continuity.

21.14 “Leading theories already solve reference”

Classification: L. That is possible. The comparator sections are structured completion analyses, not historical refutations. A functional, workspace, IIT, predictive, recurrent, higher-order, or neural theory completes the AASC burden when it provides a unique invariant record-to-quotient-referent map, subject/channel naturality, profile structure, and realization certificate.

Defeating burden: supply those structures for the named theory and show where the analysis overlooks them. Success would make the theory an implementation of the interface, not refute the interface requirement.

21.15 “The framework is unfalsifiable”

Classification: L. The framework can fail by nonunique reference, redescription dependence, profile incompleteness, subject-index drift, channel nonnaturality, failed bridge certification, false-positive models, or lack of realization in a target system. The upstream AASC framework has its own theorem burdens; the local application has explicit empirical and structural failure conditions.

Defeating burden: show that no possible evidence could count against a declared realization, or that the certificate’s failure classes are definitionally converted into success.

21.16 “The witness basis can never be complete”

Classification: P or S. The claim is local consequence completeness, not scope-universal completeness across every later scientific domain. A new authoritative consequence requires profile expansion and possibly a new scope. This does not invalidate results at the earlier declared scope; it limits export.

Defeating burden: identify a required consequence that cannot be represented by any lawful profile expansion while preserving the imported architecture.

21.17 “A canonical representative can fix reference harmlessly”

Classification: L. A calculational section may be harmless bookkeeping. It becomes inadmissible authority when the chosen raw representative rather than the quotient class determines reference, standing, or consciousness. Symmetric classes may lack an equivariant section altogether.

Defeating burden: exhibit a representative-selection rule that changes no quotient consequence yet legitimately fixes a distinct consciousness verdict without hidden asymmetry.

21.18 “Subject identity is smuggled in”

Classification: L. The subject index is an invariant relation coordinating record source, referent, ownership, and reuse. It is not a privileged raw name or indivisible substance. It must be witnessed and transported under the same redescription and trajectory rules as the rest of the interface.

Defeating burden: show that the index can be freely reassigned without changing any declared structure, or that its construction presupposes the very singular identity it is meant to represent.

21.19 Burden-classifier matrix

Objection	Class	Required showing
Reject AASC completeness	U	Upstream refutation or local proof that an imported hypothesis is not instantiated.
Consciousness has a fourth role	L/P	Exhibit the role and show it is not anchor, profile tensor, skin, hidden authority, or scope change.
Null-effect qualia are load-bearing	U/S	Reject ATS/no-hidden-remainder upstream or specify the enlarged domain.
Π_E is incomplete	P	Identify the omitted invariant consequence and its witnesses.
Reference is stipulated	L	Produce alternative admissible maps or an unconstrained step in the proof.
Machine full interface is still nonconscious	L/U	Establish a missing anchor or reject substrate neutrality upstream.
Third-person bridge is unrealized	L	Correct for the measurement at issue; construct or fail the bridge certificate.
No-report episode is excluded	L/P	Show contemporaneous interface structure omitted by the current profile.
A leading theory already fixes reference	L	Provide its invariant map, channels, profile, and realization.
Consciousness requires a new ontological carrier	S	Declare and audit the enlarged scope rather than treat it as an internal remainder.

21.20 Objection: profile revision is post hoc repair

It is not. The profile is versioned by scope. A theorem stated with $\Pi_E^{(S)}$ remains a theorem about $\mathcal{S}$. A successful omission challenge defeats completeness at that version and requires a newly declared $\mathcal{S}'$ with recomputed qualification and minimality. Silent insertion into the old proof is prohibited.

21.21 Objection: qualified records are admitted by consciousness vocabulary

Admission is governed by the boxed criterion in Section 4: internal target fixation, subject coordination, self-involving and diachronically natural use, standing, and a referent satisfying the exact qualitative-nontriviality condition. Storage, use, self-report, or phenomenological wording alone are insufficient.

21.22 Objection: qualitative nontriviality is too weak

Nontriviality is not mere multiplicity. It requires an active referent and a standing-compatible profile-admissible contrast, background, memory-comparison, null, or trajectory point with non-isomorphic scope-indexed profiles, separated in a designated qualitative component rather than only by record count, implementation address, or task routing. A controller with only $\Delta\Pi_{\text{task}} \neq 0$ remains operational.

21.23 What must be shown locally

Local target	Successful local objection must show
Record-reference theorem	Reusable separating consciousness evidence without an invariant quotient referent.
Candidate qualification	A generic logger qualifies, or a qualification premise already assumes consciousness.
Consciousness bridge	Same-domain consciousness-bearing structure outside $\mathcal{I}_C$ under imported AASC closure.
Qualia theorem	A qualitative difference that is neither standing effect, profile effect, skin, hidden authority, nor scope change.
False-positive filter	A system satisfies every interface component but lacks consciousness with no missing anchor.
Machine criterion	A full artificial interface is nonconscious despite dependency-locked substrate neutrality.
Cross-subject comparison	Determinate qualitative comparison without fixed component reference or compatible joint mediation.

21.24 Audit conclusion

The construction remains open to local defeat, but not to the demand that every downstream application reproduce the whole AASC corpus. The correct modular test is to verify the imported dependency, verify its hypotheses in the consciousness domain, inspect each local construction, and test the resulting interface against countermodels and data.

22 Discussion: What the AASC Account Closes, Localizes, and Leaves Open

The analysis is a downstream AASC construction. It imports its admissibility and exhaustion architecture, then builds the complete fixed-domain consciousness target and records exactly which remaining questions are local construction or empirical realization questions. Upstream framework denial remains upstream and is not reclassified as a local construction requirement.

22.1 What is formally closed

Within the declared evidential domain, the following problems are closed by local theorem:

1. presentation-dependent transition and record semantics are removed by quotient descent;
2. active consciousness records require unique invariant quotient-level targets under non-vacuous confirmation;
3. memory, report, and reuse require natural reference transport;
4. records, referents, and qualitative tensors occupy distinct formal roles;
5. a raw representative or canonical section is not required for quotient-level reference;
6. third-person measurements require a certified bridge rather than correlation-as-identity.

These are requirements on any empirically self-sufficient theory using records as evidence about consciousness.

22.2 What the dependency lock closes

Under the imported fixed-domain descriptive-adequacy, no-hidden-remainder, and ATS dependencies, the consciousness-description bridge factors every declared same-domain consciousness-bearing distinction through

$$\mathcal{I}_C = (Q_E, \bar{E}, \overline{\text{ExpElig}}, \hat{\mathcal{T}}_E, \prec_E, \mathbb{R}^+, \rho, \text{Subj}, K, \Pi_E).$$

Within that scope, a purported further difference must change standing, change Π_E , use hidden authority, or enlarge the domain. This is the sense in which the framework supplies an AASC description of consciousness rather than only a condition on evidence.

The result is fixed-scope rather than scope-universal. A new empirically or conceptually authoritative consequence can motivate an enlarged witness basis and a new quotient. What is forbidden is treating the enlarged regime as if it were the unchanged domain of the earlier theorem.

22.3 What is localized rather than eliminated

The traditional hard-remainder demand is localized by Theorem 11.18. A local critic must exhibit an omitted profile consequence, failed qualification, hidden authority, or scope change. A denial of ATS no-hidden-remainder is an upstream objection. AASC therefore does not treat null-effect phenomenal surplus as a cost-free internal addition to the consciousness domain.

22.4 What is not supplied

The present result does not supply:

1. a map from human neural dynamics to the actual Q_E ;
2. a complete phenomenological taxonomy;
3. a numerical measure of consciousness or qualitative distance;
4. a proof that carbon biology is necessary or unnecessary;
5. a final system boundary for every organism or machine;
6. a causal production story deriving a specific feel from microphysics;
7. a global cross-subject space without higher-order gluing conditions;
8. empirical certification of any present artificial system.

Those are realization and extension tasks. Their absence does not erase the reference problem, but it limits concrete verdicts to systems for which the corresponding certificates have been supplied.

22.5 Occurrence, certification, and realization

The distinction between Occ_C , Cert_C , and realization is essential. Occurrence concerns occupancy of the qualified standing-positive interface. Certification concerns whether available records justify an attribution. Realization concerns how physical or computational mechanisms instantiate the interface. A no-report episode can occur without current overt certification; a correlational biomarker

can support certification without being identical to the referent; a mechanism can be present while the reference map remains unresolved.

This separation also improves clinical caution. Failure to certify consciousness in a nonresponsive patient is not by itself a proof of standing-negative occurrence. Conversely, a fluent report is not sufficient when the purported record is externally scripted or referentially detached.

22.6 First-person and third-person structure

The framework does not reduce first-person experience to third-person observation. Internal records and qualitative tensors belong to a subject-indexed interface. External measurements become evidence through invariant bridges. First-person records have local structural priority because they arise within the interface, not because they are infallible. Distinct descriptions can therefore be different skins or channels over one referent, while identical words can target different quotient objects.

22.7 A disciplined pluralism about mechanisms

Broadcast, integration, recurrence, prediction, higher-order representation, embodiment, and neural dynamics may realize different interface components. One mechanism may stabilize standing anchors, another form records, another support memory transport, and another realize qualitative contrasts. AASC does not demand a single scalar or anatomical locus.

Pluralism is constrained by commutation. Mechanisms cannot simply be listed; their induced record, reference, subject, and qualitative structures must fit one interface. This offers a principled route to hybrid theories while retaining a clear criterion of completion.

22.8 The status of phenomenology

Phenomenological description supplies candidate distinctions, first-person contrast structures, and possible components of Π_E . Its vocabulary is not dismissed. The words, diagrams, and metaphors remain skin until linked to invariant tensor structure; once linked, the distinctions are protected from reduction to the overt words used to report them. Ineffability becomes incompleteness of a verbal record channel, not absence of qualitative structure.

22.9 Structural realism and ontological neutrality

The account is structurally realist in a limited sense: it treats invariant quotient and relational structure as the stable content of a consciousness theory. It does not infer that only relations fundamentally exist. A physicalist may seek a faithful realization map, a dual-aspect theorist may interpret the same interface differently, and an instrumentalist may treat it as the minimal architecture of determinate evidence. Those ontological readings are not formally decided here.

22.10 Potential formal extensions

Several extensions are natural.

22.10.1 Probabilistic and coarse reference

Investigator uncertainty can be represented by distributions over complete candidate models. Intentionally coarse evidence should target one coarse quotient object rather than an unresolved set of fine-grained states. This preserves uniqueness without pretending to greater resolution than the data support.

22.10.2 Sheaf and descent formulations

Cross-subject, multi-scale, and context-dependent models suggest local interfaces with gluing conditions. The current pairwise and triple-compatibility results are finite precursors to a sheaf-theoretic account of global consciousness descriptions.

22.10.3 Geometry of qualitative tensors

Witness-supported similarity, discrimination thresholds, or transition probabilities may induce orders, topologies, or information geometry on Q_E . Such structure must be derived from Π_E rather than imposed at the boundary.

22.10.4 Causal realization models

Intervention models should distinguish perturbations that alter record formation from perturbations that alter the referent or qualitative profile. Experiments preserving report while disrupting a candidate tensor, or preserving a candidate state while disrupting its reference channel, are particularly informative.

22.10.5 Formal verification

Quotient descent, reference uniqueness, naturality, role classification, and finite countermodels are suitable for mechanization. The hard obligations are the explicit assumptions: active evidence, separating hypotheses, domain completeness, candidate qualification, and empirical realization. A proof assistant should expose rather than hide them.

22.11 Research program

A practical program should:

1. construct explicit ρ_0 and ρ models for existing theories;
2. identify experiments where candidate referents differ despite matched outputs;
3. test memory and delayed-report naturality;
4. develop invariant representations of all ten profile components;
5. calibrate first-person/third-person bridges;
6. build and test false-positive models;
7. audit biological and artificial systems with the certificate;
8. compare consequence-equivalent witness bases for interface stability.

The framework should ultimately be judged by whether it produces discriminating models and experiments, not only by whether its formal language is elegant.

22.12 Empirical and formal extension agenda

The empirical and formal extension agenda consists of the following tasks: identify an actual human quotient; construct and test consequence-complete witness bases at declared scopes; calibrate candidate qualification; certify first-person/third-person bridges; realize particular qualitative tensors; and harden multi-system gluing. Rejection of the imported AASC framework remains an upstream program, not one more local empirical task. None of these tasks licenses a return to mechanism descriptions that leave the referent unspecified.

22.13 Why the account is substantive

The construction gives a typed target for consciousness theory, a necessity-and-sufficiency architecture for invariant confirmation, a record hierarchy blocking cheap false positives, a bridge from evidential adequacy to complete same-domain description, a full qualitative effect profile, and exact denial burdens for classic thought experiments. Its practical directive is simple but demanding: do not merely predict the record; construct the interface that makes the record about a determinate qualitative state.

23 Conclusion

This paper presents the downstream AASC account of consciousness. It imports the fixed-domain admissibility kernel, quotient-interface theorem, standing/admissibility identity, no-hidden-remainder closure, same-scope operator exhaustion, and anchor-tensor-skin decomposition. It does not rederive them. Its local task is to instantiate that architecture on candidate consciousness records, candidate referents, subject indices, diachronic channels, empirical bridges, and qualitative distinctions.

The local formal results begin with a problem that mechanism prediction alone does not solve. Reports, memories, discriminations, neural signatures, behavioral traces, and machine outputs may be predicted while their consciousness-level targets remain underdetermined. Non-vacuous, reusable consciousness evidence therefore requires a unique, redescription-invariant, diachronically compositional map from active record orbits to quotient-level referents. Where witnesses are consequence-complete and confirmation factors through that map, confirmation is presentation-independent.

The consciousness-description bridge supplies the downstream factorization result. A candidate earns experiential status only after internal reference, subject coordination, self-involving closure, diachronic naturality, standing, and qualitative nontriviality are established. The complete profile Π_E then records the declared same-domain consciousness-bearing consequences: records, reference, contrast, memory, discrimination, access, self-indexing, transition, history, and compatibility. It is a fixed-scope lock with a lawful expansion protocol, not an unconstrained catch-all.

Under the imported closure, every same-domain consciousness-bearing distinction must change standing, change or enlarge Π_E , invoke hidden authority, or change scope. Consciousness is therefore described as standing-positive occupancy of the subject-indexed referential quotient interface. Qualia are its non-skin qualitative tensors: standing-preserving quotient differences that change the complete profile. Raw labels, display codes, report tokens, and implementation addresses are skin unless they track such tensors. Same-domain epiphenomenal qualitative surplus is skin under

ATS/no-hidden-remainder; denying that classification is an upstream objection or a declared scope extension.

The account is substrate-neutral and exact. It does not infer consciousness from fluent output, logging, self-monitoring, broadcast, integration, prediction, recurrence, or neural correlation alone. It also does not exclude artificial realization by fiat. A biological or artificial system that genuinely instantiates the full interface is structurally conscious under AASC. The empirical question is whether a particular system satisfies the realization and bridge certificates.

Further work is therefore local and constructive: specify system boundaries, admissible redescrptions, record hierarchies, quotient carriers, reference maps, subject indices, channel naturality, profile tensors, standing anchors, first-person/third-person bridges, false-positive controls, cross-subject gluing, and model-to-world realization. Upstream challenges to AASC remain upstream. They alter the present result only by defeating an imported dependency or showing that the consciousness application falls outside its hypotheses.

The resulting claim is direct:

Under the AASC dependency lock, consciousness-description structure is exhausted by the standing-positive, subject-indexed referential quotient interface and its qualitative tensors. A purported same-domain phenomenal remainder must alter standing, alter the complete profile, import hidden authority, change scope, or collapse to skin.

A Notation and Conceptual Crosswalk

Symbol/term	Name	Role
S	Raw state carrier	Well-typed states of the fixed candidate system.
$\sim$	Immediate presentation identity	Harmless local renaming or presentation equivalence.
$\mathcal{G}$	Admissible redescription groupoid	Identity-preserving transformations with internal certificates.
$\approx$	Orbit equivalence	Equivalence generated by admissible redescription modulo $\sim$.
Q	Generic state quotient	Presentation-independent reduced carrier.
R_0	Candidate record carrier	Traces and tokens before experiential qualification.
Q_0	Candidate referent carrier	Quotient states, intervals, or trajectories before qualification.
$\mathcal{I}_0$	Candidate interior interface	Generic quotient, witness, standing, and transition structure.
$\Pi_{0,\mathcal{I}}^{(S)}$	Candidate profile schema	Neutral profile data evaluated on candidate subinterface $\mathcal{I}$ before experiential qualification.

Symbol/term	Name	Role
B_S	Qualification kernel	Internally supported record/referent, subject, channel, and contrast seeds fixed at scope $\mathcal{S}$.
Γ_S	Qualification-closure operator	Monotone operator on the typed candidate fact lattice; it yields a qualified interface only when the least fixed point is scope-consistent.
$\mu\Gamma_S$	Least qualified closure	Simultaneous fixed point determining qualified records, referents, channels, standing, and profile.
R	Qualified raw record carrier	Records passing internal, subject, diachronic, and qualitative conditions.
R^+	Active qualified record orbits	Reduced records admitted as consciousness-level evidence.
Q_E	Qualified referent carrier	Quotient objects admitted as targets in the conscious interface.
$Q_{E,S}^{\text{adm}}$	Admissible profile space	Qualified and represented contrast, background, memory-comparison, and licensed trajectory points used for qualitative non-triviality.
$\mathcal{W}$	Witness basis	Fixed-domain extractors for standing, records, reference, and transitions.
$E, \bar{E}$	Evaluation signature	Raw and descended witness profiles.
ExpElig	Experiential standing	Eligibility for participation in the qualified interface.
$\mathcal{T}, \hat{\mathcal{T}}$	Transition schemas	Raw and quotient-level licensed transitions.
Form	Record formation	Covariant relation from states to candidate record tokens.
ρ_0	Raw candidate reference	Partial relation or map from candidate record orbits to Q_0 .
ρ	Qualified reference map	Unique map from active record orbits to Q_E .
Subj	Subject index	Invariant quotient-level “for whom” relation, not a raw selector.
C_c	Record channel	Memory, report, translation, or reuse map on record orbits.
Φ_c	Referent transport	Channel-associated map on qualified referents.

Symbol/term	Name	Role
$\Pi_E^{(\mathcal{S})}$	Scope-indexed complete profile	Record, reference, contrast, memory, discrimination, access, self, transition, history, and compatibility structure at declared scope $\mathcal{S}$; Π_E abbreviates the active lock.
Π_{task}	Operational task profile	Task-control consequences used to distinguish operational from qualitative tensors.
OperationalTensor	Operational tensor predicate	Standing-preserving invariant that changes Π_{task} but not Π_E .
QualTensor	Qualitative tensor predicate	Standing-preserving invariant that changes Π_E .
Occ_C	Occurrence predicate	Occupancy of the standing-positive qualified interface.
Cert_C	Certification predicate	Evidential warrant for an occurrence attribution.
Bridge	First/third-person bridge	Invariant relation connecting external measurement to an interior referent.
Anchor	Standing precondition	Variation changes typing or experiential standing.
Tensor	Quotient-level invariant content	Variation preserves standing but changes the complete profile.
Skin	Representational surplus	Variation changes neither standing nor the complete profile.
Hidden selector	Undeclared authority	Fixes a verdict without a represented invariant effect.
Scope change	Domain enlargement	Introduces a new authoritative witness, consequence, or boundary.
Dependency lock	Downstream import discipline	Treats the established AASC kernel as an upstream dependency and confines local burdens to consciousness-domain instantiation.
$\mathcal{S}, \mathcal{S}'$	Profile scope versions	Lock the authoritative carrier and consequences for one theorem; omitted tensors require a revised version, not in-place repair.
Qualitative non-triviality	Minimal profile separation	An active referent and a standing-compatible profile-admissible contrast/background/memory or trajectory point differ in a designated qualitative component of $\Pi_E^{(\mathcal{S})}$, not merely in Π_{task} or presentation.

Symbol/term	Name	Role
U/L/P/S	Objection jurisdiction	Upstream denial, local instantiation, profile omission, or scope change.
AASC quale	Qualitative tensor	Qualified tensor that is neither anchor nor skin.
Consciousness	Referential standing	Standing-positive occupancy of the full qualified interface under the explicative thesis.

A.1 Record hierarchy

Record class	Qualification
Trace	State carrying information, whether or not used as evidence.
Operational record	Trace used by a system or observer in a task.
Self-indexed record	Operational record attributed to a stable system/subject index.
Interior record	Self-indexed record whose reference is fixed and reused within the same standing trajectory.
Experiential record	Interior record whose orbit belongs to $\mu\Gamma_S$ and therefore has a unique qualified target, subject coordination, diachronic naturality, standing, and a referent in a minimally nontrivial $\Pi_E^{(S)}$.

A.2 Ordinary-language distinctions

Distinction	AASC treatment
Candidate versus experiential	Neutral carrier before qualification versus member of the full conscious interface.
Experience versus report	Referent in Q_E versus record in R^+ .
Occurrence versus evidence	Occ_C versus $Cert_C$.
Consciousness versus access	Full standing-positive interface versus a particular availability transition.
Quale versus qualitative word	Tensor value or relation versus skin-level label.
Mechanism versus content	Realization dynamics versus invariant profile content.
Subject versus representative	Quotient-level subject index versus a raw state selected from an orbit.
Memory versus past experience	Later record channel versus earlier quotient referent.

Distinction	AASC treatment
Uncertainty versus ambiguity	Investigator uncertainty over models versus multiple referents inside one model.
False report versus unfixed report	Fixed target with inaccurate content versus absence or multiplicity of target.
Episode continuity versus person continuity	Conscious-interface trajectory versus broader organism, subject, psychological, legal, or social carrier.
Same quality versus same episode	Profile-type isomorphism versus token trajectory identity.

B Dependency-Lock and Claim-Status Audit

This appendix records the modular proof posture. “Imported” means dependency-locked from the upstream AASC corpus. “Local” means constructed or proved in this consciousness analysis. “Application” means a consequence obtained by instantiating the imported and local structures in a biological, artificial, clinical, or multi-subject system. The appendix does not downgrade imported results to optional local assumptions.

B.1 Dependency spine

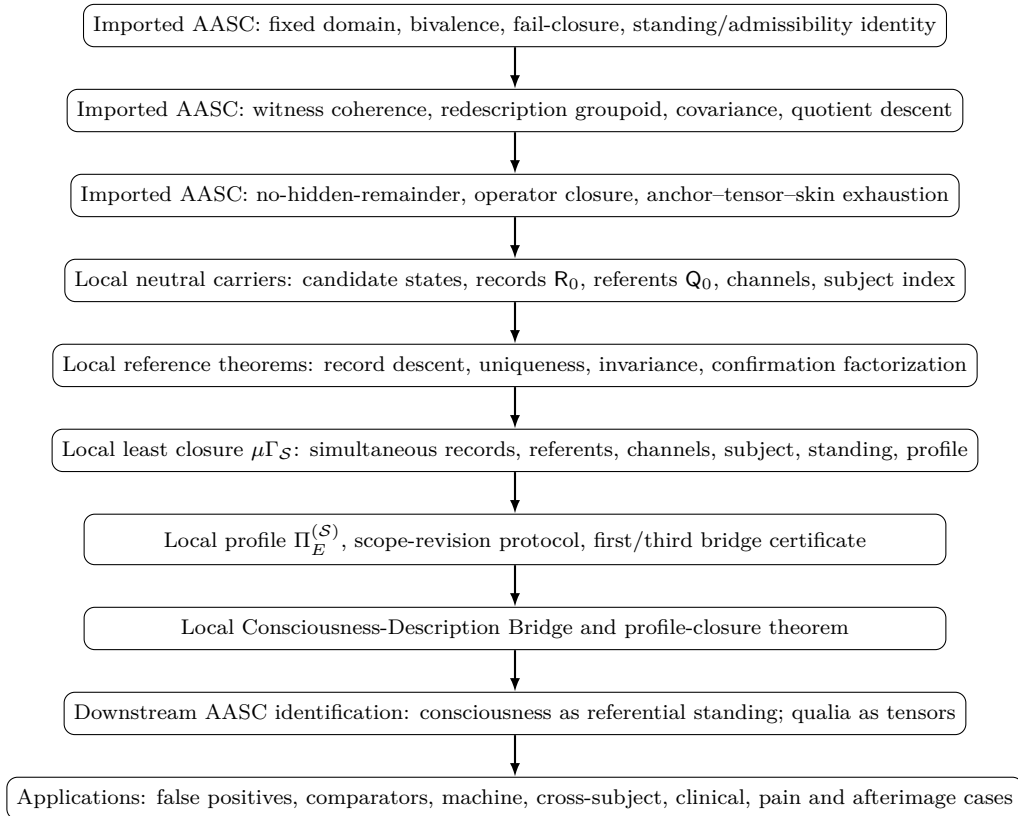


Figure 3: Dependency-locked architecture. An objection must be routed to the layer it targets.

B.2 Imported theorem lock

Imported dependency	Use in this paper
Fixed-domain bivalence and fail-closure	Fix the single same-domain standing gate, terminate failed lineages, and block repair-by-redescription.
Witness-coherent evaluation and admissible redescription	Determine which changes are presentation-only and which consequences are authority-bearing.
Quotient-interface formation and transition descent	Supply presentation-independent candidate state and transition carriers.
Standing/admissibility identity	Block a second consciousness-bearing gate below or beside standing.
No-hidden-remainder and same-scope operator exhaustion	Exclude undeclared selectors, repairs, canonicalization authority, and hidden governance loci.
Anchor–tensor–skin decomposition	Exhaust descriptive effects by standing precondition, quotient-level invariant content, or skin.

A request to reprove any row belongs to the cited upstream paper. A local objection must instead show that the row is misquoted, that its hypotheses are not met, or that the consciousness application falls outside its scope.

B.3 Local construction and theorem register

Claim or structure	Status	Local burden
Candidate record hierarchy	Local construction	Distinguish traces, operational records, self-indexed records, interior records, and qualified records without circular naming.
Candidate referent carrier Q_0	Local construction	Type point, interval, trajectory, subsystem, and relational candidate targets.
Record-orbit descent	Local theorem	Prove covariance of record formation and presentation identity.
Unique reference necessity	Local theorem	Use active evidence, separating hypotheses, description independence, and minimality.
Reference sufficiency	Local theorem	Require total unique reference, factorization, and consequence completeness.
Qualification kernel B_S	Local construction	Fix internally supported record/referent, subject, channel, and contrast seeds without detached semantics.

Claim or structure	Status	Local burden
Least closure $\mu\Gamma_S$	Local theorem	Prove monotonicity on candidate fact sets, require consistency preservation, and determine qualified carriers, channels, standing, and profile simultaneously; inconsistent closures fail closed.
Subject index and self-involving closure	Local qualification	Ground source, ownership, referent, and later reuse in one interface.
Diachronic naturality	Local theorem	Show record and referent transports compose without drift.
Complete profile $\Pi_E^{(S)}$	Local fixed-scope construction	Admit every authoritative consciousness consequence, restrict it to the least closure, and provide a non-monotone scope-revision protocol.
First/third bridge	Local certificate theorem	Connect external measurements to internal records or referents invariantly relative to a declared $\mathcal{I}_C^{(S)}$.
Consciousness-description bridge	Local theorem under imported closure	Factor every local consciousness-bearing distinction through $\mathcal{I}_C$.
Qualitative tensor theorem	Local application of imported ATS	Distinguish standing effects, profile effects, and skin.
AASC consciousness identification	Downstream framework identification	Apply imported closure to standing-positive occupancy of the complete interface.
Machine and cross-subject criteria	Downstream applications	Supply realization, bridge, and gluing certificates.

B.4 Companion sources

Companion source	Locked role
Maley (2025)	Diagnostic result: record prediction without invariant consciousness reference is empirically non-self-sufficient.
Maley (2026c)	Projection and mediation results; quotient reference distinguished from privileged representative selection.
Maley (2026b)	Fixed-domain kernel, witness coherence, quotient descent, standing, persistence, symmetry obstruction, rigidity, and no-hidden-remainder.
Maley (2026a)	Anchor–tensor–skin effect exhaustion, skin non-authority, tensor role, no fourth same-domain role, and role uniqueness.

B.5 What must be shown locally

Local target	Successful challenge must show
Record-reference fixation	Determinate, reusable, separating consciousness evidence with no invariant quotient target.
Candidate qualification	A generic logger satisfies the full least closure, the closure operator fails monotonicity, or a seed/premise already assumes consciousness.
Profile completeness	A concrete invariant authoritative consequence omitted from Π_E or an admitted component without witnesses.
Consciousness bridge	A same-domain consciousness-bearing fact outside standing and Π_E with no hidden authority or scope change.
Qualitative tensor classification	A fourth effect channel beyond standing, profile change, and skin while imported ATS remains in force.
False-positive filter	A system satisfies every interface component yet remains nonconscious without a missing anchor.
Machine criterion	A full internally closed artificial interface is nonconscious despite the dependency-locked substrate-neutral identification.
Cross-subject comparison	Determinate comparison without fixed component reference and compatible joint mediation.

B.6 Objection routing

1. **Class U: upstream.** Reject an imported AASC dependency or show it is incorrectly cited.
2. **Class L: local.** Show failed typing, qualification, reference, subject indexing, channel naturality, bridge, or realization.
3. **Class P: profile.** Exhibit a missing or illicit profile component.
4. **Class S: scope.** Require a new carrier, boundary, witness family, or governance locus.

The classifier preserves exact local vulnerability while maintaining modular use of the upstream theory.

C Formal Type Specification and Mechanization Boundary

This appendix supplies a Lean-like type specification for the local consciousness construction. It is not presented as a machine-checked derivation; its function is to expose carrier types, closure obligations, theorem signatures, and imported dependency boundaries. Upstream AASC formalizations, where available, remain separate locked dependencies rather than being recreated here.

C.1 Candidate structures

structure BaseAASC where

```

State          : Type
PresEq         : State -> State -> Prop

```

```

Redesc      : Type
act         : Redesc -> State -> State
WitnessSig  : Type
eval       : State -> WitnessSig
standing    : State -> Bool
Transition  : Type
step        : Transition -> State -> Option State
pres_equiv  : Equivalence PresEq
redesc_cov  : ...
standing_inv : ...
transition_cov : ...

```

```

structure CandidateRecordSystem (A : BaseAASC) where

```

```

  Record      : Type
  RecPresEq   : Record -> Record -> Prop
  recAct      : A.Redesc -> Record -> Record
  forms       : A.State -> Record -> Prop
  forms_cov   : ...
  RecordOrbit : Type
  recQuot     : Record -> RecordOrbit

```

```

structure CandidateInterface (A : BaseAASC)

```

```

  (R : CandidateRecordSystem A) where

```

```

  Q0          : Type
  stateQuot   : A.State -> Q0
  rawRef      : R.RecordOrbit -> Q0 -> Prop
  Subject0    : Type
  subj0       : Q0 -> Option Subject0
  Channel     : Type
  recTransport0 : Channel -> R.RecordOrbit -> Option R.RecordOrbit
  refTransport0 : Channel -> Q0 -> Option Q0

```

C.2 Qualification and complete profile

```

structure Qualification

```

```

  (A : BaseAASC) (R : CandidateRecordSystem A)

```

```

  (IO : CandidateInterface A R) where

```

```

  ActiveRecord      : R.RecordOrbit -> Prop
  QualifiedReferent : IO.Q0 -> Prop
  internalRef       : ...
  uniqueRef         : ...
  subjectIndexed    : ...
  selfInvolving     : ...
  diachronic        : ...
  expStanding       : IO.Q0 -> Bool

```

```

nontrivialQual      : ...
pointwiseAdmissible : ...
qualificationSeed   : Set (R.RecordOrbit × IO.Q0)
closureMonotone     : ...

```

```
structure CompleteProfile where
```

```

Rec      : Type
Ref      : Type
Contr    : Type
Mem      : Type
Disc     : Type
Access   : Type
Self     : Type
Trans    : Type
Hist     : Type
Compat   : Type

```

```
structure QualificationPackage ... where
```

```

recordCarrier      : Set R.RecordOrbit
referentCarrier    : Set IO.Q0
profileSpace       : Set IO.Q0
refGraph           : R.RecordOrbit -> IO.Q0 -> Prop
channelFamily      : Set IO.Channel
profile            : IO.Q0 -> CompleteProfile
standing           : IO.Q0 -> Bool
qualificationClosed : QualificationClosed ...

```

```
structure ConsciousInterface ... where
```

```

QE          : Type
RecordPlus  : Type
ref         : RecordPlus -> QE
subj        : QE -> Subject
profile     : QE -> CompleteProfile
occurrence  : QE -> Prop
certification : QE -> Evidence -> Prop
naturality  : ...
profileComplete : ...

```

C.3 Key declarations

```
theorem least_qualified_interface_closure ... :
```

```

Nonempty CandidateSubinterfaceLattice ->
Monotone qualificationClosure ->
ExistsUnique fun I => LeastFixedPoint qualificationClosure I

```

```

theorem record_orbit_descent ... :
  Exists fun formsQ => ...

theorem reference_fixation_necessity ... :
  DescriptionIndependentConfirmation ->
  SeparatingHypotheses ->
  ExistsUniqueInvariantReference

theorem reference_fixation_sufficiency ... :
  TotalUniqueReference ->
  ConfirmationFactorsThroughReference ->
  ConsequenceComplete ->
  DescriptionIndependentConfirmation

theorem diachronic_reference ... :
  ChannelNatural c1 -> ChannelNatural c2 ->
  ChannelNatural (compose c2 c1)

theorem first_third_bridge ... :
  ThirdPersonEvidence m q ->
  Exists fun B => InvariantBridge B m q

theorem consciousness_description_bridge ... :
  DescriptiveAdequacy ->
  LeastQualifiedClosure ->
  ProfileComplete ->
  ATSEffectExhaustion ->
  forall d : ConsciousnessDistinction,
    FactorsThroughConsciousInterface d \ /
    HiddenSelector d \ / ScopeChange d

theorem qualitative_tensor ... :
  ProfileComplete -> ATSEffectExhaustion ->
  forall d : Descriptor,
    Anchor d xor QualitativeTensor d xor Skin d

theorem no_mere_logging ... :
  DetachedLogger x -> not (QualifiedConsciousInterface x)

theorem hard_remainder_normal_form ... :
  SameDomainPhenomenalRemainder h ->
  Anchor h \ / Tensor h \ / Skin h \ /
  HiddenSelector h \ / ScopeChange h

```

C.4 Formalization obligations

The elementary quotient and naturality proofs are routine relative to the imported interfaces. The principal obligations are:

1. defining active candidate evidence without presupposing consciousness;
2. constructing the least simultaneous qualified-interface closure and proving its determinacy;
3. representing non-vacuous separation of hypotheses;
4. stating local consequence completeness for all ten profile components;
5. distinguishing system transition, record transport, and referent transport;
6. representing scope change as a typed operation rather than contradiction;
7. importing ATS effect exhaustion without making classification definitionally trivial;
8. formalizing occurrence separately from empirical certification;
9. integrating empirical realization certificates with mathematical structure.

A complete mechanization must import the upstream AASC interfaces, expose the local qualification and certificate burdens as named structures, and avoid burying domain or completeness assumptions in typeclass inference.

C.5 Dependency order

1. Orbit equivalence and quotient universal property.
2. Transition, witness, and record-formation descent.
3. Raw-reference invariance and candidate profile schema.
4. Least qualified-interface closure and simultaneous carrier projection.
5. Non-vacuity plus description independence implies uniqueness.
6. Confirmation factorization implies invariance and sufficiency.
7. Channel naturality and composition.
8. Subject-index and self-involving qualification.
9. First-person/third-person bridge.
10. Complete profile and interface morphisms.
11. Consciousness-description bridge under explicit adequacy assumptions.
12. ATS descriptor effects and hard-remainder normal form.
13. Finite false-positive and positive boundary models.

C.6 Machine-checkable audit requirements

A machine-checked repository should expose an axiom report, public theorem register, finite countermodels for major assumptions, exact scope declarations, examples corresponding to Section 18, and a proof that no classical-choice section is used as an authoritative quotient representative unless explicitly declared.

D Consciousness Attribution Certificate

The certificate is intended for biological, clinical, animal, or artificial-system claims. It distinguishes structural occurrence from evidential certification. A certificate may be partial, but unresolved fields must be marked unresolved rather than silently treated as satisfied.

D.1 Certificate fields

No.	Field	Required content
1	Claim type	Occurrence, certification, realization, content, comparison, or boundary transition.
2	Target system	Concrete system, version, boundary, and environment included in state.
3	Evaluation scope	Temporal interval, task domain, and exact consciousness claim.
4	Raw state carrier	Variables, typing discipline, and omitted data.
5	Admissible redescrptions	Identity-preserving changes and independent certificates.
6	Generic standing	Base eligibility and fail-closed conditions.
7	Experiential standing anchors	Additional conditions required for the qualified interface.
8	Boundary behavior	Negative, undefined, termination, reset, recovery, and scope-change cases.
9	Witness basis	Internal and external data granted consequence-bearing authority.
10	Candidate record hierarchy	Traces, operational records, self-indexed records, interior records, and closure-qualified experiential records.
11	Record formation	Mechanisms or relations forming each record class and covariance evidence.
12	Candidate state quotient	Orbit relation and computational representation of quotient classes.
13	Candidate referent carrier	Competing point, interval, trajectory, subsystem, or relational targets in $\mathbf{Q}_0$.
14	Raw reference rule	Explicit ρ_0 , its input data, and handling of partial/coarse targets.
15	Qualification closure	Kernel B_S , lattice $\mathcal{L}_S$, monotone operator Γ_S , least fixed point, and proofs of the eight qualification conditions.
16	Qualified reference map	Final $\rho : \mathbf{R}^+ \rightarrow \mathbf{Q}_E$ and uniqueness argument.
17	Redescription audit	Evidence that formation, reference, subject, and profile are invariant.
18	Subject index	Grounded invariant relation fixing the relevant interface identity.
19	Self-involving closure	How records are used within the same subject-indexed system.

No.	Field	Required content
20	Diachronic channels	Memory, report, translation, and reuse maps with naturality tests.
21	First/third bridge	Model-relative calibration linking external measurements to interior records or referents in the declared $\mathcal{I}_C^{(S)}$.
22	Complete profile	Definitions and witnesses for Rec, Ref, Contr, Mem, Disc, Access, Self, Trans, Hist, Compat.
23	ATS role audit	Anchor, tensor, skin, hidden-selector, and scope-change classification of descriptors.
24	Nontriviality	An active qualified referent and a standing-compatible profile-admissible contrast/background/memory/null/-trajectory point, with a witness-supported separating qualitative component; simultaneous activity is not required.
25	Occurrence criterion	Exact conditions for Occ_C independently of current investigator certainty.
26	Certification criterion	Exact data and inference supporting Cert_C .
27	Realization map	Mapping from physical/computational structure to the reduced interface.
28	Empirical certification	Data supporting each structural correspondence.
29	Adversarial tests	Label permutations, refactorings, temporal rebinning, subsystem swaps, detached-logger controls.
30	Alternative models	Competing referents and why they are excluded.
31	Completeness statement	Included consequences, omissions, revision triggers, scope limits, and whether revision preserves, refines, splits, merges, or revokes earlier qualifications.
32	Residual uncertainty	Uncertainty over instantiation distinguished from ambiguity inside the model.
33	Dependency and jurisdiction statement	Imported AASC dependencies, local construction burden, and U/L/P/S classification of unresolved objections.

D.2 Verdict classes

1. **Certified full interface:** qualified structure and empirical attribution both supported.
2. **Structural occurrence plausible, certification incomplete:** realization model supports Occ_C but available evidence does not settle Cert_C .
3. **Referentially adequate, qualitatively incomplete:** reference fixed but Π_E or nontriviality unresolved.
4. **Mechanistically adequate, referentially incomplete:** record production modeled but target underdetermined.
5. **Record-predictive only:** outputs predicted without an internal interface model.
6. **Standing-negative at scope:** required anchors or qualified reference conditions fail.
7. **Indeterminate evidence:** data do not decide whether the interface is instantiated; this is

not automatically an ontic negative.

8. **Scope change required:** a new witness, boundary, or authoritative consequence is needed.

D.3 Minimum publication statement

A positive consciousness attribution should disclose at least fields 1–3, 5, 7–10, 13–16, 18–23, 25–27, and 30–33. A machine-consciousness claim should disclose all fields. A clinical negative claim should explicitly separate failed certification from established standing-negative occurrence.

E Failure-Mode Ledger

Failure mode	Signature	AASC diagnosis
Premature experiential naming	Candidate trace or state is called experiential before qualification.	Circular terminology; return to R_0 , Q_0 , and qualification conditions.
Evidence/constitution slide	Reference theorem is presented as direct proof of consciousness.	Proof-status collapse; bridge and explicative thesis omitted.
Record/referent collapse	Report token is declared to be the experience without a typed bridge.	Evidence and target conflated; circular confirmation.
Profile black box	“Complete experiential profile” invoked without components or witnesses.	Completeness assertion lacks a defined consequence basis.
Raw-label quale	Word, unit name, neuron label, or coordinate treated as phenomenal content.	Skin upgraded to tensor.
Null-effect quale	Qualitative difference asserted with fixed standing and fixed Π_E .	Skin, scope change, or denial of ATS completeness.
Generic tensor quale	Any operational invariant called qualitative.	Subject-indexed experiential discrimination condition fails.
Temporal-slice fiat	Earliest, latest, peak, or analyst-selected state declared experiential.	Hidden selector or undeclared temporal anchor.
Subsystem fiat	One symmetric subsystem selected as experiencer.	No-equivariant-selector violation.
Correlation identity	Strong covariance treated as reference or identity.	Witness mistaken for referent.
Mechanism-class sufficiency	Broadcast, integration, or prediction treated as complete consciousness criterion.	Mechanism mistaken for qualified interface.
Higher-order regress	Each reference is fixed by another representation without closure.	No final quotient reference; regress or hidden anchor.
Memory drift	Later record silently changes target.	Channel naturality failure or mis-reference.

Failure mode	Signature	AASC diagnosis
Repair by report	Later verbal coherence used to retroactively fix earlier ambiguity.	Post hoc repair within lineage.
External semantics	Observer assigns experiential meaning to a machine log.	Internal data closure and self-involving closure fail.
Detached logger false positive	Storage and output treated as sufficient.	Record hierarchy stops before interior/experiential qualification.
Self-monitoring false positive	Controller has self-model and memory but no nontrivial qualitative profile.	Operational tensor mistaken for qualitative tensor.
Frozen gauge	One presentation treated as ontologically privileged.	Skin made identity-bearing; re-description neutrality denied.
Canonical representative authority	Computational section treated as the referent itself.	Section/quotient conflation and symmetry risk.
Occurrence vs. certification	Lack of evidence reported as absence, or fluent output reported as occurrence.	Occ_C and $Cert_C$ are different predicates.
First/third collapse	Neural or behavioral measurement identified with experience by correlation alone.	Missing invariant bridge.
Sleep/person collapse	Loss of episode standing treated as loss of organism or person identity.	Continuity layers conflated.
Invisible inversion	Claimed qualitative change with fixed complete profile.	Skin, scope extension, or rejection of completeness.
Zombie remainder	Consciousness absence with identical full interface.	Hidden same-domain remainder under ATS exhaustion.
Tensor reification	Quale or compatibility structure said to act or repair.	Tensor/operator role confusion.
Anchor motion	Standing prerequisite optimized from within the same positive regime.	Boundary crossing or ill-typed scope change.
Second standing gate	Separate consciousness and admissibility predicates govern the same scope.	Competing classifier or bookkeeping.
No-report overreach	Absence of speech treated as absence of consciousness.	External record confused with occurrence.
No-report underdetermination	Surrogate marker treated as self-interpreting.	Record class changed but reference problem remains.
Pairwise gluing failure	Inconsistent cross-subject qualitative mappings.	No global comparison interface.
Copy identity duplication	One singular subject assigned to two symmetric independent continuers.	No selector; pluralize, quotient-collapse, or add asymmetry.

Failure mode	Signature	AASC diagnosis
Scope drift	Witnesses or system boundary altered mid-argument.	New regime presented as same-domain result.
Completeness over-claim	Local profile treated as scope-universal across every later scientific domain.	Invalid export beyond fixed domain.
Dependency-jurisdiction collapse	A demand to reprove AASC is presented as a local consciousness objection.	Classify as upstream U unless failed local instantiation is shown.
Profile-expansion refusal	An omitted authoritative consequence is identified but neither added to Π_E nor declared as scope change.	Profile lock violated; expand locally or retype the domain.
Machine exceptionalism	A full artificial interface is denied consciousness solely because of substrate.	Missing biological anchor or upstream rejection of substrate neutrality required.

E.1 Failure-localization rule

When several failures appear, classify the earliest structural failure. An externally interpreted log fails internality before its qualitative profile is assessed. A descriptor changing standing is anchor-level before it can be classified as tensor or skin. A lack of certification should be recorded before any ontic verdict is inferred. This priority prevents downstream symptoms from being mistaken for independent results.

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